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Neural means and kernel corrections for operator learning

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Abstract

A neural network mean followed by an exact Matérn kernel regression has, on two public emulation problems with published baselines, two distinct uses. On the structural-mechanics benchmark of de Hoop et al., where kernel methods and neural operators are competitive, the kernel corrects the residual of a stacked neural mean: the pipeline reaches $4.572 \pm 0.010\%$ test error over ten seeded replicates, below every published method except PARA-Net at 4.55%, a tie within the test block's sampling error, and $5.433 \pm 0.093\%$ in the 1250-sample protocol against a published 6.49%. On the OCO-2 radiative-transfer problem of Lamminpää et al., where the same kernel on the raw atmospheric state trails the network tenfold, the kernel belongs on the network's learned features: there it overtakes the network, and a per-coordinate combination improves on the published Gaussian-process emulator on both of that problem's error metrics on all three spectral bands, at ten seeds per band. The replicates also show why ensembles saturate on structural mechanics: the residuals of every architecture we train correlate above 0.83 at every seed and their shared component is flat in diversity and sample size, consistent with a floor set by the benchmark data. The supporting bounds are checked numerically.

Keywords: operator learning, kernel ridge regression, deep kernel learning, ensembles, conformal prediction, emulation

1 Introduction

Operator learning constructs fast surrogates for the solution maps of parametric partial differential equations and physical forward models.[1] Neural operators (Kovachki et al., 2023; Lu et al., 2021; Li et al., 2021) learn their own representations; kernel and Gaussian-process methods (Batlle et al., 2024; Mora et al., 2025; Owhadi and Scovel, 2019) come with exact solves and error bounds and a strong record on the same benchmarks. Placing a neural network in the mean of a kernel or Gaussian-process regression is an established way of combining the two (Mora et al., 2025), and so is putting a kernel on a network's learned features (Wilson et al., 2016; Calandra et al., 2016). This paper takes a plain version of that pairing — a neural mean, trained in the metric the application reports, followed by an exact Matérn kernel regression fitted after the network — and asks, on two public problems with published baselines, what the kernel stage is for. The answer differs between the two

1. Code, data pointers, and the per-run summaries behind every number reported here are at `https://github.com/yspennstate/neural-means-kernel-corrections`.

Table 1: The two problems studied in depth. OCO-2 rows compare against the Gaussian-process emulator of Lamminpää et al. (2025), scored from its own stored predictions on the same 2000 test points, in both of that paper's error metrics (Section 5); ours are mean $\pm$ standard deviation over ten seeds per band. Structural-mechanics rows quote the best published mean relative L^2 test error at each training size (de Hoop et al., 2022; Mora et al., 2025), reported without uncertainties in the original work; ours are mean $\pm$ standard deviation over ten seeds, and the 20000-sample comparison is a tie within the test block's own sampling error (Section 4.4).

Problem	Metric	Published	This work
OCO-2, O2 band	reduced	16.89%	$4.12 \pm 0.05\%$
	radiance	0.0448%	$0.0294 \pm 0.0020\%$
OCO-2, weak CO_2	reduced	24.06%	$16.28 \pm 0.06\%$
	radiance	0.0599%	$0.0507 \pm 0.0052\%$
OCO-2, strong CO_2	reduced	16.14%	$8.08 \pm 0.01\%$
	radiance	0.1147%	$0.0584 \pm 0.0041\%$
Structural mechanics, 20000 samples	relative L^2	4.55% (PARA-Net)	$4.572 \pm 0.010\%$
Structural mechanics, 1250 samples	relative L^2	6.49% (FNO-mean GP)	$5.433 \pm 0.093\%$

problems, and the difference is the subject of the paper. Table 1 collects the published numbers and ours.

On the structural-mechanics benchmark of de Hoop et al. (2022) the two families are competitive: the published neural operators and the kernel method of Batlle et al. (2024) sit within 0.65 points of one another, and our own kernel ridge reproduces the published kernel number. There the kernel's role is to correct the residual of a stacked neural mean, and the corrected pipeline reaches $4.572 \pm 0.010\%$ under the 20000-sample protocol, replicated at ten independently seeded runs: below every published method but PARA-Net's 4.55%, and within one standard error of that one once the test block's own sampling error is counted, which we read as a tie. Under the 1250-sample protocol of Mora et al. (2025) it reaches $5.433 \pm 0.093\%$ against a published best of 6.49%. The published numbers on this benchmark carry no uncertainties, and replication is what lets gains of a few hundredths of a point be resolved at all; it also shows why ensembles saturate there. The residuals of every architecture we train correlate between 0.83 and 0.97 at every seed; their shared component concentrates at the corners of the loaded edge, where the finite element solution is least reliable, and is flat in ensemble diversity and in training-set size; and the spread of our own pipeline across seeds, 0.010 points, is some sixty times smaller than the span of the published architectures, so the published cluster near 4.5% is not run-to-run noise. That evidence is consistent with a floor set by the benchmark's data rather than by any surrogate class, but it is circumstantial: the experiment that would decide it — re-solving a set of the hardest cases on a refined mesh or with higher-order elements and comparing the solver discrepancy with the shared residual — has not been run, and Section 4.6 says what each measurement does and does not establish.

On the OCO-2 radiative-transfer emulation problem of Lamminpää et al. (2025) the same components separate by an order of magnitude, and the kernel's role changes. Per spectral band of the satellite, the task maps a reduced atmospheric state to a reduced radiance spectrum, and the baseline is that paper's kernel-flow emulator, whose stored predictions we score on the same test points. An exact Matérn regression on the raw state reaches 40% on the reduced metric and the network 4.4%; the same regression on the trained network's features reaches 4.1%, past the network it feeds on, at ten of ten seeds on each band. The comparison is controlled — the same kernel, tuning routine and budget on both inputs — and it locates the difference: the effective dimension of the two Gram matrices is essentially the same, while the target's squared native-space norm falls by a factor of about forty, which is what the pullback identity of Proposition 6.17 says a learned feature map can do and a rescaling of the input cannot. Combined per output coordinate, across networks trained in the flat and in metric-weighted losses and the kernel heads on their features, the surrogate improves on the emulator on both of that problem's error metrics on all three bands, at ten seeds per band; one band's radiance margin is thin, and Section 5 gives the per-seed counts.

What this paper adds to the prior constructions is therefore not the pairing but what it is used to measure: an exact post-hoc residual correction at $n = 19000$, one dense Cholesky factorization rather than an approximation; the controlled raw-input versus learned-feature comparison above; the per-coordinate treatment of OCO-2's two metrics, which trade off coordinate by coordinate and had been won by different models; the residual-correlation diagnosis of why ensembles saturate on the structural-mechanics benchmark; and a replicated audit, at ten seeds, of every stage. The theory of Section 6 supports the deployed estimators rather than idealizations of them, and is kept in proportion: a symmetrization inequality behind the reflection averaging; a second-moment identity that predicts the stacked risk from measured residual correlations and brackets how close any convex ensemble sits to its floor; capacity bounds for the fitted stacks and the grid-tuned correction, with constants measured on the runs; the residual-norm bound of Theorem 6.18, which is the one statement we call a certificate, and the pullback identity; an informal rate calculation (Proposition 6.16) for what an input metric can buy a kernel, whose rate half the OCO-2 sweep does not bear out; and a distribution-free coverage statement whose hypotheses the calibration protocol satisfies. The last matters because the design factor in Theorem 6.18, though a valid bound, ranks the pointwise error poorly on the benchmark, and its split-conformal rescaling is the one uncertainty signal that survives our tests (Section 4.8). Each statement is checked numerically (Appendix C); the two-member stacking rule is scored prospectively on every member pair of every seed and reported as it came out, reliable outside a margin band set by its own estimation noise and no better than chance inside it (Section 5.2).

Section 2 describes the problems, protocols, published results and related work, and includes a survey of the same recipe on the remaining problems of the operator-learning suite, which is uncontrolled and reported as such. Section 3 specifies the pipeline. Sections 4 and 5 report the two studies. Section 6 states the supporting theory, with proofs in Appendix A and their numerical verification in Appendix C, and Section 7 discusses limitations.

2 Benchmark, protocol, and related work

2.1 Problem and data

The dataset originates in the cost-accuracy study of de Hoop et al. (2022) and is the one distributed with Batlle et al. (2024) and Mora et al. (2025). An isotropic elastic plate occupies $D = (0, 1)^2$; the displacement field w satisfies $\nabla \cdot \sigma = 0$ with a fixed constitutive law, displacement conditions on the bottom and lateral parts of the boundary, and a prescribed normal traction $\bar{t}$ on the top edge $\Gamma_t = [0, 1] \times \{1\}$. The quantity of interest is the von Mises stress field σ_v on D. The learning task is the map $G : \bar{t} \mapsto \sigma_v$. Loads are drawn from the Gaussian field $\mathcal{GP}\big(100,\ 400^2(-\Delta + 3^2 I)^{-1}\big)$ with homogeneous Neumann boundary conditions for the Laplacian; outputs are finite element solutions interpolated to a regular 41×41 grid, and the load is sampled at 41 points. The distributed file contains 40000 input/output pairs; the input array stores the load broadcast along the second grid coordinate, which we verified is an exact copy (maximal deviation 0 across all 40000 samples), so all our methods consume the load as a vector in $\mathbb{R}^{41}$.

Errors are mean relative L^2 errors over the test set,

$$\frac{1}{N_{\text{test}}} \sum_n \frac{\|\widehat{G}(u_n) - G(u_n)\|_2}{\|G(u_n)\|_2},$$

computed on grid values in double precision. Quadrature weighting (trapezoidal instead of plain vector norms) changes our reported numbers by less than 0.02 percentage points, and we report the plain-norm convention of the prior work.

2.2 Protocols and published results

Two regimes appear in the literature, and we follow both. In the *high-data* protocol the first 20000 samples are available for training and the last 20000 form the test set; de Hoop et al. (2022) report, at 20000 training samples, 4.55% (PARA-Net), 4.67% (PCA-Net), 4.76% (FNO) and 5.20% (DeepONet), and Batlle et al. (2024) report 5.18% for their optimal-recovery kernel method (Matérn/rational quadratic; 27.11% for the linear kernel) on the same task. Within the 20000-sample budget we hold out the last 1000 samples (of a fixed permutation) for model selection and stacking, and train on the remaining 19000, so no method of ours sees more than the 20000 training samples used by the baselines. In the *low-data* protocol of Mora et al. (2025), 1250 samples are available and the same 20000-sample test set is used; their table gives 8.70% (DeepONet), 6.62% (FNO), 6.95% (the kernel method of Batlle et al. 2024), 6.74% (their zero-mean GP), 7.12% (DeepONet-mean GP) and 6.49% (FNO-mean GP). In this regime we carve the validation split (250 samples) out of the 1250, so training uses 1000 samples and every choice made by the pipeline is informed by the 1250 available labels only.

2.3 The remaining problems of the suite: an uncontrolled survey

For orientation we also ran one fixed recipe, unchanged, on the other problems of the operator-learning suite that Batlle et al. (2024) assembled from de Hoop et al. (2022) and Lu et al. (2022): a Matérn kernel at the median length scale with the fit capped at 6000 points, a residual MLP mean, outputs reduced by PCA, and the stage selected on validation.

Table 2: The remaining problems of the suite: an uncontrolled survey. Best published mean relative L^2 test error on each problem (per-problem methods tuned by their authors), against one recipe run unchanged and once per problem, without seed replication, on training splits and grids that differ from the baselines' in the cases noted in the text.

Problem	Map	Best published	This work (single run)
Burgers	initial condition $\to$ solution	1.93% (FNO)	2.38%
Darcy flow	permeability $\to$ pressure	2.32% (POD-DeepONet)	2.01%
Advection I	smooth pulse $\to$ solution	$\approx 0\%$ (kernel)	—
Advection II	discontinuous pulse $\to$ solution	11.28% (linear kernel)	11.29%
Helmholtz	wavespeed $\to$ field	1.00% (kernel)	1.75%
Navier–Stokes	vorticity $\to$ vorticity	0.12% (kernel)	0.51%

Table 2 lists the results next to the best published number on each problem. The comparison is not controlled, and we draw nothing from it beyond orientation: the published numbers are per-problem methods tuned by their authors; our rows are single runs without seed replication; Burgers and Darcy use slightly smaller training splits than their baselines (800 and 824 pairs) and Darcy a different grid, from the copies we could obtain; and we report no number for Advection I. What the survey shows is the sorting one would expect: where the map is smooth and data are plentiful (Helmholtz, Navier–Stokes, Darcy) the kernel stage wins the recipe's internal selection and lands within a factor of two of the tuned per-problem kernels, and on Burgers the corrected mean sits near the Fourier neural operator. The two problems studied in depth, with replication, are the subject of the rest of the paper.

2.4 A data-driven check of the mirror symmetry

The continuous problem is invariant under $x_1 \mapsto 1 - x_1$: the domain and boundary partition are symmetric, and the input law has a symmetric covariance. On the grid, reflecting the load (S) should reflect the stress field along the first grid coordinate (T), i.e. $G(Su) = TG(u)$. Rather than assume this, we test it on the data. For each of the first 200 samples we searched all 40000 loads for the nearest neighbor of the reflected load Su_i and compared the corresponding outputs. For the five best-matching pairs, the input mismatch $\|Su_i - u_j\|/\|u_j\|$ ranges over 0.27–0.32, while the output mismatch $\|Tv_i - v_j\|/\|v_j\|$ ranges over 0.085–0.14; reflecting along the second coordinate instead gives output mismatches of order one. Outputs of near-mirror inputs are far closer than the inputs themselves, which is what equivariance plus a Lipschitz solution map predicts, and the effect singles out the correct output reflection axis. A complementary check on the input law: the empirical mean and covariance of the training loads are invariant under reflection to within 1.1% and 1.5% respectively, as required for Proposition 6.1. Consistently with all of this, reflection averaging at test time improves twenty-nine of the thirty reflected members of the seed campaign (Section 4), which is what Proposition 6.1 predicts: it constrains the expected loss, so a finite test sample is free to go the other way, and at one seed it does.

2.5 Related work

The benchmark sits at the meeting point of three lines of work. *Neural operators* learn maps between function spaces: DeepONet (Lu et al., 2021) with its branch/trunk factorization, the Fourier neural operator (Li et al., 2021) and the broader neural-operator framework (Kovachki et al., 2023), and the reduced-basis PCA-Net and PARA-Net architectures assembled for the cost–accuracy study of de Hoop et al. (2022). These methods are fast and mesh-flexible but do not by themselves come with error control, and it is their numbers on this problem that define the plateau we start from. *Kernel and Gaussian process methods* for operator learning approach the same maps through reproducing kernels: the optimal-recovery framework of Batlle et al. (2024), with its convergence theory and a-priori bounds, is the most direct comparison, and it is competitive with neural operators on most of the benchmarks it considers. Data-adapted kernels learned by cross-validation (Darcy et al., 2023) and vector-valued kernel formulations extend the reach of this family. *Hybrids* that place a neural network and a kernel in the same estimator are the closest relatives of our pipeline: Mora et al. (2025) use a neural operator as the mean of a Gaussian process and fit the two together, and this is the work we most directly build on, differing in that we fit the mean first and the kernel after, tune by cross-validation rather than marginal likelihood, and solve the correction exactly at nineteen thousand points. Kernels on learned representations are likewise established: deep kernel learning (Wilson et al., 2016; Calandra et al., 2016) places a Gaussian process on the features of a network and trains through it. Our feature-space stage is the post-hoc form of that construction, with the exact solve and validation tuning used everywhere else in the pipeline, and what we add is not the construction but the controlled comparison of the same kernel on the raw input and on the learned features under one tuning routine and budget (Section 5).

Two further threads inform the design. The kernel-flow method (Owhadi and Yoo, 2019) learns a kernel from data by minimizing a half-sample cross-validation loss, and its use as a regularizer for the inner layers of a network (Yoo and Owhadi, 2021) is exactly the term we analyze in Proposition 6.2; kernel flows have since been used at scale as emulators for physical forward models, for instance in atmospheric radiative transfer retrievals (Lamminpää et al., 2025) and in the inference of convective-storm structure from satellite observations (Prasanth et al., 2021), where presenting the input in its physical parametrization and adapting the kernel to data are what make the emulator accurate. Our pipeline follows the same instinct, with the smoothing of the target performed by a neural ensemble rather than by a hand-chosen reduction. Finally, the effective-dimension reading of the kernel solve (Lemma 6.21) draws on the random-matrix description of kernel Gram spectra (Koltchinskii and Giné, 2000) and on the classical Marčenko–Pastur (Marčenko and Pastur, 1967) and spiked-covariance (Baik et al., 2005) laws; the same effective dimension governs the generalization of kernel ridge regression (Caponnetto and De Vito, 2007).

3 Method

The pipeline has three stages: neural means, stacking, and kernel corrections. All components operate on the load vector $u \in \mathbb{R}^{41}$ (standardized by training statistics) and produce stress fields in $\mathbb{R}^{41\times 41}$; all networks are trained with the reported metric as the loss, i.e. the per-sample relative L^2 error in original units, which we found mildly but consistently better than mean squared error on normalized targets.

3.1 Neural means

Residual MLP. The primary mean is a residual multilayer perceptron: three or five residual SiLU blocks of width 1024–1536 mapping $\mathbb{R}^{41} \to \mathbb{R}^{1681}$ directly. This is essentially PARA-Net (de Hoop et al., 2022) with a modern training recipe, and it also supplies the penultimate features used by the feature-space kernel stage.

Kernel-conditioned refiner. The second mean is the same residual network given an extra input channel: the kernel method's predicted stress field for the same load, concatenated with the load. It therefore learns to correct the kernel predictor rather than to reproduce the map from scratch, and it is the strongest single model in our study. During training the refiner reads *out-of-fold* kernel predictions (four-fold, so the kernel channel is never fit on the target sample), and at test time the full-data kernel prediction, so the channel never sees its own target.

Other instances. The mean slot is not tied to a particular architecture. We also use a Fourier neural operator (Li et al., 2021) that consumes the broadcast load as a field and an MSE-trained variant of the MLP; both are drop-in means, both enter the ensemble of Section 4.6, and their pairwise error correlations are one of the measurements of this paper. A UNet on the same field representation was part of the pre-specified configuration but completed at only two of the ten campaign seeds, so it is absent from the reported ensemble (Section 4.1). We additionally implement an encoder–decoder transformer (Vaswani et al., 2017; Dosovitskiy et al., 2021) that tokenizes the 41 load samples and decodes the field by cross-attention at the grid points; it is a natural further architecture family, but we were unable to train it to the level of the others on the hardware available for this study and report the ensemble without it (Appendix B).

All means are trained with AdamW and a cosine schedule, batches of 128–256, for up to a few hundred epochs (Appendix B lists every hyperparameter). During training each batch is reflected with probability 1/2 (load reversed, target field reflected along the first grid coordinate); at test time each model is evaluated as $\frac{1}{2}(f(u) + Tf(Su))$. Proposition 6.1 shows the test-time step cannot hurt on average, and the ablation confirms small consistent gains from both steps. Optionally, the training loss carries a kernel-flow term $\beta\, e_2$ computed from the batch (Section 6.2) with a Gaussian kernel on pooled penultimate features whose log-bandwidth is trained jointly; this variant appears in the ablation.

3.2 Stacking

With M trained models $f_1, \ldots, f_M$ we form the convex combination $m(u) = \sum_m w_m f_m(u)$, $w \in \Delta^{M-1}$, minimizing the validation relative error; the weights are initialized at the simplex minimizer of the measured second-moment matrix of Proposition 6.8 and polished by a short search on the 1000 held-out samples (250 in the low-data protocol). A per-pixel variant allows the weights (plus an intercept) to vary over the grid, ridge-fitted on half the validation split and accepted only when it beats the global weights on the other half; it is the variant the final pipeline uses. Stacking on a held-out split rather than uniform averaging costs nothing, guards against a weak member (Wolpert, 1992), and is statistically almost free at this size (Proposition 6.3).

3.3 Kernel corrections

The stacked mean is then corrected by kernel ridge regression of its residuals. Let $X = (u_1, \dots, u_n)$ be the training loads (standardized coordinatewise), $R \in \mathbb{R}^{n\times 1681}$ the matrix of training residuals $v_i - m(u_i)$, and k a Matérn-5/2 kernel $k(u, u') = \kappa\big(\|u - u'\|/s\big)$. The correction is

$$\widehat{r}(u) = k(u, X)\,\alpha, \qquad \alpha = (K + n\lambda I)^{-1}R,$$

with (s, λ) chosen on the validation split from a small grid (s as a multiple of the median pairwise distance, $\lambda \in \{10^{-6}, 10^{-5}, 10^{-3}\}$; the grid is searched on an 8000-sample subsample and the chosen pair is refit on all of X). The corrected surrogate is $m+\widehat{r}$. A second corrective stage repeats the construction with the kernel evaluated on deep features, the concatenated penultimate activations of the ensemble members (reflection-averaged, standardized), fitted to the residuals of $m + \widehat{r}$; it contributes a smaller improvement and is included when it helps on validation. At $n = 19000$ the exact solve is a single Cholesky factorization of a 19000×19000 matrix, under half a minute in double precision on a laptop CPU (measured in Section 4.10), and prediction is two matrix products; no inducing points, preconditioners, or stochastic solvers are involved. The Gaussian process reading of the same formulas supplies the posterior standard deviation $P_\lambda(u)$ of (2) at negligible extra cost (one triangular solve per evaluation batch), which is the design factor in the residual bound of Theorem 6.18 and the uncertainty signal studied in Section 4.8.

Two remarks on scope. Nothing in the construction uses properties of this particular PDE beyond the verified reflection symmetry, so the recipe (native input parametrization, symmetrized accurate means, validated kernel correction of the residual, a conformal band built on the posterior standard deviation) transfers to other operator learning problems with low-dimensional input parametrizations. And the pipeline degrades gracefully: dropping the feature stage, the stacking, or the corrections recovers progressively simpler methods whose individual numbers appear in the ablation table.

4 Results on the structural-mechanics benchmark

4.1 What is replicated, and at how many seeds

Every structural-mechanics number in this section is reported over ten seeds unless it is marked otherwise. A seed here is not a reinitialization of one network: it redraws the 1000-sample validation split from the training pool and reinitializes every trainable member, so the seed spread contains both the model-selection noise and the optimization noise. The test block of 20000 samples is fixed by the benchmark's convention and never moves, at any seed.

Five members are available at all ten seeds: the kernel ridge on loads, the residual MLP trained on the reported metric, the same architecture trained on normalized mean squared error, the kernel-conditioned refiner, and a Fourier neural operator. The stack and the residual correction are run at all ten seeds from those five, and so are the second-moment measurement and the conformal calibration.

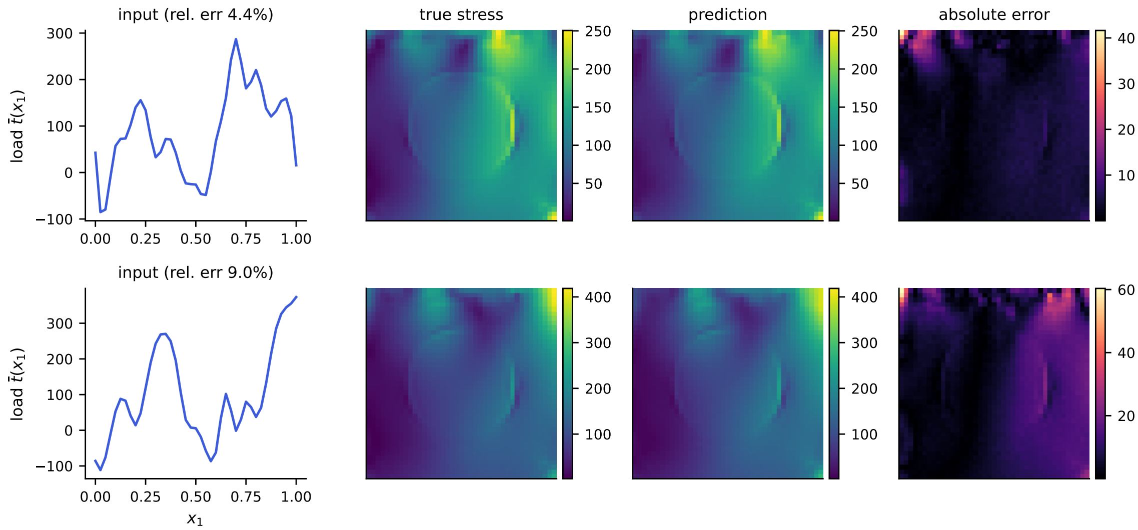

Figure 1: Example predictions of the full pipeline. Top: a median-error test case; bottom: a case at the 98th error percentile. Columns: input load, true von Mises stress, prediction, and pointwise absolute error (note the smaller scale). Errors concentrate in the high-traction corner, as also reported by Mora et al. (2025).

Two runs are incomplete, and their stopping rules are stated here once. The FNO was stopped at every seed by the campaign's 36-hour per-task wall clock and is finalized from the best-validation checkpoint its trainer wrote before the stop, which falls between epochs 10 and 70 of a scheduled 200, so its cosine schedule never annealed; nothing is retrained, and Appendix B records how far each run got. The one full-schedule run of this architecture we have scored 4.705% against the 4.754% the checkpoints average, so the truncation costs the member accuracy; the tables report the truncated member because it is what the campaign produced. A sixth member, a UNet, was scheduled after the FNO and completed at two of the ten seeds before the wall clock, so the six-member configuration that included it is not reported; as a preliminary observation only, at those two seeds the corrected six-member pipeline scored 4.539% and 4.537%.

The five-member set is not a configuration searched for after seeing results: it is the pre-specified six-member configuration minus the member the campaign could not supply. Whether five members are preferred to four is decided on validation, per seed, by the same rule that governs every other stage choice here, and it prefers five at ten seeds out of ten (Section 4.3). The four-member pipeline is reported alongside throughout, because the FNO's admission is a measurement and the reader should see what it was measured against.

The low-data pipeline and the OCO-2 combination have the seed coverage stated where they appear, which is ten in both.

4.2 Main comparison

Table 3 reports the high-data protocol. The published numbers are quoted from de Hoop et al. (2022) (as tabulated by Batlle et al. 2024) and from Batlle et al. (2024); our rows are

Table 3: Structural mechanics, high-data protocol (20000 training samples, 20000 test). Mean relative L^2 test error. Rows above the rule are published results on the same data, quoted without uncertainties because none were reported. Rows below are ours: mean $\pm$ standard deviation over ten seeds. The four-member row is the same pipeline with the FNO removed, kept because the FNO's admission is a measurement rather than a design choice (Section 4.3). TTA denotes reflection averaging at test time.

Method	Error (%)	Parameters
DeepONet (de Hoop et al., 2022)	5.20	–
FNO (de Hoop et al., 2022)	4.76	–
PCA-Net (de Hoop et al., 2022)	4.67	–
PARA-Net (de Hoop et al., 2022)	4.55	–
Optimal-recovery kernel (Batlle et al., 2024)	5.18	–
Kernel ridge on loads (ours)	5.197 ± 0.003	–
Residual MLP + reflection TTA	4.836 ± 0.018	4.9M
+ affine stack over five members	4.595 ± 0.008	22.9M
+ residual kernel correction	4.572 ± 0.010	22.9M
the same pipeline without the FNO	4.611 ± 0.005	16.5M

computed under the split of Section 2, which gives our methods strictly no more training information than the baselines had. The kernel ridge row reproduces the published kernel result almost exactly (5.197% against 5.18%, and it is the most reproducible row in the table: its seed standard deviation is 0.003 points, three thousandths of a point, because the only randomness it carries is the validation draw). We take that agreement as evidence that the protocols are aligned, so the gap opened by the rest of the pipeline is not an artifact of evaluation choices.

The replicated pipeline reaches $4.572 \pm 0.010\%$, the uncertainty being the spread over seeds. That is below every published method except PARA-Net, and it is not separable from PARA-Net: the difference is 0.022 points against an unpaired standard error of 0.020 (Section 4.4), so about one. We claim a tie there and nothing more. Against PCA-Net (4.67%) the separation is 0.098 points, five standard errors, and the margins over the FNO (4.76%), DeepONet (5.20%) and the optimal-recovery kernel (5.18%) are larger still.

One correction to an earlier version of this work belongs here. That version reported this configuration at 4.55% from a single run and presented the number as matching PARA-Net. Rebuilt across ten seeds, first without the FNO and then with it, the pipeline gives $4.611 \pm 0.005\%$ and $4.572 \pm 0.010\%$: the four-member figure sits three standard errors above PARA-Net and the five-member figure is level with it, so the earlier claim survives only at the configuration that carries the spectral member, and only as a tie. Section 4.6 presents evidence, consistent with a floor in the benchmark data rather than a coincidence of tuning, for why the published cluster sits where it does, and says what would settle it.

Table 4: Low-data protocol (1250 training samples, 20000 test). Published rows from Mora et al. (2025). Our pipeline sees only the 1250 labels (250 of them used as the validation split). The pipeline row is mean ± standard deviation over ten seeds. Which stage the validation split selects varies with the seed — the input-space kernel correction at six seeds, its two-stage form at three, the feature-space one at the remaining seed — so the row is the error of the selected pipeline, not of a fixed final stage. The kernel-ridge row is a single run at the multiscale setting.

Method	Error (%)
DeepONet	8.70
GP, optimal recovery (Batlle et al., 2024)	6.95
Zero-mean GP (Mora et al., 2025)	6.74
FNO	6.62
FNO-mean GP (Mora et al., 2025)	6.49
Kernel ridge on loads (ours, multiscale, one seed)	6.66
Full pipeline (ours, ten seeds)	5.433 ± 0.093

4.3 Members and the stack, across seeds

Table 5 gives the five replicated members and the stages built on them. Four features of it are worth naming.

The ordering of the members is stable at the top and not below it, and the seeds are what separate the two cases. Scored on the reflection-averaged validation predictions that the stack consumes, the normalized-MSE network is the most accurate single member at all ten seeds — not nine, ten. Second place is not stable: the refiner takes it at six seeds and the FNO at four, and on test the two are separated by 0.024 points against the FNO's own seed spread of 0.031. So the first rank survives ten independent redraws of the validation split and ten reinitializations and is not a coincidence of tuning; the second is a coin toss that a single-seed table would have reported as a fact. That contrast is the argument for the seeds, in one measurement.

The FNO is the interesting member, and not because it is accurate. At $4.754 \pm 0.031\%$ it is third on average, behind both residual-MLP variants, though it passes the refiner at three seeds of the ten; its seed spread is three to five times theirs — a consequence of checkpoints early-stopped anywhere between epoch 10 and 70 (Appendix B). Yet removing it costs the finished pipeline 0.039 ± 0.011 points, and validation prefers keeping it at ten seeds out of ten. Accuracy is not what a member contributes to a stack; the second-moment analysis of Section 4.6 says what is, and the simplex weights there give the FNO the largest single share despite its being third on the metric.

The stages are more reproducible than the members they are built from, and the way the seed spread moves along the chain is itself informative. Equal weighting, which fits nothing, takes the members' spread down to 0.005 points on test — ordinary variance reduction. Each subsequent stage fits something, and each gives a little of that back: fitted global weights 0.006, the affine stack 0.008, the correction 0.010. The end of the chain is both the

Table 5: Members and stages under the high-data protocol, mean ± standard deviation over ten seeds. Member errors are reflection-averaged where the member has a reflection; the kernel ridge and the refiner train with the symmetrization already applied. "Fitted" weights are solved on the validation split and read out once on test; equal weights involve no fitting and are reported on both splits.

Member or stage	Family / loss	Test error (%)
Kernel ridge on loads	Matérn-5/2	5.197 ± 0.003
Residual MLP	MLP, metric loss	4.836 ± 0.018
Residual MLP	MLP, normalized MSE	4.712 ± 0.006
Kernel-conditioned refiner	MLP on $(u, \mathrm{KRR}(u))$	4.730 ± 0.006
Fourier neural operator	spectral, metric loss	4.754 ± 0.031
Equal-weight ensemble	no fitting	4.648 ± 0.005
Fitted-weight ensemble	solved on validation	4.627 ± 0.006
Affine stack (per-pixel)	selected on a held-out half	4.595 ± 0.008
+ residual kernel correction	–	4.572 ± 0.010

most accurate stage and the least reproducible one. Fitting is not free, and a stack that is tuned harder is not automatically a stack that reproduces better.

Every stage gain is small and every one is resolved. Pairing within each seed on the test set, equal weighting beats the best single member by 0.064 ± 0.007 points, fitted weights beat equal weighting by 0.021 ± 0.003, the affine stack beats fitted global weights by 0.032 ± 0.005, and the kernel correction beats the stack by 0.024 ± 0.005. All four improve at ten seeds out of ten, as does the end-to-end figure of 0.140 ± 0.014 points off the best member the pipeline contains. Each is a few hundredths of a point, which is precisely the resolution a single seed cannot deliver.

Three selection rules were exercised rather than assumed, and all three are decided without touching the test block. Per-pixel affine weights are fitted on half the validation split and accepted only if they beat global convex weights on the other half: accepted at all ten seeds with five members, where with four they were declined at one. The correction is accepted on validation at all ten. And the member set itself is chosen the same way — validation prefers five members to four at ten seeds out of ten, which is what admits the FNO. The test-set consequence of that last decision, 0.039 ± 0.011 points, is read out afterwards and plays no part in making it.

4.4 Which uncertainty is the binding one

Seeds are not the only source of uncertainty in Table 3, and on this benchmark they are not the largest. Every seed is scored on the same fixed 20000-sample test block, so the seed standard deviation measures training and selection variability and nothing else. The test block is itself a sample. Its contribution is the standard error of the mean relative error over those 20000 cases, and at a per-sample standard deviation of 0.0169 that is 0.012 points, slightly larger than the pipeline's seed spread of 0.010. Treating the two as independent, the uncertainty on our reported 4.572% is about 0.016 points, and the seeds contribute a little

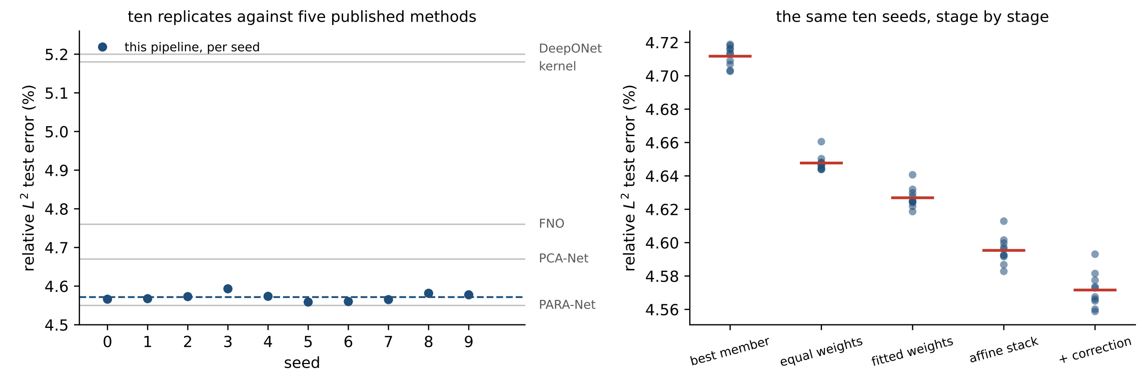

Figure 2: Left: the corrected pipeline's test error at each of the ten seeds (points), their mean (dashed), and the five published single numbers on the same axis (grey). The scatter of one pipeline across seeds is a small fraction of the differences the benchmark is used to adjudicate, so the published cluster cannot be attributed to run-to-run noise. Right: the same ten seeds followed down the pipeline, each stage scored on test, with the stage mean marked. Fitting per-pixel weights lowers the mean and widens the scatter; the correction lowers both.

under half of it. A study that replicates seeds and stops has measured the smaller half and is liable to quote it as though it were the whole.

This decides what the table can and cannot support. Comparisons against a published number are unpaired, because we do not hold those methods' per-sample predictions, and so carry the test-block error twice: about 0.020 points in quadrature. The separation from PCA-Net, 0.098 points, is five of those and real. The separation from PARA-Net is 0.022 points, about one, and is therefore not a separation at all. Both figures are conservative, since scoring on identical test points makes the two errors positively correlated in a way we cannot exploit without the baselines' predictions.

Comparisons inside the pipeline are paired, and pairing is worth far more here than seeds are. Evaluating the corrected surrogate and the best single member on the same cases removes the shared difficulty of each case, and the gain of 0.140 points is then resolved at roughly a hundred standard errors at every seed. The same holds for the FNO's admission: 0.039 points, paired, is overwhelming on any one seed, while the interesting quantity — whether it survives retraining — is the across-seed spread of ±0.011 and the count of ten seeds out of ten. The practical rule the budget implies is the ordinary one, and the campaign is what made it visible: replicate seeds to learn whether a design choice survives retraining, pair on the test set to learn whether it helps, and never use either number to answer the other's question.

4.5 Ablation

Table 6 builds the pipeline up one component at a time. The kernel ridge on loads and the reflection-averaged neural mean start at 5.197% and 4.836%. Stacking is trivial with a single high-data mean and leaves it unchanged; the value of the combination appears once

Table 6: Building the high-data pipeline. Mean relative L^2 test error under the 20000-sample protocol; "+" rows are cumulative. All rows are mean ± standard deviation over ten seeds.

Stage	Test error (%)
Kernel ridge on loads	5.197 ± 0.003
Reflection-averaged neural mean	4.836 ± 0.018
+ refiner and MSE variant, affine stack	4.628 ± 0.009
+ residual kernel correction	4.611 ± 0.005
+ FNO member, affine stack	4.595 ± 0.008
+ residual kernel correction	4.572 ± 0.010

there are several means to combine and again in the correction stage. The feature-space and second input-space corrections did not improve validation error beyond the first correction with a single mean and so were not selected; we report the stages that the validation split actually chose.

Two smaller effects are worth isolating. Reflection test-time averaging lowers the metric-trained MLP's test error by 0.16 points, consistent in direction with Proposition 6.1, which fixes the sign of the expected change but not its size; the effect is much smaller on the normalized-MSE network (0.02 points) and absent by construction from the refiner, which trains symmetrized. Adding the kernel-flow regularizer (Section 6.2) to the low-data MLP did not help here (it moved test error from 5.64% to 5.87% at a single seed, a comparison the seed campaign did not repeat and which is smaller than the low-data chain's own seed spread of 0.09 points, so we record it as a null result rather than an effect); Proposition 6.2 says what the term estimates, not that estimating it is useful on every problem, and on this smooth-output benchmark the plain metric loss was already a good proxy. Nor does a richer correction kernel help: replacing the single tuned Matérn with a sum of Matérn kernels at several bandwidths leaves the validation error unchanged to two decimals, which is consistent with the residual being close to kernel-irreducible in the input geometry, and is why the boosted correction stages of Section 3.3 were not selected. The correction's own grid confirms the same reading from the other side: at all ten seeds the tuner selected the largest nugget on its grid, $\lambda = 10^{-3}$, at nine of the ten (the tenth took 10^{-5}), and a length scale of twice the median at five, of the median itself at four and of four times it at one — the most heavily regularized correction available to it, chosen independently at nearly every seed. The length scale is the less stable of the two choices, and the four-member ablation concentrates more tightly than this (eight seeds at twice the median), so the spread here belongs to the five-member configuration.

One positive lesson emerges from the refiner variants. Conditioning the refiner on the kernel method's prediction (test error 4.730%) beats conditioning it on another network's prediction (4.84% at one seed), even though the two conditioning fields have almost the same accuracy; the kernel field carries information complementary to the network's inductive bias, whereas a second network's field is largely redundant. Enlarging the ensemble beyond the residual family is the subject of Section 4.6, where the returns of each addition turn out to be predictable in advance from the correlation structure of the members' errors.

4.6 Architectural diversity and the shared residual

The natural reading of Table 3 is that the remaining error is a modeling problem: different architectures make different mistakes, so a more diverse ensemble should stack to a lower number. We tested this directly, and at every seed the measurements point the other way.

Figure 3 shows the correlation of per-sample normalized residuals between members. Over the ten seeds every pair of members correlates between 0.83 and 0.97, and the kernel predictor — the most alien member by construction, an exact solve on the raw loads rather than a trained network — still correlates 0.86–0.88 with the most accurate network. The mean pairwise correlation is 0.913 ± 0.005 across seeds. The members are not making different mistakes. They are making the same mistake with small private variations, and how small does not depend on the seed.

The second-moment identity of Proposition 6.8 makes the consequence quantitative before any weights are fitted, and the campaign lets us check the prediction prospectively rather than illustrate it once. At each seed we measure the matrix S of normalized residual second moments on the validation split, solve for the optimal simplex weights, and compare the predicted root-mean-square relative error $\sqrt{w^\top S w}$ with what those weights realize. The prediction is $4.940 \pm 0.057\%$ and the realized mean relative error is a factor 0.938 ± 0.003 of it. That factor is the dispersion between a root-mean-square and a mean, and its stability to three thousandths across ten independent replicates is the substantive claim: S alone, measured without ever evaluating the metric, predicts the stack.

The weights it assigns are as informative as the risk it predicts, and they correct something. The optimum spreads across all five members at essentially every seed — refiner 0.31 ± 0.04, FNO 0.34 ± 0.06, normalized-MSE network 0.24 ± 0.08, kernel 0.06 ± 0.03, metric-trained MLP 0.06 ± 0.04 — with the kernel receiving a nonzero share at ten seeds out of ten and the plain MLP at nine. (An earlier version of this work reported zero weight for those two members; its solver took clipped gradient steps rather than projecting onto the simplex and stopped at near-vertex points 0.8–3.6% above the true minimum.) Solved properly the answer is a spread, not a vertex: every member earns some weight.

That the FNO takes the largest single share while ranking only third on the metric is the whole content of the diversity argument. S is a matrix of second moments, so what it rewards is a residual that is both small and unlike the others', and the spectral member is the only architecture here that is structurally unlike the residual-MLP family. The pairwise admission test of part (i) — a member of error e_2 helps a reference of error e_1 when their correlation falls below e_1/e_2 — admits it at all ten seeds, though narrowly: correlation 0.95–0.97 against a threshold of 0.987–1.001. The kernel is admitted more comfortably, 0.86–0.88 against 0.905–0.918, and also enters the deployed pipeline twice more, through the refiner it conditions and through the affine stack, where weights are not constrained to be nonnegative.

Part (ii) of the proposition puts the infinite-ensemble floor of an equicorrelated family at $\bar{e}\sqrt{\bar{\varrho}}$, which at these measurements is $4.922 \pm 0.056\%$. That number and the pipeline's 4.572% are not on the same footing, and the comparison has to be made carefully for the reason Remark 6.9 gives: the floor is a root-mean-square quantity and 4.572% is a mean, so the mean lies below the floor by Jensen's inequality whatever the estimator does, and a comparison across the two would establish nothing. Converted into the floor's own metric with this pipeline's measured dispersion factor, 4.572% corresponds to about 4.87%, which sits within 0.056 of the floor with the sign undetermined. We therefore claim no margin below the floor here. What can be said without a conversion is the qualitative point, and it is unaffected: the floor bounds convex combinations of the members, while the deployed estimator fits an affine stack per grid point and then corrects its residual with a kernel, neither of which is a convex combination. What the floor measures is how little the convex operation can buy, and the answer, at every seed, is a few hundredths of a point.

Where does the common mistake live? Write r_m for member m's residual field on a validation case and $c = \frac{1}{M}\sum_m r_m$ for the across-member mean, the component that no amount of uniform averaging removes. Measured over the validation split, the shared component carries 90% of the average residual energy (Figure 4). Three of its properties matter. Spatially, its energy concentrates at the two corners of the loaded edge, where the traction boundary condition meets the lateral supports; relative to the local field scale it is three to four times larger there than the grid average, and these are precisely the locations where the finite element solution is least accurate and where interpolation to the 41×41 grid is most strained. Spectrally, it is enriched in high radial frequencies relative to the stress fields themselves: the fields put 97.5% of their energy below radial wavenumber 4, the shared residual only 69%. And its magnitude is statistically independent of everything we can compute from the input: the correlation of $\|c\|$ with the output norm is 0.01, with the load norm 0.01, with the load's total variation 0.01. A component that no architecture avoids, that no input statistic predicts, that lives at the stress concentrations, and that fixed-scale additive noise would reproduce is consistent with noise of the data-generating process itself — finite element and grid-interpolation error at the singular corners — and harder
demonstration: every measurement here is made on surrogates. The experiment that would decide it is to re-solve a set of the hardest cases on a refined mesh or with higher-order elements and compare the solver discrepancy, case by case, with c; we have not run it, and the remaining measurements in this section should be read with that in mind.

Three further measurements are consistent with the same reading. First, the residual kernel correction, which regresses the stack's residual on the load, improves the stack by 0.024 points at $n = 19000$ whatever the kernel scale, and its tuner reaches for the smoothest kernel and the largest nugget on its grid: at this sample size the shared residual is not a function of the load in any way a Matérn RKHS on $\mathbb{R}^{41}$ can see. Second, the error stops responding to data. Under identical recipes at a single seed, the MLP's test error is 4.95% with 3500 training samples, 4.86% with 8500, and 4.86% with 19000 (Figure 6); the ten-seed value at the largest size is $4.836 \pm 0.018\%$, so the last five-and-a-half-fold increase in data buys nothing that seed noise does not already cover. The kernel ridge curve still falls, at roughly $n^{-0.11}$, but toward the same region. Third, and this is what ten seeds add, the error does not respond to reseeding either, and the scale of the comparison is what makes it an argument: the five published architectures in Table 3 span 0.65 points, while ten reseedings of our own pipeline span 0.034, about five percent of it. A modeling limitation would be expected to yield to capacity, diversity, data, or reseeding; within the range we could test, this error yields to none of them, which is the pattern a floor in the data would produce.

4.7 Where the remaining error is

The seed replicates make it possible to ask not just how large the error is but where it sits, and whether the ensemble touches it. Both questions have the same answer at every seed.

Per sample, the error is concentrated and the concentration is not where an ensemble can help. Ranked by relative error on the validation split, the twenty hardest cases sit at 10.24% against a median of 4.44%, a factor of 2.31. On exactly those cases the five-member ensemble reaches 10.18% and the mean over the individual members is 10.28%: averaging five members, including the architecturally distinct one, recovers a tenth of a point against the average member and six hundredths against the best one, on cases whose error is more than twice the median. Everywhere else the ensemble buys what Section 4.6 says it should; on the hard tail it buys almost nothing, because the members fail there together — and adding the spectral member, which helps everywhere else, does not change that. Within this ensemble, then, the hard tail is not a shortage of member diversity; whether a surrogate class we did not train would fail there too is not something five architectures can settle, but the location of these cases, at the loaded edge and the supports, is where the shared residual of Section 4.6 concentrates.

Spatially, the error is a boundary phenomenon. The root-mean-square pointwise error over the validation samples is 8.99 on the perimeter of the 41×41 grid against 5.64 in the interior, a ratio of 1.59; both figures are stable across seeds to under a percent of themselves. Averaging lowers them to 8.87 and 5.59, so the ensemble takes about a tenth off a boundary error of nine and half that off an interior error of five and a half: it shaves the concentration slightly but does not touch its shape. The location agrees with the anatomy of the shared component above and with the error maps of Figure 1: it is the loaded edge and the supports, where the traction condition meets the geometry, not a diffuse modeling deficit.

4.8 Uncertainty quantification

Theorem 6.18 certifies the corrected surrogate's error by $\|G - m\|_K \, P_\lambda(u)$, and it is tempting to read the design factor P_λ as a pointwise error bar. On this benchmark that reading fails, in an instructive way. Table 7 confirms the operator factor behaves as the theory predicts: the computable squared RKHS norm of the fitted correction is about $271\times$ smaller when the kernel regresses ensemble residuals than when it regresses raw stress fields, so an accurate mean genuinely leaves a smoother remainder and a tighter certificate (the bound, linear in the norm, by the square root of that). But the design factor P_λ correlates only weakly with the corrected surrogate's absolute error (Pearson 0.08), and *negatively* with the benchmark's relative error. The cause is measurable: P_λ is large for loads far from the training set, those loads tend to have large amplitude, large-amplitude loads produce large-norm stress fields, and dividing by the output norm makes their relative error small. Indeed P_λ correlates 0.65 with the output norm (Figure 5). The error is dominated by the neural mean, whose mistakes are not a function of the input-space geometry that P_λ sees, so a purely input-space power function cannot rank them.

The obvious alternative fares no better, and it fails in the same direction, which is the part worth noticing. Ensemble disagreement, the spread of the members' predictions, is the standard deep-ensemble error signal. Its correlation with the corrected surrogate's absolute error is 0.074 ± 0.006 over the ten seeds, and with the benchmark's relative error it is

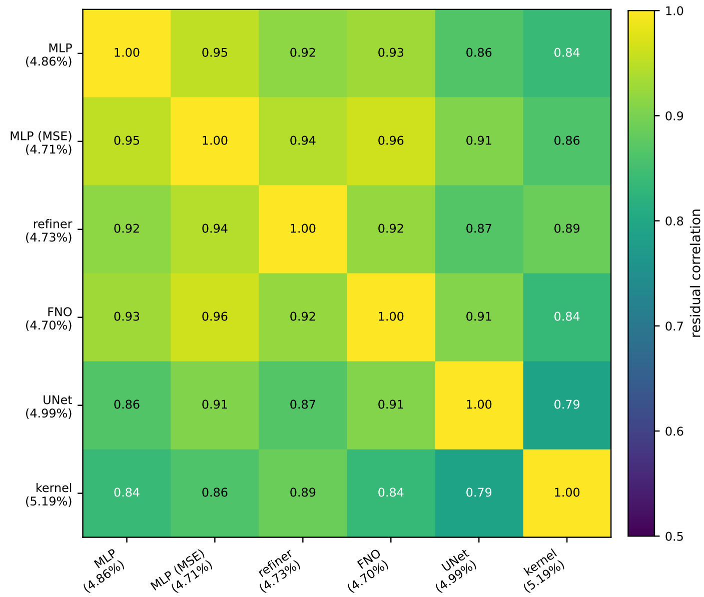

Figure 3: Correlation of per-sample normalized residuals on the validation split, drawn at one seed from the original single-run study, which also trained a UNet member (test-set values agree to two decimals). Across the ten replicated seeds, whose five members all appear here, every pair correlates between 0.83 and 0.97, with a mean of 0.913 ± 0.005; the kernel predictor, the most alien member by construction, still correlates 0.86–0.88 with the most accurate network at every one of them.

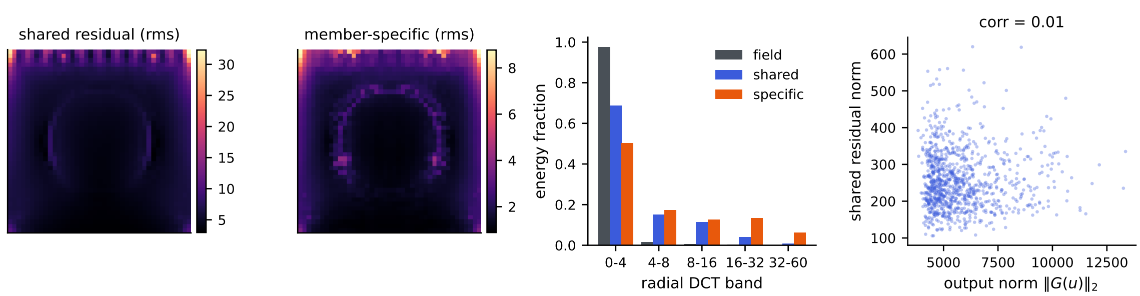

Figure 4: Anatomy of the shared residual c (across-member mean of the residual fields), at one seed. Left to right: rms of c over the grid; rms of the member-specific remainders; radial DCT band energies of the stress fields, of c, and of the specific parts; and the norm of c against the output norm, whose correlation is 0.01.

-0.25 ± 0.04: weakly positive against the quantity the theory speaks about, negative against the quantity the benchmark reports, exactly as for P_λ, and for the same amplitude reason. Two uncertainty signals of quite different provenance — one a Gaussian-process design factor computed from the inputs alone, the other a spread across trained models — fail on this problem in the same way and to the same degree. Adding the spectral member does not repair it: the same correlation is 0.083 on the four-member ensemble, so the member

Table 7: Squared RKHS norm of the fitted kernel interpolant, $\mathrm{tr}(\alpha^\top K\alpha)$, when the same validated kernel and design regress raw stress fields versus ensemble residuals.

Kernel regresses	$\|\widehat{r}\|_K^2$
Raw stress fields ($m = 0$)	1.39×10^{10}
Ensemble residuals	5.14×10^{7}
Ratio	$271\times$

whose admission is worth 0.039 points of accuracy buys nothing as an uncertainty signal. Section 4.6 says why it cannot: disagreement measures the member-specific components of the error, which carry under a tenth of the residual energy, while the ranking signal for the shared component is invisible to any within-ensemble statistic, because the members agree precisely where they are jointly wrong. Section 4.7 is the same fact seen on the hard tail.

What does hold is coverage, and it holds at every seed. We take the exact split-conformal quantile of the corrected surrogate's error score on a calibration subset drawn from the test block — 1000 calibration cases against 19000 evaluation cases, neither used for training, checkpointing, stacking, or kernel tuning — and evaluate coverage on the disjoint remainder. Conformal validity needs no correlation between the score and the error, only exchangeability, which this protocol supplies exactly (Section 6.7). At the nominal 90% level the observed coverage is $90.2 \pm 0.6\%$, and the right comparison is not with 90% but with the exact finite-sample law: with $m = 1000$ calibration points the coverage of a split-conformal band is $\mathrm{Beta}(901, 100)$ distributed, whose central 95% interval is $[88.1\%, 91.8\%]$. All ten seeds fall inside it. At the 95% level the band is $[93.6\%, 96.3\%]$ and the observed coverage is $95.0 \pm 0.6\%$, again with all ten inside. Scaling the score by ensemble disagreement, which the correlations above say should not help, indeed does not: it changes mean coverage by three tenths of a point and leaves every seed inside the same band. The lesson is narrow but real: a valid pointwise bound (Theorem 6.18) is not automatically a useful pointwise indicator, the standard uncertainty signals can both fail on the same problem, and a relative-error metric has to have its normalization accounted for before a posterior standard deviation means what it appears to.

4.9 Spectra and effective dimension

Figure 7 shows the spectrum of the Matérn Gram matrix on the loads and the effective dimension $d_{\mathrm{eff}}(\lambda)$ computed from it by the exact identity of Lemma 6.21. The spectrum decays quickly: a few hundred eigenvalues carry essentially all the mass, the tail falling many orders of magnitude below the leading modes. At the cross-validated nugget the effective dimension is a few hundred out of $n = 19000$. This is a statistical statement, not a computational one — the dense factorization costs its full n^3 operations regardless — but it is why the correction generalizes despite interpolating nineteen thousand points: the problem the kernel solves has the statistical dimension of the retained spectrum, not of the sample. The cross-validated λ sits at the shoulder of the d_{eff} curve, past the leading modes and into the rapidly decaying tail, matching the reading of Lemma 6.21.

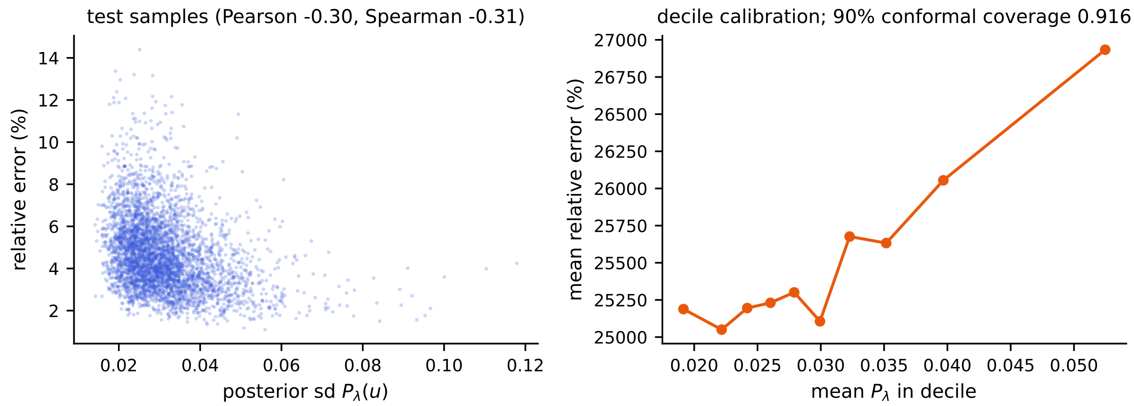

Figure 5: Both panels are one seed. Left: the Gaussian process posterior standard deviation P_λ against the corrected surrogate's relative error on the test set; the association is weak and, because of the output-amplitude confound, slightly negative. Right: decile calibration of P_λ against absolute error, the theory-consistent quantity. Across all ten seeds the distribution-free conformal band attains $90.2 \pm 0.6\%$ coverage at the nominal 90% level, inside the exact finite-sample interval at ten seeds out of ten.

4.10 Cost and reproducibility

The pipeline was built on a single laptop and replicated on four cloud CPU instances. The kernel stages are exact: one Cholesky factorization of the 19000×19000 Matérn Gram matrix in double precision takes 24 seconds on the laptop's CPU (16 threads, OpenBLAS; the matrix itself is 2.9 GB), and between 18 and 32 seconds on the campaign instances, on which it ran ten times, once per seed. Prediction is two matrix products; there are no inducing points, stochastic solvers, or preconditioners, and no GPU is needed for the correction. The factorization performs the full $n^3/3 \approx 2.3 \times 10^{12}$ floating-point operations — numerical low rank does not reduce the work of a dense Cholesky — and the measured rate, about 100 gigaflops per second, is simply what a current multicore CPU delivers; exactness at this n costs half a minute, not a cluster. What the low effective dimension of Section 4.9 buys is statistical rather than computational: it describes where the solve's information concentrates at the working nugget, not its flop count.

Building the Gram matrix, not factoring it, is the stage that needs care at this size: forming it through a full squared-distance matrix and its elementwise transforms allocates several 2.9 GB temporaries at once, which exhausted a 31 GB instance during the campaign. Filling it in row blocks into a single preallocated array holds the peak at the matrix itself and is what the released code does. The neural means are small by the standards of the literature (a few million parameters) and train in the metric itself. Every split is fixed by a published permutation seed, the input is consumed as the 41-dimensional load after the exact broadcast check of Section 2, and every model is selected on the validation split and evaluated once on the test set; we release the splits, the code, and the per-seed result record behind every stage of every run, so any row of any table can be rechecked against the records

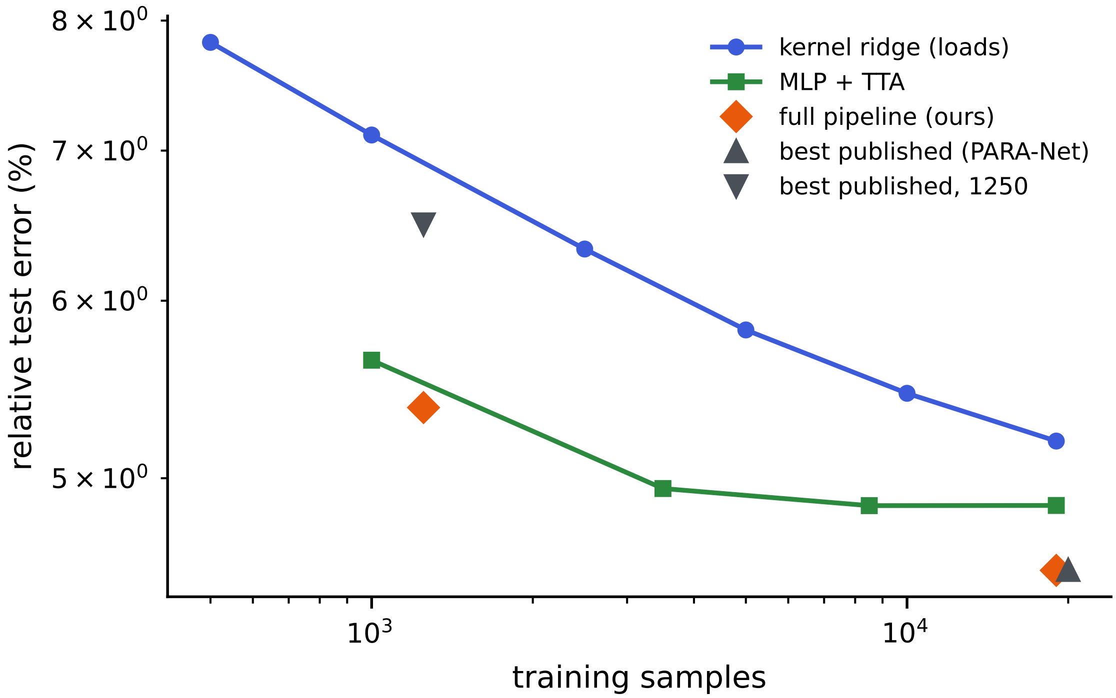

Figure 6: Relative test error against training set size for the kernel ridge on loads (log-log), with the full pipeline and the best published results marked in both regimes.

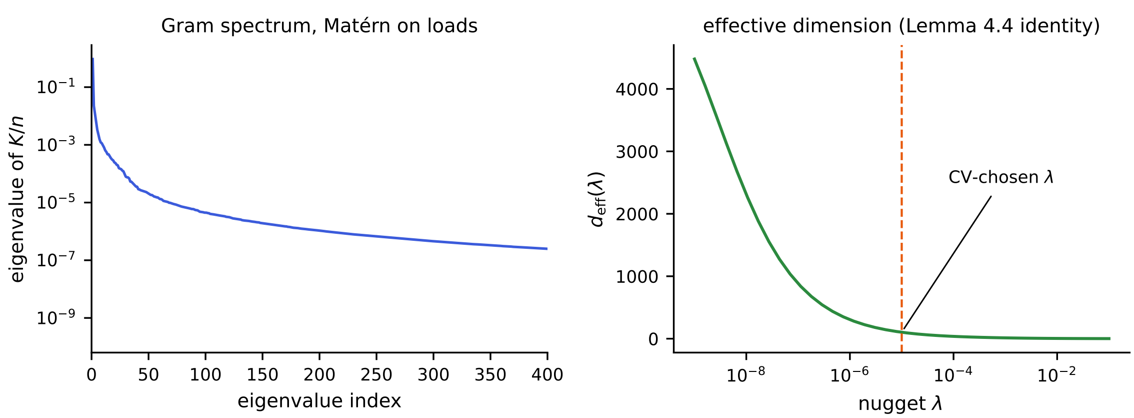

Figure 7: Left: eigenvalues of the Matérn Gram matrix K/n on the loads (log scale); the spectrum decays by many orders of magnitude within a few hundred modes. Right: the effective dimension $d_{\text{eff}}(\lambda)$ from the exact identity of Lemma 6.21, with the cross-validated nugget marked.

or regenerated end to end by rerunning the pipeline. The prediction arrays and checkpoints themselves are not distributed, so regenerating a row means retraining it.

5 The OCO-2 radiative-transfer emulator

The structural-mechanics benchmark put the neural mean and the kernel in the balanced regime. To see the other regime we apply the same components to the radiative-transfer emulation problem of Lamminpää et al. (2025): per spectral band of the OCO-2 instrument, learn the map from a reduced atmospheric state (20–24 dimensions after the dimension reduction of that paper) to the reduced radiance (40 PCA coefficients of the monochromatic spectrum). The OSF project (`u2t8a`) publishes a training pool of 20000 pairs and, separately, 2000 test states together with the kernel-flow emulator's own predictions on them; we verified the two sets share no point. We split the pool 18000/2000 into training and validation, and every choice in our pipeline — the network checkpoints, the kernel head's length scale and nugget, and the per-coordinate winners of the combination below — is made on that validation split; the 2000 public test states are used once, for the final numbers. The comparison is therefore computed on identical test points against the published emulator itself rather than against our reimplementation of it.

Two error metrics matter, and they disagree in an instructive way. The *reduced* metric is the relative L^2 error on the 40 standardized coefficients. The *radiance* metric maps predictions back to the monochromatic spectrum through the stored PCA projection and norms before comparing; because the projection is orthogonal, the error *numerator* on the reduced side is exactly the diagonally weighted norm $\|s_z \odot (\hat{z} - z)\|$, whose weights concentrate almost entirely on the first few coefficients (the denominator is the full reconstructed-radiance norm, which adds the reconstruction offsets). It is this diagonal weighting of the residual that the per-coordinate combination exploits.

Table 8 carries four findings, and Figure 8 shows the two central ones. First, the representation ladder: the same exact kernel machinery scores 40% on the raw input, 30% after rescaling each input coordinate by its measured sensitivity, and 4.1% on the trained network's features, past the network's own head. The raw-input kernel was limited by its features, not by anything kernel-shaped, and the deep kernel head, an exact solve on learned features, is the strongest single model on the reduced metric. Its margin over the network it feeds on ($4.11 \pm 0.05\%$ against $4.43 \pm 0.05\%$) is small — a third of a point — but it is not a seed accident: the head wins at ten of ten seeds on this band and at ten of ten on each of the other two, paired within seed. We still lean on the ladder's order of magnitude rather than on that gap. Learning the metric rather than setting it by hand does not change this reading. A per-dimension metric fit by the kernel-flows cross-validation loss of Owhadi and Yoo (2019) reaches the same 30% on the raw state, and a low-rank Mahalanobis metric with a thousand more parameters does not improve on it; on the features, where the network has already conditioned the representation, a learned metric gives no gain over the isotropic kernel. The metric is a constant-factor lever, and the representation is the order-of-magnitude one.

5.1 What the metric buys, measured

The heuristic rate calculation of Proposition 6.16 splits a metric's advantage into a constant σ_1^s and, only under an approximate-rank gap, a rate. The seed campaign scores that split directly: both raw-input kernel rows are refit at five nested training sizes over a sixteenfold range, at ten seeds, on each of the three bands, with the sensitivity map estimated causally from the first n rows only so that no row sees data a smaller sample would not have had.

Table 8: OCO-2 O2 band, scored on the 2000 public test states, which are disjoint from the training pool; all model selection happened on a validation split of the pool. Every trained row is mean ± standard deviation over ten seeds at a matched 250-epoch budget; the kernel-flow row is the emulator of Lamminpää et al. (2025), scored from its own stored predictions on fixed test points, and is identical at every seed. "Features" means the penultimate activations of the trained network; the kernel head is the same exact Matérn solve as everywhere else in this paper, and the two raw-input rows are fit by that same routine under the same budget as the feature heads: the same scale grid $\{0.5, 1, 2, 4\}$ times the median distance, the same nugget grid $\{10^{-8}, 10^{-6}, 10^{-4}\}$, the same 6000-sample tuning subsample, and the winner refit on all 18000 training points. An earlier version of this work fit those rows with a lighter diagnostic script and said so; the rows here are the matched ones, which is why the selected hyperparameters vary from seed to seed. The ARD radiance cell is left empty because validation, which scores the reduced metric, ties between two hyperparameter configurations whose radiance differs by a factor of 2.2 (0.150% at seven seeds, 0.329% at three); a mean over that mixture would describe the selection noise, not the model. Table generated from the campaign artifacts by `campaign/gen_oco_table.py`.

model	reduced	radiance
Matérn kernel, raw input, one length scale	$40.57 \pm 0.10\%$	$0.233 \pm 0.003\%$
Matérn kernel, raw input, sensitivity-scaled (ARD)	$29.98 \pm 0.15\%$	—
kernel-flow emulator (Lamminpää et al., 2025)	16.89%	0.0448%
residual MLP, flat metric	$4.43 \pm 0.05\%$	$0.195 \pm 0.014\%$
residual MLP, weighted-coefficient loss	$49.32 \pm 0.40\%$	$0.0442 \pm 0.0044\%$
residual MLP, exact radiance loss	$59.02 \pm 0.76\%$	$0.0571 \pm 0.0057\%$
Matérn head on the flat network's features	$4.11 \pm 0.05\%$	$0.0793 \pm 0.0072\%$
Matérn head on the weighted network's features	$21.70 \pm 0.52\%$	$0.0319 \pm 0.0020\%$
Matérn head on the radiance network's features	$20.48 \pm 0.61\%$	$0.0379 \pm 0.0037\%$
per-coordinate combination	$\mathbf{4.12 \pm 0.05\%}$	$\mathbf{0.0294 \pm 0.0020\%}$

Two things come out, and one of them corrects a number reported in an earlier version of this work. The approximate rank of the sensitivity map is 12 of 20 on O2 and 13–14 of 24 on the two CO_2 bands, counting the singular values that hold 99% of the squared energy. The participation ratio is a softer count and agrees on O2 (11.99 against 12) while sitting a third of a unit below the threshold count on the two CO_2 bands (13.63 against 14, 13.33 against 13); the rank ratio is between 0.54 and 0.60 on either reading, which is what the argument uses. That is a rank ratio between 0.54 and 0.60, not the near-full rank previously stated for O2, and it is the regime in which the calculation would permit a rate gain and not merely a constant.

The rate gain does not appear. Table 9 gives the paired difference of the two fitted exponents at each band. On O2 the two exponents are indistinguishable across seeds; on the CO_2 bands the adapted exponent is steeper, but by 0.011 and 0.052 against isotropic exponents of 0.086 and 0.203, where the rank ratios above would license something several times larger. What is stable instead is the constant: at the largest training size the isotropic error exceeds the adapted error by a factor of 1.353 ± 0.008, 1.118 ± 0.002 and 1.514 ± 0.114

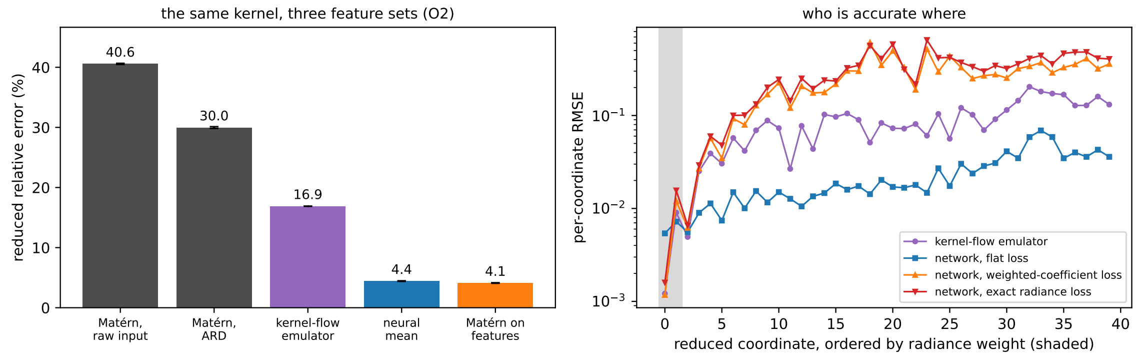

Figure 8: Both panels are drawn from the ten-seed campaign by `campaign/make_oco2_fig.py`: bars and error bars in the left panel are the mean and standard deviation over the ten seeds, curves in the right panel are per-coordinate test RMSE averaged over the same ten. Left: the representation ladder on the O2 band. The same exact Matérn machinery moves from 40% on the raw input to 4.1% on the trained network's features; the kernel-flow emulator and the network sit in between. Right: per-coordinate test RMSE with the reduced coordinates ordered by their radiance weight (shaded). Networks are named by the loss they were trained on. The per-output-tuned kernel-flow emulator is most accurate exactly on the heavily weighted leading coefficient (0.0012 against the flat network's 0.0054), while the flat network is nearly four times more accurate across the trailing thirty coordinates that the radiance barely sees (0.0276 against 0.1037). Both weighted losses buy the leading coefficient at the price of the tail, and by almost the same amount (0.0012 and 0.0016), which is the measurement behind the reading below; the per-coordinate combination takes each coordinate from whichever model wins it.

Table 9: The raw-input kernel refit at nested training sizes, ten seeds per band. "Rank" counts singular values of the causally estimated sensitivity map holding 99% of the squared energy, out of the input dimension. The exponent difference is paired within seed; its standard deviation is over seeds. The last column is the isotropic-to-adapted error ratio at $n = 18000$.

Band	Rank	−slope iso	−slope adapted	difference	ratio at n=18000
O2	12/20	0.221 ± 0.007	0.215 ± 0.017	$+0.006 \pm 0.015$	1.353 ± 0.008
weak CO_2	14/24	0.086 ± 0.005	0.097 ± 0.006	-0.011 ± 0.009	1.118 ± 0.002
strong CO_2	13/24	0.203 ± 0.029	0.255 ± 0.020	-0.052 ± 0.021	1.514 ± 0.114

on the three bands, and those ratios are flat in n over the whole sweep. The reading we take is the one the measurement supports and no more: a metric is worth a constant here, the constant is between 1.1 and 1.5, and the tenfold gap to the feature kernel is not something a metric closes. Both fitted exponents are also steeper than the heuristic's own s/d, so the absolute rates are not being tested here — only the difference between them, which is what the split concerns.

At its largest size the sweep recovers the raw-kernel rows of Table 8 to the digit, seed by seed and including the selected hyperparameters. We record that as a consistency check on the two code paths and nothing more: both call the same tuning routine on the same standardized inputs, so the agreement is a re-execution of one deterministic computation rather than corroboration by an independent measurement. One protocol note: the table's matched 250-epoch budget is deliberately short of convergence for the network rows. An earlier version of this work trained longer and reached 3.99% (flat network) and 3.82% (feature head) on this band, and a single 750-epoch run of the seeded code reproduces those levels (3.82% and 3.71%, with the combination at 3.71% reduced and 0.0174% radiance). The budget moves the levels of the trained rows together; at the diagnostic seed it reverses none of the orderings discussed below.

Second, the training metric behaves as a budget, and the campaign now measures which part of the metric carries the budget. The second network in Table 8 is trained on a diagonally weighted loss over the reduced coefficients, $\|s_z \odot (\hat{z} - z)\|/\|s_z \odot z\|$. That is a surrogate for the radiance-relative error, not the error itself: the reported radiance metric maps predictions through the stored PCA reconstruction and divides by the reconstructed-radiance norm, and the surrogate omits the reconstruction. An earlier version of this work described that row as radiance-trained, which overstated it, and drew from the comparison the rule "train in the metric you report". The campaign adds the row that tests the rule — the third network is trained in the exact radiance metric, reconstruction and denominator included — and the rule does not survive. Trained in its own reported metric, the exact-loss network scores *worse* there than the surrogate-trained one: $0.0571 \pm 0.0057\%$ against $0.0442 \pm 0.0044\%$ on O2 and $0.093 \pm 0.008\%$ against $0.075 \pm 0.003\%$ on SCO2, the surrogate ahead at ten of ten seeds on both bands, with WCO2 the reverse at eight of ten ($0.064 \pm 0.015\%$ against $0.072 \pm 0.007\%$). The kernel heads built on their features repeat the pattern. Tripling the training budget narrows the O2 gap (0.0212% against 0.0201% at 750 epochs, one seed) without reversing it, so slow optimization of the exact loss is at most part of the story. The reason the surrogate gives nothing away is the orthogonality noted above: the two losses share their numerator exactly, and differ only in the per-sample denominator, which reweights states, not coordinates. What the metric buys is coordinate weighting, and either loss buys it — the weighted networks give up the trailing coordinates the radiance barely sees and win the reported metric by a factor of three to four over the flat network, whose per-coordinate profile shows the gap directly: the kernel-flow emulator (lengthscales tuned per output) predicts the leading coefficient to 0.0012 root mean square against the flat network's 0.0054, while the flat network is three to four times more accurate on the trailing thirty coordinates. The practical rule that survives the measurement is narrower than the one we first drew: train in a loss that concentrates where the reported metric concentrates. Matching the metric's weighting is what pays; matching its algebra is not.

Third, the same-class floor of the structural-mechanics study reappears here, one level down, and the campaign lets us test it rather than illustrate it. Proposition 6.8 is stated for the squared-metric risk, so the measurement is made there: for each band we form the matrix S of the ten seeded copies of the flat network on the test set, in the relative metric, and read the floor off it. Three things follow, and the third corrects this paper's earlier reading.

The identity itself holds numerically. At uniform weights $w^\top Sw$ and the directly measured risk of the averaged prediction agree to 7×10^{-18}, 3×10^{-17} and 1×10^{-17} on the three bands — the proposition's content, checked against real predictions rather than assumed.

The floor is where the proposition puts it. Over all 45 seed pairs per band the residual correlations are 0.787 ± 0.010 (O2), 0.971 ± 0.001 (WCO2) and 0.961 ± 0.001 (SCO2), tight enough that the equicorrelated case of part (ii) is a fair description. It predicts a ten-member value of 4.419%, 17.997% and 9.267% against realized 4.419%, 17.997% and 9.267%, and infinite-seed floors of 4.360%, 17.970% and 9.248%. Ten seeds have already taken essentially all of what seed-ensembling has to give: the remaining distance to the infinite-seed floor is six hundredths of a point on O2 and less on the other two bands.

What the floor does not do is fall to a change of member. An earlier version of this work reported that the Matérn head on learned features passed below it. That comparison was invalid: the floor is a squared-metric quantity and the head's error was quoted from Table 8, which reports the mean relative error $\mathbb{E}\,\ell$. Measured in the same metric as the floor, the head reaches 4.575%, 18.11% and 9.326% — better than an individual seeded network (4.915%, 18.238%, 9.433%) and worse than that network's ten-seed ensemble. The ranking is the same in the mean metric, where the seed ensemble reaches 3.916% against the head's 4.111% on O2. At this budget a different kind of member buys less than ensembling ten copies of the plain one, and we say so; the order-of-magnitude result of this section is the representation ladder, which does not depend on this comparison. Ensembling within an architecture class saturates by the same second-moment law on both problems, and the two ways of spending a fixed compute budget — more seeds of one member, or one member of a better kind — are closer here than we previously claimed.

The floor is a property of the architecture class, not of the problem, and it can be probed directly. Different architectures decorrelate: on WCO2 two residual MLPs of different activation correlate at 0.84, a wide shallow network against the residual MLP at 0.40, and a random-Fourier-feature network against it at 0.08 to 0.17, all far below the 0.971 ± 0.001 that ten seeds of one architecture reach on this band. Yet the floor did not move, because its other input is accuracy: by part (i) of Proposition 6.8 a member of error e_2 helps a reference of error e_1 only when their correlation falls below e_1/e_2, and across a bandwidth sweep the Fourier network was either accurate and correlated (error 31%, correlation 0.54, just above the 0.53 threshold) or decorrelated and inaccurate (error above 140%). No architecture we tried was both accurate and decorrelated enough to lower the floor, which is why the residual MLP is the member of record: not that diversity is impossible, but that on this map no more accurate diverse member was found. Figure 9 shows both halves of this: the correlations that make the floor class-specific, and the accuracy-decorrelation trade-off that keeps it in place. Adding the Fourier members to the per-coordinate combination changes the reported error by less than a hundredth of a point, in either direction, on all three bands: the half-split selection gives them no weight.

Fourth, the regimes of Section 6.4 are decided by these numbers before any stack is fit. Against the flat network the raw-input kernel has $\varrho = 0.14$ and $e_n/e_k = 0.109$ (the network's 4.43% over the kernel's 40.57%): the condition of Proposition 6.8 fails, by a margin that ten seeds do not put in doubt, and the kernel is dropped — in contrast to structural mechanics where the same test keeps it. Because the radiance metric is diagonal in the reduced coordinates, the per-coordinate combination of Proposition 6.13 is optimal for both

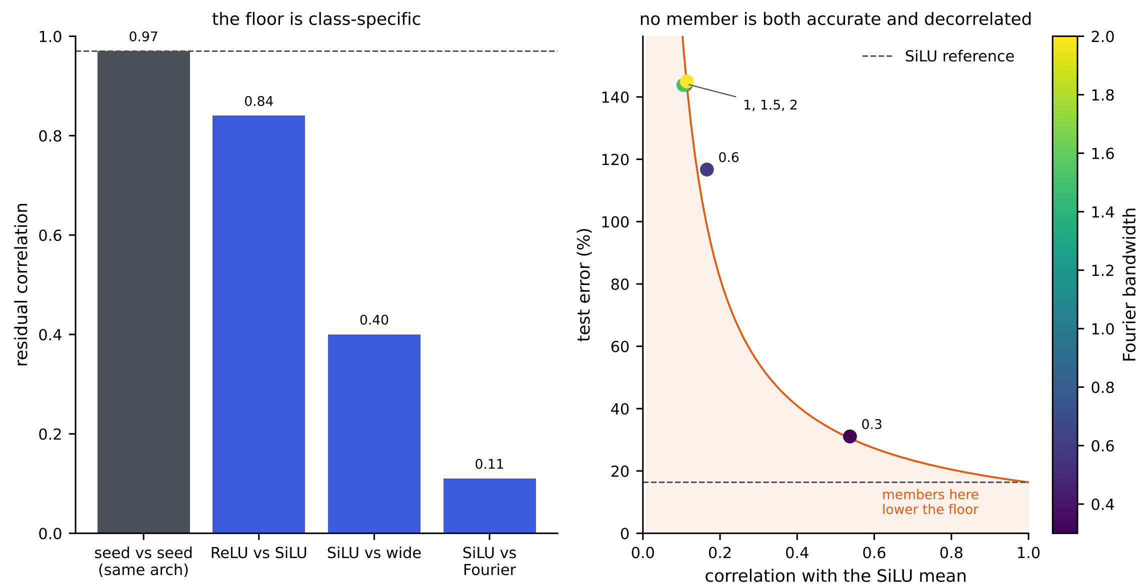

Figure 9: The ensembling floor on WCO2; the architecture comparison in the left panel is drawn from one seed of each, and the seed-pair correlation it shows is the campaign's mean over 45 pairs. Left: residual correlations. Two seeds of one architecture correlate near the value that sets the floor (dashed); different architectures correlate far less, so the floor is a property of the architecture class. Right: the Fourier-feature member across bandwidths. A member helps the ensemble only in the shaded region, below the e_{ref}/ϱ curve of Proposition 6.8(i); the accurate settings are too correlated and the decorrelated settings too inaccurate, so none lands there with room to spare.

metrics simultaneously. It yields a single surrogate that beats the kernel-flow emulator on both metrics at once, now on all three bands: O2 reaches $4.12 \pm 0.05\%$ against 16.89% reduced and $0.0294 \pm 0.0020\%$ against 0.0448% radiance, SCO2 $8.08 \pm 0.01\%$ against 16.14% and $0.0584 \pm 0.0041\%$ against 0.1147%, and WCO2 $16.28 \pm 0.06\%$ against 24.06% and $0.0507 \pm 0.0052\%$ against 0.0599%. Counted by seed, the combination beats the emulator on both metrics at ten of ten seeds on O2 and SCO2 and at nine of ten on WCO2, the single exception missing on the radiance metric by 0.0015 points.

WCO2 was a stated exception in an earlier version of this work, where the two best heads split the metrics and no single model won both. What removed it was not a better fit but a wider pool: the combination selects per coordinate from six members here, the flat, weighted and exact-radiance networks and the Matérn head on each of their features, where previously the two radiance-loss members were absent. The emulator's own predictions are not among the members, so the comparison remains against an outside baseline. The band that needed the extra members is the one whose metrics disagreed, which is what the per-coordinate argument predicts; but a nine-of-ten margin of 0.0015 points is thin, and we report it as a band that has just crossed rather than as a comfortable win. The code releases the per-band numbers alongside this paper.

5.2 Where the two-member rule can be trusted

Part (i) of Proposition 6.8 is the one result in this paper used as a decision rule rather than as a description: it says a weaker member helps exactly when $\varrho < e_1/e_2$, and we drop members on that test. A rule of that kind deserves to be scored the way it is used, so we decide it on the validation split and check it on the test block, for every pair of the eight members at every band-seed — $\binom{8}{2} = 28$ pairs at each of the thirty seeds, 840 in all.

Two separate things are being tested by this, and separating them is the point. The first is whether the proposition is true. Evaluated with both sides computed on the same block, the criterion $\varrho < e_1/e_2$ agrees with whether the optimal convex mix actually beats the better member on all 840 pairs, without exception. That is the proposition's own content rather than evidence about the world — it is an identity, and this is the numerical check that we implemented it correctly — but 840 instances is a strong form of that check, and it locates the difficulty elsewhere.

The second is whether the rule survives being used, which requires ϱ and e_1/e_2 to be estimated on data the decision does not see. Decided on validation and checked on test, the rule is correct on 62.7% of the 840 pairs. The average is the wrong summary: sorted by the margin $|\varrho - e_1/e_2|$ that the rule must resolve, the outcome is 105 of 255 below 0.02, 94 of 217 from 0.02 to 0.05, 79 of 86 from 0.05 to 0.10, 217 of 250 from 0.10 to 0.25, and 32 of 32 above 0.25 — that is 41%, 43%, 92%, 87% and 100%. These counts are not independent observations: the 28 pairs of a band-seed share members, and the ten seeds of a band share one test block, so the bins should be read as describing a trend across margin and not as binomial samples. Nothing about the proposition degrades in the small-margin pairs; what degrades is the transfer of two estimates from one block to another.

That transfer is measurable, and it is what sets the band. The signed gap $D = \varrho - e_1/e_2$ moves between validation and test with standard deviation 0.099 across the 840 pairs, essentially unbiased (mean -0.0005). This is an order of magnitude larger than the seed-to-seed scatter of D within a fixed pair, which is 0.010: replication across seeds is not what limits the rule, and a study that varied only the seed would have missed this. Because the pairs are binned by the margin the decision actually sees, it is part (ii) of Proposition 6.10 that speaks to the scoreboard, and it is a joint statement: with this σ, the probability that a decision is both wrong and taken at an observed margin of at least 0.25 is at most $\exp(-0.25^2/2\sigma^2)$, about four percent of all decisions, and the corresponding bound at a margin of 0.10 exceeds one half and says nothing. The proposition does not bound the error rate within a margin bin, which would require a lower bound on how often such margins occur, so the percentages above are an empirical calibration of the rule rather than a guarantee it carries. What they show is a trend consistent with the proposition: the rule's accuracy rises with the observed margin, and at the top of the range, where the joint bound leaves room for only a few percent of confident errors among the 840 decisions, the scoreboard records none in 32.

The practical reading, from the scoreboard rather than from the bound: on these problems the rule is dependable when the two quantities are separated by more than about 0.25, suggestive between 0.05 and 0.25, and uninformative below that. This applies to our own use of it. The kernel-versus-network decision earlier in this section has $\varrho = 0.14$ against $e_n/e_k = 0.109$, a margin of 0.031, which falls in the band the scoreboard says not to trust, and we therefore do not rest the exclusion of the raw kernel on that comparison. What does settle it is the same proposition read for magnitude rather than for sign: with these two errors and this correlation, part (i) puts the unconstrained two-member optimum at 4.4299% against the network's own 4.4320%, and the optimizing weight falls just outside the simplex, so no convex mix reaches even that value. Whichever side of the threshold the pair truly falls, the entire quantity in dispute is two thousandths of a point, so the member is dropped on the size of the available gain and not on a threshold the data cannot resolve. Reporting the margin alongside the verdict is what makes that distinction visible, and we recommend it wherever this rule is used.

5.3 The error is limited by data, and yields to more of it

The structural-mechanics error was flat in the sample size, which Section 4.6 reads as evidence consistent with a floor set by the benchmark data themselves, a reading whose decisive test, re-solving the benchmark on a refined mesh, has not been run; the emulator error is not flat, and the contrast is the sharpest practical difference between the two problems. The left panel of Figure 10 fixes the O2 task and varies only the number of training pairs, scoring each on a disjoint held-out set. The corrected surrogate's reduced-radiance error falls as a clean power law in n, fitted slope -0.68, from 74% at a few hundred pairs to 4.3% at $n = 17700$, still descending at the largest sample size we can afford and approaching the 3.8% our full pipeline reaches on all 18000 pairs (Table 8). The exact kernel on the raw state improves far more slowly, so at these sizes the network is the component that turns data into accuracy. The residual on this task is sample-limited: unlike the structural-mechanics error it recedes as pairs are added, so the largest lever on this problem is simply more of them, and pairs are cheap because the forward model is a program. The complementary case is the full-resolution $58 \to 3048$ map, where the error is instead flat in n across the few hundred retrievals we hold: that task is limited by its representation, not its sample count, and the reduction the emulator applies is what moves it.

A second benchmark carries the same reading into a data range OCO-2 cannot reach, though only over part of the range we first attempted, and the reason is worth stating before the result. ClimSim (Yu et al., 2023) is a climate emulation dataset of about ten million samples that maps a 124-dimensional atmospheric state to 128 physics tendencies, with published neural baselines. An earlier version of this work swept the training size across more than three orders of magnitude here and reported a crossover between the two model families. That sweep selected the kernel's hyperparameters against test rows. We reran it under the protocol used everywhere else in this paper — selection on a held-out block drawn from the training pool, the test block scored once — and the rerun reached only the three largest sizes before the campaign ended. The small-sample points are therefore withdrawn rather than restated, the superseded script is kept in the release so the difference can be examined, and the right panel of Figure 10 plots the leak-free points alone. ClimSim publishes no canonical test split, so each seed there draws its own 20000-row test block and the spreads below include that variation.

What the leak-free points establish is the high-data half of the picture, and they establish it cleanly. The exact kernel on the state is flat: 0.390 ± 0.007, 0.391 ± 0.006 and 0.387 ± 0.007 at one, two and three and a half million samples, unmoved across a factor of three and a

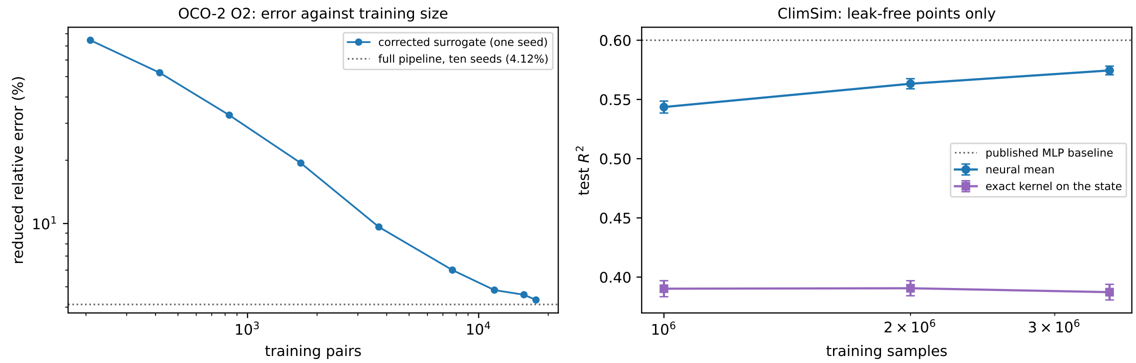

Figure 10: Error against training-set size, drawn by `campaign/make_scaling_fig.py`. Left: the OCO-2 O2 task, reduced error of the corrected surrogate at one seed, from the data-scaling run; it falls as a power law (fitted slope -0.68) to 4.3% at $n = 17700$, against the 4.12% the ten-seed pipeline reaches on all 18000 pairs (dotted). Right: ClimSim, test R^2 over the outputs with non-negligible variance, showing only the points measured under the corrected selection protocol — five seeds at one million samples and three at each of two and three and a half million, error bars over seeds. The small-sample points of the earlier version of this figure are withdrawn, so the crossing between the two families is not shown and is not claimed; over the range that remains the kernel is flat and the neural mean is far above it and still rising toward the published baseline (dotted).

half in data. The neural mean over the same range is both far above it and still climbing: 0.544 ± 0.005 at one million (five seeds), 0.563 ± 0.004 at two million and 0.575 ± 0.004 at three and a half million (three seeds each), closing on the published multilayer-perceptron baseline near 0.6. The separation is some twenty times the seed scatter at every size, so the ordering at these sizes is not a matter of which run one looks at, and the contrast between a data-limited mean and a saturated kernel is exactly the one the OCO-2 sweep shows on its own scale.

What we do not now claim is where the two families cross. That was the striking part of the earlier figure and it rested entirely on the withdrawn small-sample points; the leak-free sweep begins above any plausible crossing and so cannot locate it. The regimes of Section 6.4 still gain a data axis from what remains — an accurate mean is worth building once there is enough data to train it past the kernel, and at these sizes there plainly is — but the sample size at which that becomes true is an open number here, and rerunning the small-n ladder under the corrected protocol is the obvious next measurement.

6 Theory

Throughout this section $\mathcal{U} = \mathbb{R}^p$ denotes the (discretized) input space and $\mathcal{V} = \mathbb{R}^q$ the output space, $G : \mathcal{U} \to \mathcal{V}$ the target operator, μ the input distribution, and $(u_1, v_1), \ldots, (u_N, v_N)$ i.i.d. draws with $v_n = G(u_n)$; the data carry no sampling noise in the usual sense, coming from a deterministic solver, though Section 4.6 has something to say about solver error. We write $\ell(w, v) = \|w - v\|_2 / \|v\|_2$ for the relative error and $R(\widehat{G}) = \mathbb{E}_{u \sim \mu}\, \ell(\widehat{G}(u), G(u))$ for the risk, which is the quantity reported in Section 4. The results below are largely elementary, but each one licenses a specific design decision in the pipeline of Section 3, and each has a measurable footprint in the experiments: one statement per stage, from the symmetry handling and the kernel-flow term through stacking, the residual correction, its coverage guarantee, and the choice of nugget. Proofs are collected in Appendix A.

6.1 Symmetry averaging does not increase the risk

The elasticity problem behind the benchmark is invariant under the reflection $x_1 \mapsto 1 - x_1$: the domain, the boundary partition, and the constitutive model are all symmetric, the input distribution has a reflection-invariant covariance, and reflecting the load reflects the stress field. On the discrete grid this is expressed by two permutation matrices: $S \in \mathbb{R}^{p\times p}$ reverses the load samples and $T \in \mathbb{R}^{q\times q}$ reverses the rows of the stress field. Both are orthogonal involutions. Section 2 describes a direct data-driven check of the equivariance $G(Su) = T\,G(u)$; we treat it as an assumption here.

Proposition 6.1 (symmetrization)**.** *Assume $G(Su) = TG(u)$ for μ-a.e. u, that μ is invariant under S, and that T is an orthogonal involution ($T^2 = I$). For a measurable $\widehat{G} : \mathcal{U} \to \mathcal{V}$ define its symmetrization*

$$\widehat{G}_S(u) \;=\; \tfrac{1}{2}\Big(\widehat{G}(u) + T\,\widehat{G}(Su)\Big).$$

Then, for every loss $\ell(\cdot, v)$ that is convex in its first argument and satisfies $\ell(Tw, Tv) = \ell(w, v)$,

$$\mathbb{E}_{u\sim\mu}\,\ell\big(\widehat{G}_S(u), G(u)\big) \;\le\; \mathbb{E}_{u\sim\mu}\,\ell\big(\widehat{G}(u), G(u)\big).$$

The relative error satisfies both hypotheses (T orthogonal gives $\ell(Tw, Tv) = \ell(w, v)$). Proposition 6.1 justifies the two uses of the symmetry in the pipeline: averaging each trained model with its reflected evaluation at test time can only help on average, and training on reflection-augmented data is ordinary empirical risk minimization for the symmetrized class. The measured effect is small and, being an expectation, need not hold sample by sample: over the thirty reflected members of the seed campaign it helps twenty-nine times and hurts once, by a thousandth of a point (Appendix C). The convexity argument behind the proposition is the standard one for group-averaged predictors; Lyle et al. (2020) give a general account.

6.2 What the kernel-flow regularizer estimates

The kernel-flow (KF) loss of Owhadi and Yoo (Owhadi and Yoo, 2019), in the ℓ^2 variant used by Yoo and Owhadi (2021) to regularize deep networks, enters our ablation study as an optional term in the training of the neural means. Its motivation is often stated informally ("a kernel is good if half the data predicts the rest"). The following observation makes the statement exact. Let $z_i = \phi_\theta(u_i)$ be the features of a batch $B = \{1, \ldots, b\}$ (b even), let k be a positive definite kernel on feature space, and for $C \subset B$, $|C| = b/2$, let I_C denote the k-interpolant of $\{(z_j, v_j) : j \in C\}$. The batch KF loss is

$$e_2(\theta; C) \;=\; \sum_{i\in B} \big\|v_i - I_C(z_i)\big\|_2^2. \tag{1}$$

Proposition 6.2 (KF loss is a leave-half-out estimate)**.** *Let C be drawn uniformly among the subsets of B of size $b/2$ and assume k restricted to $\{z_i\}_{i\in B}$ is strictly positive definite. Then $e_2(\theta;C)=\sum_{i\in B\setminus C}\|v_i-I_C(z_i)\|_2^2$, and*

$$\mathbb{E}_C\big[e_2(\theta;C)\big] \;=\; \frac{b}{2}\,\mathbb{E}_{C,\,i\sim \mathrm{Unif}(B\setminus C)}\big[\big\|v_i-I_C(z_i)\big\|_2^2\big],$$

i.e. $\frac{2}{b}\,e_2$ is an unbiased estimator, over the split randomness, of the expected squared error committed on a held-out point by the kernel predictor trained on a random half of the batch. When (B,C) are resampled at every step, the stochastic gradient $\nabla_\theta e_2(\theta;C)$ is unbiased for $\nabla_\theta\,\mathbb{E}_{B,C}[e_2(\theta;C)]$ whenever the expectation and derivative commute (e.g. under local Lipschitz domination).

The training loss we minimize for a neural mean is a sum of an empirical risk term and, optionally, $\beta\,e_2$: the first term is a linear functional of the empirical measure of the batch and measures fit, while Proposition 6.2 shows the second measures the *generalization of a kernel method built on the learned features.* This is the sense in which a KF-regularized network is trained to be a good feature map for the kernel correction that follows.

6.3 Stacking on a held-out split is safe

Between the means and the correction sits one more fitted object: the convex weights of the stacked ensemble, chosen to minimize the empirical risk on the validation split. Two elementary facts justify this step. First, by convexity, a convex combination of predictors is never worse than the corresponding weighted average of their risks, so stacking cannot be hurt by a weak member beyond its weight. Second, the weights live on a low-dimensional simplex and are fitted on hundreds of samples, so the selection cannot meaningfully overfit; the following proposition quantifies this with explicit constants.

Proposition 6.3 (validation stacking)**.** *Let $f_1,\dots,f_M:\mathcal{U}\to\mathcal{V}$ be fixed maps, let $\Delta=\{w\in\mathbb{R}^M : w_m\ge 0,\sum_m w_m=1\}$, and for $w\in\Delta$ write $F_w=\sum_m w_m f_m$. Assume the members' per-sample relative errors are bounded: $\ell(f_m(u),G(u))\le B$ for μ-a.e. u and every m.*

(i) For every $w\in\Delta$, $\ell(F_w(u),G(u))\le\sum_m w_m\,\ell(f_m(u),G(u))$ pointwise; in particular $R(F_w)\le\sum_m w_m R(f_m)\le\max_m R(f_m)$, and $\ell(F_w(u),G(u))\le B$.

(ii) Let $\widehat{w}$ minimize the empirical risk $\widehat{R}(w)=\frac{1}{m}\sum_{i=1}^m \ell(F_w(\tilde u_i),G(\tilde u_i))$ over Δ, computed on an i.i.d. validation sample $\tilde u_1,\dots,\tilde u_m\sim\mu$ independent of $f_1,\dots,f_M$. Then for every $\delta\in(0,1)$, with probability at least $1-\delta$,

$$R(F_{\widehat{w}}) \;\le\; \min_{w\in\Delta} R(F_w) \;+\; \frac{8B}{m} \;+\; B\sqrt{\frac{2\big((M-1)\log\big(m(M-1)+1\big)+\log(2/\delta)\big)}{m}}.$$

With $M=5$ members, $m=1000$ validation samples and $\delta=0.05$ the right-hand excess evaluates to about $0.28\,B$; at $B=0.25$, the largest per-sample member error we observe, this caps the selection cost at roughly seven percentage points. The bound is conservative, as such bounds are: the realized validation-to-test gap of the stack in Section 4 is two orders of magnitude smaller. Its role is the scaling $\sqrt{M \log m/m}$, which says that fitting a handful of convex weights on a thousand held-out samples is statistically almost free; this is why the pipeline spends its validation data there and on the kernel hyperparameters and nowhere else.

The stack the pipeline deploys on the structural-mechanics benchmark is finer than a single convex vector: the weights are affine and vary per grid point, fitted by ridge on the validation split, and the choice between this object and the global convex stack is itself made on held-out data (fit both on one half of the split, keep whichever wins on the other half, refit the winner on the full split). Proposition 6.3 does not cover that construction; the following two statements do, in the same elementary style.

Lemma 6.4 (validated selection)**.** *Let $g_1, \dots, g_K : \mathcal{U} \to \mathcal{V}$ be fixed maps with $\ell(g_k(u), G(u)) \le B$ for μ-a.e. u and every k, and let $\widehat{k}$ minimize the empirical risk over an i.i.d. sample of size m' drawn from μ independently of everything used to build the g_k. Then, with probability at least $1-\delta$,*

$$R(g_{\widehat{k}}) \;\le\; \min_{k \le K} R(g_k) \;+\; B\sqrt{\frac{2\log(2K/\delta)}{m'}}.$$

Here $K = 2$ and $m' = 500$: comparing two candidate stacks on half the validation split costs at most a few points of risk in the worst case, and in practice far less. The refit winner is covered by the next statement, which prices the per-coordinate affine fit itself.

Proposition 6.5 (per-pixel affine stacking)**.** *Fix members $f_1, \dots, f_M$, write $D = \sqrt{M+1}$, and for each output coordinate j let $\varphi_j(u) = (f_1(u)_j, \dots, f_M(u)_j, b) \in \mathbb{R}^{M+1}$, where b bounds all member and target values: $|G(u)_j| \le b$, $|f_m(u)_j| \le b$ for μ-a.e. u and all j, m. Consider the per-coordinate affine class $\{u \mapsto w^\top \varphi_j(u) : \|w\|_2 \le W\}$ and the coordinate risks $R_j(w) = \mathbb{E}\,(w^\top \varphi_j(u) - G(u)_j)^2$, estimated on an i.i.d. validation sample of size m independent of the members.*

(i) If $\widehat{w}_j$ minimizes the empirical risk of coordinate j over the class, then with probability at least $1-\delta$,

$$\frac{1}{q}\sum_{j=1}^{q}\Big[R_j(\widehat{w}_j) - \min_{\|w\|_2 \le W} R_j(w)\Big] \;\le\; \frac{8\,Wb^2D\,(1+WD) + 2\,b^2(1+WD)^2\sqrt{\frac{1}{2}\log(4q/\delta)}}{\sqrt{m}}.$$

(ii) If instead $\widehat{w}_j$ minimizes the ridge objective $\widehat{R}_j(w) + \lambda\|w\|_2^2$ and satisfies $\|\widehat{w}_j\|_2 \le W$, the same bound holds with an additional λW^2 on the right.

This is a capacity bound, and it should be read as one. Its content is the rate: with a handful of members the per-coordinate fit is paid for by the validation split at rate $\sqrt{(M+1)/m}$ per coordinate, uniformly over coordinates, and the ridge form the pipeline uses adds only its explicit penalty ($\lambda = 10^{-3}$ here, so λW^2 is small against the coordinate risks at the fitted weight norms, which the released runs report). Its evaluated value is not informative. The bound is stated for the coordinatewise mean-squared risk in the units of the stress field, not for the relative error of the tables, and the constants b and W enter it polynomially; at the measured values (b is the largest stress value in original units, about a thousand) the right-hand side exceeds the realized excess by many orders of magnitude, as the released evaluation shows. What the statement establishes is that fitting $q(M+1)$ numbers on a thousand held-out samples is a selection problem of the ordinary $m^{-1/2}$ kind, with no dependence on q beyond the logarithm; the half-split comparison of Lemma 6.4 then guards the one discrete choice that remains.

One stage of the pipeline is still uncovered at this point: the residual correction, whose scale and nugget are chosen from a finite grid on the same validation split that fitted the stack. The next statement closes the chain. Its key observation is structural: for a fixed grid point the correction is fit to the training residual of the stack, which is affine in the stack weights, so the corrected predictor is again a per-coordinate affine function of w with a transformed feature map, and the machinery of Proposition 6.5 applies to the composite class at the cost of one further constant, a uniform bound on the correction weights, which Lemma 6.7 below supplies in closed form.

Corollary 6.6 (capacity bound for the deployed corrected surrogate)**.** *Assume the setting of Proposition 6.5, and let Γ be a finite set of correction configurations, each defining weights $c_\gamma(u) \in \mathbb{R}^{n_{\mathrm{tr}}}$ with the corrected predictor*

$$h_{w,\gamma}(u)_j \;=\; w_j^\top \varphi_j(u) + c_\gamma(u)^\top \big(Y_j - \Phi_j w_j\big),$$

where $Y_j \in \mathbb{R}^{n_{\mathrm{tr}}}$ collects the training targets of coordinate j and Φ_j the members' training predictions; $\gamma = 0$ (no correction, $c_0 \equiv 0$) is included in Γ. Suppose the correction weights are uniformly bounded, $\sup_{u,\gamma} \|c_\gamma(u)\|_1 \le \kappa$. If the deployed surrogate uses any validation-measurable choice $(\widehat{w}, \widehat{\gamma})$ with $\|\widehat{w}_j\|_2 \le W$ and $\widehat{\gamma}$ minimizing the empirical validation risk, aggregated over coordinates, among $\gamma \in \Gamma$ at the fitted $\widehat{w}$ (the grid choice is shared across coordinates, as in the pipeline), then with probability at least $1-\delta$,

$$\frac{1}{q}\sum_j R_j\big(h_{\widehat{w},\widehat{\gamma}}\big) \;\le\; \min_{\gamma\in\Gamma}\, \frac{1}{q}\sum_j R_j\big(h_{\widehat{w},\gamma}\big) + \frac{2(1+\kappa)^2\, b^2 (1+WD)\Big(4WD + (1+WD)\sqrt{\tfrac{1}{2}\log\big(4q|\Gamma|/\delta\big)}\Big)}{\sqrt{m}}.$$

Since $\gamma = 0$ is among the candidates, the oracle on the right is at most the fitted stack's own risk, and combining with Proposition 6.5(ii) bounds the deployed risk against the best affine stack itself. The constants b and W are measured and released with the runs; for kernel ridge weights, κ is supplied by the next lemma.

The hypothesis on the correction weights is not an empirical assumption. In the pipeline each $\gamma = (s, \lambda)$ is a length scale and a nugget, and the weights are those of kernel ridge regression, $c_\gamma(u) = (K_\gamma + n\lambda I)^{-1} k_\gamma(X, u)$, so that $\widehat{r}(u) = c_\gamma(u)^\top R$ in the notation of Section 3.3. Their ℓ^1 norm is bounded uniformly in u by the nugget alone.

Lemma 6.7 (uniform bound on kernel ridge weights)**.** *Let k be a positive definite kernel with $k(u,u) \le k_{\max}$ for all u, let $X = (u_1, \ldots, u_n)$ be any design with Gram matrix K, and let $\lambda > 0$. The kernel ridge weights $c_\lambda(u) = (K + n\lambda I)^{-1} k(X,u)$ satisfy, for every u,*

$$\|c_\lambda(u)\|_1 \;\le\; \sqrt{n}\,\|c_\lambda(u)\|_2 \;\le\; \frac{P_\lambda(u)}{\sqrt{\lambda}} \;\le\; \sqrt{\frac{k_{\max}}{\lambda}},$$

where $P_\lambda(u)^2 = k(u,u) - k(u,X)(K+n\lambda I)^{-1}k(X,u)$ *is the posterior variance of Section 6.6, equation* (2)*. In particular, for the residual correction of Section 3.3 — a Matérn kernel with* $k(u,u)=1$ *and the nugget grid* $\lambda \in \{10^{-6}, 10^{-5}, 10^{-3}\}$ *— the hypothesis of Corollary 6.6 holds, whatever the length scale, with*

$$\kappa_{\mathrm{an}} \;:=\; \max_{\gamma\in\Gamma} \lambda_\gamma^{-1/2} \;=\; 10^3.$$

The proof is two lines in Appendix A, and the bound is the right kind of constant for the corollary: it depends on nothing that is estimated. It is also why the corollary is a capacity bound and not a certificate. The excess term carries $(1+\kappa)^2$, so at $\kappa_{\mathrm{an}} = 10^3$ the corollary's right-hand side is, like that of Proposition 6.5 before it, vacuous at the shipped constants by many orders of magnitude. The realized scale of the weights is far smaller: on a test probe of five hundred points the largest ℓ^1 norm over the grid averages 31.2 across three seeds (Appendix C), about a thirtieth of κ_{an}, so that evaluating the display at the probe value would shrink it by a factor of close to a thousand. But a supremum over a finite probe is a lower bound on the constant the corollary needs, not the constant, and we do not use it as one. What the corollary establishes is structural: the object the pipeline ships — fitted stack, grid-tuned correction, validated final choice — is covered by an oracle inequality of the ordinary $m^{-1/2}$ form, in the coordinatewise mean-squared metric, whose every constant is either measured or given in closed form. The certificate of this paper, in the sense of a bound whose evaluated value is the quantity reported, is Theorem 6.18.

6.4 What the stack can achieve

Proposition 6.3 says fitting the weights is safe; it does not say mixing is worth anything. That is decided by a single measurable matrix. For a map f write the *normalized residual* at u as $\rho_f(u) = \big(f(u) - G(u)\big)/\|G(u)\|_2$, so that $\ell(f(u), G(u)) = \|\rho_f(u)\|_2$.

Proposition 6.8 (second-moment identity for convex stacks)**.** *Let* $f_1, \ldots, f_M$ *be fixed maps and define* $S \in \mathbb{R}^{M\times M}$ *by* $S_{mk} = \mathbb{E}_{u\sim\mu}\langle \rho_{f_m}(u), \rho_{f_k}(u)\rangle$*. For* w *in the simplex* Δ^{M-1} *let* $f_w = \sum_m w_m f_m$*. Then*

$$\mathbb{E}_{u\sim\mu}\, \ell\big(f_w(u), G(u)\big)^2 \;=\; w^\top S\, w,$$

so the squared-metric risk of every convex stack is determined by S*, and the best achievable is* $\min_{w\in\Delta^{M-1}} w^\top S w$*. In particular:*

(i) (two members) if S *has diagonal* (e_1^2, e_2^2) *with* $e_1 \le e_2$ *and correlation* $\varrho = S_{12}/(e_1 e_2)$*, mixing in the weaker member strictly helps if and only if* $\varrho < e_1/e_2$*, in which case the optimum is interior with value* $e_1^2 e_2^2(1-\varrho^2)/(e_1^2 + e_2^2 - 2\varrho e_1 e_2)$*;*

(ii) (equicorrelated family) if $S_{mm} = e^2$ *and* $S_{mk} = \varrho\, e^2$ *for all* $m \ne k$ *with* $\varrho \ge 0$*, then* $\min_w w^\top S w = e^2\big(\varrho + (1-\varrho)/M\big)$*, attained at uniform weights and decreasing to* $e^2\varrho$ *as* $M \to \infty$*.*

Part (i) is the classical two-member minimum-variance rule, and part (ii) is the equicorrelated case of the ambiguity decomposition of ensemble learning (Krogh and Vedelsby, 1995); see Ueda and Nakano (1996) for the generalization-error form and Brown et al. (2005) for a survey. Both enter here as measurable decision tools on the estimated S.

Everything in this subsection is stated for the squared metric, and the tables of Sections 4 and 5 report a different one. The distinction is small numerically and decisive logically, so we make it explicit once and refer back to it wherever the two families of numbers meet.

Remark 6.9 (the two error metrics)**.** *For a map f write $\mathcal{E}_1(f) = \mathbb{E}\|\rho_f(u)\|_2$ and $\mathcal{E}_2(f) = (\mathbb{E}\|\rho_f(u)\|_2^2)^{1/2}$, the mean and the root mean square of the per-sample relative error. The tables report $\mathcal{E}_1$; Proposition 6.8 is a statement about $\mathcal{E}_2$, since $\mathcal{E}_2(f_w)^2 = w^\top S w$ by definition. Writing $X = \|\rho_f(u)\|_2$ and $c_f = (\operatorname{Var} X)^{1/2}/\mathbb{E}X$,*

$$\mathcal{E}_2(f)^2 - \mathcal{E}_1(f)^2 = \operatorname{Var} X, \qquad \mathcal{E}_1(f)/\mathcal{E}_2(f) = (1 + c_f^2)^{-1/2} \le 1,$$

with equality only if the per-sample error is almost surely constant. Two consequences are used in this paper. The floor belongs in $\mathcal{E}_2$, because only $\mathcal{E}_2$ is determined by S: the map $w \mapsto w^\top S w$ is a quadratic form in second moments, whereas $w \mapsto \mathbb{E}\|\sum_m w_m \rho_{f_m}(u)\|_2$ depends on the full joint law of the residuals. And upper bounds transfer downward for free, since $\mathcal{E}_1 \le \mathcal{E}_2$, while lower bounds do not — so a floor read off S says nothing about a table entry without the dispersion of the same predictor. Comparing the two directly is biased in favour of the table entry by $(1 + c_f^2)^{-1/2}$, which is about six percent on the structural-mechanics stack and ten to eleven percent on the O2 objects of Section 5; the dispersion is a property of the predictor and not a constant of the problem, so a single conversion factor should not be reused across objects.

Part (i) is an exact criterion, but it is used as a decision rule, and a decision rule is applied to estimates. Both ϱ and e_1/e_2 must be formed on data the decision does not see, so what governs the rule in practice is not the criterion but the gap it has to resolve relative to the noise in that estimation. The following makes this quantitative, and is stated for any threshold comparison of this shape.

Proposition 6.10 (margin law for a threshold rule)**.** *Let $D = \varrho - e_1/e_2$ be the signed gap of Proposition 6.8(i), so that mixing the weaker member strictly helps exactly when $D < 0$, and suppose $D \neq 0$. Let $\widehat{D}$ be an estimate of D formed on a split independent of the one on which D is evaluated, and suppose the estimation error $\widehat{D} - D$ is subgaussian with variance proxy σ^2, that is $\mathbb{E}\exp\big(\lambda(\widehat{D} - D)\big) \le \exp(\lambda^2\sigma^2/2)$ for all $\lambda \in \mathbb{R}$, conditionally on D. Then, writing $\Delta = |D|$ for the margin,*

(i) on $\{D \neq 0\}$, the decision taken from $\widehat{D}$ differs from the correct one with probability at most $\exp(-\Delta^2/2\sigma^2)$;

(ii) for every $\tau \ge 0$, and with no condition on D,

$$\mathbb{P}\big(\operatorname{sign}\widehat{D} \neq \operatorname{sign} D, \quad |\widehat{D}| \ge \tau\big) \ \le\ \exp\!\left(-\frac{\tau^2}{2\sigma^2}\right).$$

Proof Suppose first $D < 0$, so $\Delta = -D$ and the correct decision is to mix. The decision taken from $\widehat{D}$ differs only if $\widehat{D} \ge 0$, which forces $\widehat{D} - D \ge -D = \Delta$. The subgaussian hypothesis and the Chernoff bound give, for every $\lambda > 0$, $\mathbb{P}(\widehat{D} - D \ge \Delta) \le \exp(\lambda^2\sigma^2/2 - \lambda\Delta)$, and optimizing at $\lambda = \Delta/\sigma^2$ yields $\exp(-\Delta^2/2\sigma^2)$. If instead $D > 0$ then $\Delta = D$, the decision differs only if $\widehat{D} \le 0$, which forces $\widehat{D} - D \le -\Delta$, and the same argument applied to

$-(\widehat{D} - D)$, which is subgaussian with the same proxy, gives the identical bound. The two cases are exhaustive because $D \neq 0$, which proves (i).

For (ii), condition on D again. If $D < 0$ the error event is $\{\widehat{D} \geq 0\}$, so on the error event intersected with $\{|\widehat{D}| \geq \tau\}$ we have $\widehat{D} \geq \tau$ and therefore $\widehat{D} - D \geq \tau - D \geq \tau$; the one-sided estimate above applies with Δ replaced by τ. If $D \geq 0$ the error event is $\{\widehat{D} < 0\}$, so on it with $|\widehat{D}| \geq \tau$ we have $\widehat{D} \leq -\tau$ and therefore $D - \widehat{D} \geq \tau + D \geq \tau$, and the one-sided estimate in the other direction applies. In both cases the bound is $\exp(-\tau^2/2\sigma^2)$, which is free of D, so taking the expectation over D removes the conditioning. ■

Three remarks on how the statement is used in Section 5.2. Part (i) is conditional on the true margin Δ, which is not available when the decision is taken; what is available is $|\widehat{D}|$, and part (ii) is the statement about that quantity. It is a joint bound: it limits the probability that a decision is both wrong and taken at an observed margin of at least τ, and it makes that event rare among all decisions once τ exceeds a few multiples of σ. It is not a bound on the error rate among the decisions whose observed margin exceeds τ — that conditional probability is the joint one divided by $\mathbb{P}(|\widehat{D}| \geq \tau)$, on which the proposition says nothing — so the accuracy within a margin bin is an empirical calibration of the rule and not a guarantee, and that is how Section 5.2 reports it. Both bounds are vacuous for margins of order σ and fall off quickly beyond, so a rule of this shape has a band around its own threshold inside which it carries no guarantee, and the band is set by σ rather than chosen. Second, σ is estimable without knowing D: the differences between $\widehat{D}$ and the value of D realized on a held-out block, taken across many pairs, have exactly this scale. Third, $|\widehat{D}|$ is observable at decision time, so a practitioner can report the observed margin next to the verdict; part (ii) says that verdicts reported at a large margin are seldom wrong, and the scoreboard of Section 5.2 says how the realized accuracy varies with that margin.

Exact equicorrelation never holds in practice, but for convex weights the floor survives one-sided bounds on the entries of S, which are what one actually measures.

Corollary 6.11 (ensembling floor)**.** *If every member satisfies $S_{mm} \geq \bar{e}^2$ and every pair satisfies $S_{mk} \geq \bar{\varrho}\,\bar{e}^2$ with $\bar{\varrho} \in [0,1]$, then every convex combination f_w obeys*

$$\mathbb{E}\,\ell\big(f_w(u), G(u)\big)^2 \;=\; w^\top S w \;\geq\; \bar{e}^2\big(\bar{\varrho} + (1-\bar{\varrho})\,\|w\|_2^2\big) \;\geq\; \bar{e}^2\,\bar{\varrho}.$$

No number of additional members with the same error level and mutual correlations moves the ensemble's root-mean-square relative error below $\bar{e}\sqrt{\bar{\varrho}}$.

The floor is a lower bound; a matching upper bound follows from the same entrywise information, and together they pin the achievable window from both sides by measured quantities alone.

Corollary 6.12 (the floor is nearly achieved)**.** *Suppose, in addition to the hypotheses of Corollary 6.11, that $S_{mm} \leq \bar{e}^2(1+\delta_e)$ for every m and $S_{mk} \leq (\bar{\varrho} + \delta_\varrho)\,\bar{e}^2$ for every $m \neq k$. Then*

$$\bar{\varrho} \;\leq\; \frac{1}{\bar{e}^2} \min_{w \in \Delta^{M-1}} w^\top S w \;\leq\; \bar{\varrho} + \delta_\varrho + \frac{1 + \delta_e - \bar{\varrho} - \delta_\varrho}{M}.$$

The width of the bracket is the measured correlation spread δ_ϱ plus a $1/M$ term: when the members' errors and pairwise correlations are tight, whatever convex ensemble one builds *sits within that window of the floor, and conversely no tuning of the weights can be blamed for more than the window.*

On the structural-mechanics benchmark the measured spreads ($\varrho \in [0.83, 0.97]$ across member pairs and seeds, at member errors within about ten percent of one another) put the window at roughly a fifth of the squared floor: the correlation spread alone is 0.14, a sixth of $\bar{\varrho} = 0.83$, and the $1/M$ term adds a few hundredths. The best convex mixture of the measured S sits at the bottom of that window, $4.940 \pm 0.057\%$ against a floor of $4.922 \pm 0.056\%$, both in the root-mean-square metric $\mathcal{E}_2$ in which the corollary is stated. The bracket is evaluated against the realized ensemble risk at every seed, and its stability is the point: the realized mean relative error $\mathcal{E}_1$ of the mixture is a factor 0.938 ± 0.003 of the predicted root mean square, the dispersion factor of Remark 6.9, so ten independent redraws of the validation split and of every network initialization move the prediction by six hundredths of a point and the ratio by three thousandths.

A second consequence of the same decomposition explains the final OCO-2 surrogate. Evaluation metrics there are diagonal in the output coordinates (the radiance metric is the weighting s_z), and diagonal metrics decouple.

Proposition 6.13 (per-coordinate combination under diagonal metrics)**.** *For weights $v = (v_{mj})$ that may depend on the output coordinate, let $f_v(u)_j = \sum_m v_{mj} f_m(u)_j$. For any diagonal metric with positive weights w,*

$$\mathbb{E}\big\|\operatorname{diag}(w)\big(f_v(u) - G(u)\big)\big\|_2^2 = \sum_j w_j^2\, \mathbb{E}\big(f_v(u)_j - G(u)_j\big)^2,$$

so the minimizing weights solve q separate problems, one per output coordinate, none of which involves w. The per-coordinate optimum is the same for every diagonal metric, and it weakly dominates every combination whose weights are shared across coordinates, for all such metrics simultaneously.

The practical content: when a problem is scored in several diagonal metrics at once, the combination stage does not need to know which one matters. Fit the best combination coordinate by coordinate on validation and the result serves all of them; only the members need metric-aware training, which is where the weighted mean of Section 5 enters.

The proposition as stated covers quadratic metrics with fixed diagonal weights, while the experiments report relative errors, which weight each sample by its output norm. The decoupling survives that weighting.

Corollary 6.14 (weighted metrics and the reported error)**.** *Let $a : \mathcal{U} \to [0, \infty)$ be measurable with $\mathbb{E}[a(u)\, \|f_v(u) - G(u)\|_2^2] < \infty$. Then, for every diagonal w,*

$$\mathbb{E}\Big[a(u)\, \big\|\operatorname{diag}(w)\big(f_v(u) - G(u)\big)\big\|_2^2\Big] = \sum_j w_j^2\, \mathbb{E}\Big[a(u)\, \big(f_v(u)_j - G(u)_j\big)^2\Big],$$

so the minimization still splits into q per-coordinate problems, none involving w. Taking $a(u) = \|G(u)\|_2^{-2}$ and $w = \mathbf{1}$ makes the left side the mean squared relative error: the combination fitted coordinate by coordinate with per-sample weights a on validation is simultaneously optimal, within its class, for the squared relative error and for every diagonal *reweighting of it. The reported metric,* $\mathbb{E}\,\|\rho_{f_v}\|_2$ *with* ρ *the normalized residual of Section 6.4, is controlled through* $\mathbb{E}\|\rho\|_2 \le (\mathbb{E}\|\rho\|_2^2)^{1/2}$*; the gap between the two sides is the dispersion factor whose measured value stays near* 0.93 *across the pipelines of this paper.*

The experiments run the combination both ways, unweighted and with the per-sample weights of the corollary, and report both.

Proposition 6.13 and its corollary describe the population optimum; the pipeline computes an empirical selection, coordinate by coordinate, on the validation split. The finite-sample cost of that step is priced by the same argument that priced the stacks.

Proposition 6.15 (empirical per-coordinate selection)**.** *Let* $f_1, \dots, f_M$ *be fixed maps,* $a : \mathcal{U} \to [0, A]$ *a bounded per-sample weight, and suppose* $a(u)\,(f_m(u)_j - G(u)_j)^2 \le L$ *for* μ*-a.e.* u*, every member* m *and coordinate* j*. On a validation sample of size* m_v *independent of the members, let* $\widehat{m}(j)$ *minimize the empirical weighted risk of coordinate* j *over the* M *members, and let the combination take coordinate* j *from member* $\widehat{m}(j)$*. Then, with probability at least* $1-\delta$*, simultaneously for every coordinate,*

$$R_j\big(\widehat{m}(j)\big) \;\le\; \min_{m\le M} R_j(m) \;+\; 2L\sqrt{\frac{\log(2Mq/\delta)}{2m_v}}, \qquad R_j(m) = \mathbb{E}\big[a(u)\big(f_m(u)_j - G(u)_j\big)^2\big],$$

and consequently, for every diagonal metric w *at once,* $\sum_j w_j^2 R_j(\widehat{m}(j)) \le \sum_j w_j^2 \min_m R_j(m) + 2L\big(\sum_j w_j^2\big)\sqrt{\log(2Mq/\delta)/(2m_v)}$*.*

On the OCO-2 bands the deployed combination is exactly this estimator ($M = 6$ members, $q = 40$ coordinates, $m_v = 2000$), in the unweighted form and, for the squared relative error, with $a(u) = \|G(u)\|_2^{-2}$; the proposition prices both at the same rate, and the selection cost it bounds is what the seed experiments measure directly.

The corollary is the quantity this paper keeps measuring, at two different levels. Across architecture families on the structural-mechanics benchmark the pairwise correlations exceed 0.86 at member errors near 4.7%, which places the floor at $4.922 \pm 0.056\%$ in the root-mean-square metric it is stated in; Section 4.6 makes the comparison with the deployed pipeline in that metric rather than across the two. Across random seeds of one architecture on the OCO-2 bands the correlations are 0.78, 0.97 and 0.96 at member errors of 4.4%, 16.4% and 8.2%; measured in the squared metric of the proposition itself, over all 45 seed pairs per band, the floors are 4.360%, 17.970% and 9.248% and the realized ten-seed ensembles sit at 4.419%, 17.997% and 9.267% (Section 5). When an ensemble is at its floor the corollary says where improvement cannot come from. The reported tables use the mean rather than the root mean square of $\|\rho_f\|_2$, and the two are not interchangeable: Remark 6.9 gives the relation, the ratio runs from about 0.89 to 0.94 across the objects here, and every comparison in this paper between a quantity read off S and a table entry is made after converting one to the other. The proposition frames the diversity experiment of Section 4.6: what a new member is worth is legible in the row it adds to S, before any weights are fitted. On this benchmark the rows all look alike, pairwise correlations of 0.83–0.97 among trained networks of three architecture families and two losses, 0.86–0.88 against the kernel predictor, so the floor sits just below the best single member, and the measured stack lands on it. Part (i) is the same computation specialized to two members; it correctly predicts, from measured

(e_1, e_2, ϱ) alone, which members receive weight in the fitted stack and which are dropped. Section 5 applies the same test on a problem where the answer comes out the other way: there the kernel's error is inflated by a factor the next subsection quantifies, the threshold e_1/e_2 collapses, and the kernel is dropped even at a residual correlation of 0.14.

6.5 Metric and representation: when a fixed kernel loses to a network

The structural-mechanics benchmark has the neural mean and the isotropic kernel within half a point of each other. On the OCO-2 emulation problem of Section 5 the same fixed kernel is an order of magnitude behind the same network. Two mechanisms can produce such a gap, and they separate cleanly.

Suppose the operator factors as $G(u) = g(Au)$ with $A \in \mathbb{R}^{d\times d}$ nonsingular and g of unit H^s norm, and that μ has a density bounded above and below on a compact domain. Order the singular values $\sigma_1 \geq \cdots \geq \sigma_d$ of A; they are the rates at which G varies along the input directions. Suppose A has *approximate rank* r at the sample size, meaning the trailing singular values fall below the achievable resolution, $\sigma_{r+1} \lesssim n^{-1/r}$, so the image $A\mathcal{U}$ is effectively r-dimensional at scale $n^{-1/r}$. The idealization beneath the display also takes the leading-r projection of the image design to keep a density bounded above and below, the trailing directions to be exact zeros at the working resolution, and the interpolation limit $\lambda \to 0$; Appendix A records the argument at that level, as a sketch rather than a complete proof. The statement is accordingly a heuristic rate calculation, and it is labelled informal: nothing else in the paper rests on it, and its role is to say what kind of gain a metric can and cannot produce.

Proposition 6.16 (isotropic and adapted kernel rates, informal)**.** *Let $\widehat{G}_n^{\mathrm{iso}}$ be kernel ridge regression with a Matérn kernel of smoothness s and a validation-optimal single length scale, and $\widehat{G}_n^{\mathrm{ard}}$ the same construction in the metric $u \mapsto Au$. Up to constants depending on (σ_i), g, s and the density bounds, and up to logarithmic factors in n, with high probability over n i.i.d. samples,*

$$\mathbb{E}\big\|G - \widehat{G}_n^{\mathrm{iso}}\big\| \;\lesssim\; \sigma_1^s\, n^{-s/d}, \qquad \mathbb{E}\big\|G - \widehat{G}_n^{\mathrm{ard}}\big\| \;\lesssim\; n^{-s/r},$$

so the ratio of the two bounds is of order $\sigma_1^s\, n^{s(1/r-1/d)}$. When A is close to full rank ($r \approx d$) the two rates coincide and only the constant σ_1^s separates them.

The calculation (Appendix A) is the scattered-data estimate $\|G - \widehat{G}_n\| \lesssim h_X^s \|G\|_{H^s}$ applied to the two designs: one length scale must fill all d directions, so $h_X \asymp n^{-1/d}$ and the norm carries σ_1^s; the adapted metric, when A has a spectral gap, confines the design to r resolved directions and improves the fill distance. Both displays are upper bounds under the stated idealization — the exact metric A, a fixed g, and the length-scale selection folded into the logarithmic factors — and we claim no matching lower bound; the heuristic is used here to calibrate what a metric can and cannot buy, not as a rate theorem. It isolates the *metric* part of a network's advantage, the part an anisotropic kernel would recover, and it says this part is a rate gain only under an approximate-rank gap; without one it is the constant σ_1^s alone. The rest is *representational*: a stationary kernel of fixed smoothness reaches only its native space, at any metric, while the network adapts its features to the map. The two parts are separated experimentally by handing the kernel the anisotropic metric and seeing how much of the gap closes. On the OCO-2 bands of Section 5 the metric closes a factor of 1.1 to 1.5 out of a tenfold gap (Section 5.1); the remainder is representational, and its direct evidence is that the same kernel machinery applied to the network's learned features recovers the network's accuracy and slightly exceeds it.

The rate half of the heuristic is tested rather than assumed, and the test is the more interesting for not going our way. Both kernel rows are refit at nested training sizes over a sixteenfold range, at ten seeds per band. The sensitivity map's approximate rank is 0.54 to 0.60 of the input dimension, a gap wide enough that the calculation would permit a rate gain; the measured difference between the two empirical exponents is nevertheless small on all three bands and consistent with zero on one. The displays are upper bounds with no matching lower bound, so this does not contradict them — the adapted rate is free to beat its own bound — but it does mean that on these problems the rank gap is not converted into a rate, and a reader should take from Proposition 6.16 only the part the measurement supports: an anisotropic metric is worth a constant. The controlled rank-gap target of the verification suite, where the construction is exactly the idealization the calculation assumes, does show the adapted exponent improving; the distance between that and the OCO-2 bands is the distance between the idealization and a real sensitivity map.

The representational term has a precise reading through the same optimal-recovery bound that Section 6.6 makes exact. That bound factors the error as $\|G - m\|_{\mathcal{H}} \cdot P_\lambda$, a product of the target's norm in the kernel's native space and a design factor set by the Gram spectrum. Changing the features changes both in principle, but on the O2 band only one moves. The effective dimension of the Matérn Gram, the quantity Lemma 6.21 controls, is nearly identical on the raw input and on the learned features (at $n = 4000$ and a matched median length scale, 3995 against 3994 at a 10^{-8} nugget, with equal leading eigenvalue mass), so the design factor is essentially fixed. The native-space norm of the target is not: interpolating the same outputs through the two kernels, the raw-input kernel carries the target at squared norm 1.64×10^6 and the feature kernel at 3.87×10^4, a factor of 42. The network does not condition the kernel better; it moves the target to a low-norm corner of a native space of the same size, where the exact solve reaches it.

This is not an accident of the O2 band; it is what a pulled-back kernel does. Writing φ for the feature map and k for the base kernel, the deep kernel head uses $k_\varphi(x, x') = k(\varphi(x), \varphi(x'))$, the construction studied as deep kernel learning (Wilson et al., 2016; Calandra et al., 2016).

Proposition 6.17 (feature pullback)**.** *The native space of k_φ is $\mathcal{H}_{k_\varphi} = \{f \circ \varphi : f \in \mathcal{H}_k\}$ with $\|g\|_{\mathcal{H}_{k_\varphi}} = \min\{\|f\|_{\mathcal{H}_k} : f \circ \varphi = g \text{ on } \mathcal{X}\}$. In particular, if the target factors through the features as $G = h \circ \varphi$ with $h \in \mathcal{H}_k$, then $\|G\|_{\mathcal{H}_{k_\varphi}} \leq \|h\|_{\mathcal{H}_k}$, and the optimal-recovery bound of Theorem 6.18 for the feature-kernel regression is governed by $\|h\|_{\mathcal{H}_k}$ rather than by the norm of G in the raw-input space.*

The proof (Appendix A) is the standard pullback identity for reproducing kernels (Saitoh, 1997; Paulsen and Raghupathi, 2016). Its content here is the reading: a fixed kernel on the raw input must carry the whole warped map G, whose native-space norm is large when G has fine structure; the same kernel on the features carries only the unwarped h, and training drives φ toward exactly the factorization that makes h simple. The measured factor of 42 is that norm gap, and it is the representational advantage that an anisotropic metric, which rescales the input but cannot re-express the map, leaves untouched (recovering only the factor 1.4).

6.6 An error bound for the residual kernel correction

The final stage of the pipeline is kernel ridge regression of the ensemble residual. The bound below is a vector-valued, residual form of the classical power-function estimate for kernel interpolation (Wendland, 2004) and of the optimal-recovery viewpoint of Owhadi and Scovel (2019); we include the short argument in Appendix A to keep the two factors explicit. Fix the mean $m : \mathcal{U} \to \mathcal{V}$ (in the pipeline, the stacked ensemble; in the statement below m is any fixed map independent of the correction sample) and write $r = G - m$ for the residual operator with components r_j, $j = 1, \ldots, q$. Let k be a positive definite kernel on $\mathcal{U}$ with RKHS $\mathcal{H}_k$, and assume $r_j \in \mathcal{H}_k$ for all j with

$$\|r\|_K^2 := \sum_{j=1}^{q} \|r_j\|_{\mathcal{H}_k}^2 < \infty.$$

Given inputs $X = (u_1, \ldots, u_n)$, Gram matrix $K = k(X, X)$ and nugget $\lambda \geq 0$, the correction is $\widehat{r}_\lambda(u) = k(u, X)\,(K + n\lambda I)^{-1} R$, $R_{nj} = r_j(u_n)$, and the corrected predictor is $m + \widehat{r}_\lambda$. Let

$$P_\lambda(u)^2 \;=\; k(u, u) - k(u, X)\,(K + n\lambda I)^{-1} k(X, u) \tag{2}$$

denote the Gaussian process posterior variance with the same nugget (for $\lambda = 0$ this is the classical power function of the point set X).

Theorem 6.18 (certified pointwise bound)**.** *For every $u \in \mathcal{U}$ and every $\lambda \geq 0$,*

$$\big\|G(u) - m(u) - \widehat{r}_\lambda(u)\big\|_2 \;\leq\; \|G - m\|_K\, \widetilde{P}_\lambda(u) \;\leq\; \|G - m\|_K\, P_\lambda(u),$$

where $\widetilde{P}_\lambda(u)^2 = P_\lambda(u)^2 - n\lambda\, \|(K + n\lambda I)^{-1} k(X, u)\|_2^2$.

Three comments. First, the inequality is algebraic: it holds pathwise for every fixed m, every dataset, and every u, with no probabilistic assumptions. In the pipeline m is itself fit on the same training set; this does not affect the validity of the bound, only the reading of $\|G - m\|_K$ as a random (realized) quantity rather than an a priori one. Second, the bound factorizes into a quantity that depends only on the residual operator ($\|G - m\|_K$) and a quantity that depends only on the kernel and the design (P_λ), and the second factor is exactly the posterior standard deviation that a Gaussian process interpretation of the correction would report. This is the sense in which the bound is certified: whatever error the corrected surrogate commits at u is controlled by the reported $P_\lambda(u)$ times a constant that does not depend on u. We stress that this is a statement about a valid upper bound, not about the usefulness of P_λ as a pointwise error *ranking*. Section 4.8 finds that on this benchmark P_λ ranks the error poorly, because the neural mean supplies most of the error and the relative metric is confounded by output amplitude; a distribution-free conformal rescaling of P_λ nonetheless attains its nominal coverage on the test set. Third, the theorem quantifies why one should regress residuals rather than raw targets: the design factor P_λ is the same in both cases, so the gain of the neural mean is the drop from $\|G - \bar{v}\|_K$ (kernel-only, mean $\bar{v}$)

to $\|G - m\|_K$. These norms are not directly observable, but the RKHS norms of the *fitted* interpolants, $\|\widehat{r}_\lambda\|_K^2 = \operatorname{tr}(\alpha^\top K\alpha)$ with $\alpha = (K + n\lambda I)^{-1}R$, are computable lower-bound proxies, and they drop by a factor of about 271 when the kernel is moved from raw targets to ensemble residuals (Table 7). The neural mean does not merely reduce the size of the residual in L^2; it leaves behind a residual operator that is genuinely smoother as seen by the kernel.

6.7 Distribution-free coverage for the reported band

Theorem 6.18 controls the error by $\|G - m\|_K\, P_\lambda(u)$, but the constant $\|G - m\|_K$ is not known a priori, and Section 4.8 shows that P_λ alone ranks the error weakly. What the surrogate reports as uncertainty is therefore not P_λ itself but a rescaling of it, $q\,P_\lambda(u)$, whose multiplier q is calibrated on the held-out split by the split-conformal rule

$$q = Q_{1-\alpha}\Big(\big\{s_i := \|e(\tilde{u}_i)\|_2 / P_\lambda(\tilde{u}_i)\big\}_{i=1}^{m}\Big), \qquad e(u) = G(u) - m(u) - \widehat{r}_\lambda(u), \tag{3}$$

where $Q_{1-\alpha}$ is the $\lceil (1-\alpha)(m+1)\rceil$-th smallest value of the m validation scores. The point of this construction is that its coverage needs neither the bound to be tight nor P_λ to be a good error ranking; it needs only exchangeability, which the fixed splits provide.

Proposition 6.19 (finite-sample coverage)**.** *Fix the mean m, the correction $\widehat{r}_\lambda$ and the design X, and let $P_\lambda(\cdot) > 0$. Suppose the validation and test inputs $\tilde{u}_1, \ldots, \tilde{u}_m, u^\star$ are exchangeable (in particular i.i.d. from μ) and drawn independently of $m, \widehat{r}_\lambda, X$. Let q be the conformal multiplier* (3) *and define the band $C(u) = \{v : \|v - m(u) - \widehat{r}_\lambda(u)\|_2 \le q\,P_\lambda(u)\}$. Then*

$$1 - \alpha \;\le\; \mathbb{P}\big(G(u^\star) \in C(u^\star)\big) \;\le\; 1 - \alpha + \frac{1}{m+1},$$

the upper bound holding when the scores are almost surely distinct.

Remark 6.20 (the exact law of realized coverage)**.** *When the calibration and test scores are i.i.d. with a continuous distribution F — as in the campaign's protocol, where calibration and evaluation sets are a random split of an i.i.d. test block — the coverage probability conditional on the calibration sample is F evaluated at the k-th smallest calibration score, $k = \lceil (1-\alpha)(m+1)\rceil$. Since $F(s_i)$ are i.i.d. uniform, this is the k-th order statistic of m uniforms, with distribution exactly* $\mathrm{Beta}(k,\, m+1-k)$ *(Vovk, 2012); its mean $k/(m+1)$ recovers Proposition 6.19. Conditional on the calibration sample the evaluation indicators are i.i.d. Bernoulli, so the coverage observed on N evaluation points follows the Beta–Binomial law with these parameters. Each seed's observed coverage is therefore placed against the central band of an exact finite-sample law, whose per-seed width at* $m = 1000$ *is close to $\sqrt{\alpha(1-\alpha)/m}$.*

The two-sided statement is the standard guarantee of split conformal prediction (Vovk et al., 2005; Lei et al., 2018), transported to the functional-output setting by taking the nonconformity score to be the relative residual norm $s = \|e(u)\|_2 / P_\lambda(u)$; the proof, a rank argument on the exchangeable scores, is in Appendix A. Three points make it the right closing statement for the uncertainty analysis. It is close to the quantity we compute: the reported coverage of $90.2 \pm 0.6\%$ at the nominal $1 - \alpha = 90\%$ in Section 4.8 realizes this proposition with $m = 1000$ and $\alpha = 0.1$. Any gap to nominal is not the $1/(m+1)$ slack, which is only a tenth of a point; it is the finite-sample fluctuation of conditional coverage for a single calibration draw, whose standard deviation $\sqrt{\alpha(1-\alpha)/m}$ is about a point at these sizes. Remark 6.20 gives the exact law of that fluctuation, and the ten seeds let it be checked rather than invoked: the observed spread across seeds is 0.6 points against the 0.9 points the law predicts, and every seed falls inside the central 95% band of the exact distribution at the 90% level and at the 95% level alike. The hypotheses hold exactly as stated: calibration uses a random subset of the test block that no training, checkpointing, stacking, or tuning step ever touched, and coverage is evaluated on the disjoint remainder (Appendix B), so the exchangeability the proposition assumes is exact, with no low-complexity selection argument needed. It is agnostic to everything the earlier results leave uncertain: the neural mean may be biased, P_λ may correlate with the error weakly or with the wrong sign, and the bound of Theorem 6.18 may be loose, yet the band still covers at the stated rate. And it is where the uncertainty analysis ends: among the candidate uncertainty signals we examine, the conformally rescaled P_λ is the only one that carries a guarantee, so it is the one the surrogate reports.

6.8 Effective dimension from the Gram spectrum

The remaining design choice is the nugget λ, selected by cross-validation in the pipeline. The following exact identity connects that choice to the spectrum of the Gram matrix and, through it, to random matrix descriptions of the design.

Lemma 6.21 (effective dimension and the empirical Stieltjes transform)**.** *Let K be a symmetric positive semidefinite $n \times n$ matrix with eigenvalues $\lambda_1, \ldots, \lambda_n \geq 0$, let $\widehat{m}(z) = \frac{1}{n}\sum_{i=1}^{n}(\lambda_i/n - z)^{-1}$ be the Stieltjes transform of the empirical spectral distribution of K/n. Then for every $\lambda > 0$ the effective dimension of ridge regression with nugget λ satisfies*

$$d_{\mathrm{eff}}(\lambda) \;:=\; \operatorname{tr}\bigl(K(K + n\lambda I)^{-1}\bigr) \;=\; n\bigl(1 - \lambda\, \widehat{m}(-\lambda)\bigr).$$

In particular, if the empirical spectral distribution of K/n converges weakly to a limit law F as $n \to \infty$, then $d_{\mathrm{eff}}(\lambda)/n \to 1 - \lambda\, m_F(-\lambda)$ with m_F the Stieltjes transform of F.

The identity is unconditional; the random-matrix content enters through the choice of F. For the benchmark's simplest reference model, isotropic linear features, the limit can be evaluated in closed form.

Corollary 6.22 (effective dimension under a Marčenko–Pastur limit)**.** *Let $K = XX^\top$ with $X \in \mathbb{R}^{n\times p}$ having i.i.d. entries of mean zero and variance σ^2, and let $p/n \to \gamma \in (0,1]$. Then*

$$\frac{d_{\mathrm{eff}}(\lambda)}{n} \;\longrightarrow\; \gamma\bigl(1 - \lambda\, m(-\lambda)\bigr), \qquad m(-\lambda) = \frac{\sqrt{\bigl(\lambda + \sigma^2(1-\gamma)\bigr)^2 + 4\gamma\sigma^2\lambda} - \bigl(\lambda + \sigma^2(1-\gamma)\bigr)}{2\gamma\sigma^2\lambda},$$

almost surely, where m is the Stieltjes transform of the Marčenko–Pastur law with ratio γ and scale σ^2. In a simulation with $n = 3000$, $p = 900$, $\sigma^2 = 1.7$, the formula agrees with the empirical $d_{\mathrm{eff}}(\lambda)/n$ to four decimal places across $\lambda \in [10^{-3}, 10]$.

Two uses. Quantitatively, the formula makes the nugget–capacity trade-off explicit for the bulk of a feature Gram matrix: eigenvalue mass of Marčenko–Pastur type is absorbed or discarded by λ according to a single closed-form curve, and outlying (spiked) eigenvalues x_s simply add their terms $x_s/(x_s + n\lambda)$ on top. Qualitatively, it separates the two regimes visible in Figure 7: the Matérn Gram matrix on the loads is nothing like a Marčenko–Pastur bulk (its spectrum decays exponentially, which is why d_{eff} is a few hundred out of 19000), whereas the Gram matrices of learned penultimate features do show a bulk-plus-spikes shape, and for them the corollary describes how much of the bulk the cross-validated nugget retains. For the feature Gram matrices of the trained networks we observe the familiar picture of a bulk that is well fitted by a Marčenko–Pastur law (Marčenko and Pastur, 1967) together with a small number of outlying eigenvalues carrying the regression signal, as in spiked covariance models (Baik et al., 2005); for the Matérn Gram matrix on the 41-dimensional loads the spectrum decays rapidly and d_{eff} is small ($\approx$ a few hundred at the cross-validated λ, out of $n = 19000$; Figure 7). This gives a post-hoc reading of the cross-validated nugget: λ lands where $d_{\text{eff}}(\lambda)$ has absorbed the outlying eigenvalues and the leading bulk, and further decreasing λ buys capacity precisely where the spectrum carries little signal. It also explains why the exact solve is affordable: the correction is numerically a problem of dimension d_{eff}, not n.

The effective dimension is also exactly the total budget of the design factor from Theorem 6.18. Writing $A = K + n\lambda I$, the posterior variances at the training inputs sum to

$$\sum_{i=1}^{n} P_\lambda(u_i)^2 = \operatorname{tr}(K) - \operatorname{tr}(KA^{-1}K) = \operatorname{tr}\big(n\lambda\, KA^{-1}\big) = n\lambda\, d_{\text{eff}}(\lambda),$$

using $KA^{-1}K = K - n\lambda\, KA^{-1}$. So the same d_{eff} that reads the nugget off the spectrum also fixes, up to the factor $n\lambda$, the total posterior-variance mass the correction distributes over the design; the two halves of this section are one quantity seen from two sides.

7 Discussion

Two regimes, one pairing. The two problems bracket what a kernel stage after a neural mean can do. On structural mechanics the families tie, and the kernel corrects the residual of a stacked mean; the improvement over the published band decomposes into pieces that ten replicates can size. Accurate neural means trained on the reported metric and averaged over the reflection symmetry do most of the work, from the kernel's 5.197% to the best single network's 4.712%. Combining members adds what their measured residual correlations permit and no more — 0.064 points from equal weighting, 0.021 from weights fitted on validation, 0.032 from letting those weights vary over the grid — and the kernel correction contributes 0.024 ± 0.005 points, bringing with it the residual bound of Theorem 6.18 and the coverage of Proposition 6.19. Each contribution is smaller than the difference between two published architectures, and only the seeds make them separable from noise. The largest single lever is which members the stack contains: removing the spectral member costs 0.039 ± 0.011 points, more than any stage above, although it ranks third of five on the metric. What a member is worth to a stack is set by how its errors covary with the others', not by how small they are, and a stack of near-duplicates cannot be repaired by tuning the combination harder.

On OCO-2 the kernel on the raw state trails the network by an order of magnitude, a stack gains nothing measurable from it, and its value moves inside the network as an exact head on the learned features; the pipeline's job becomes metric management, training each member in a loss that concentrates where the reported metric concentrates and combining per coordinate. In both regimes the same two measurements decide the design before any stack is fit: the pairwise residual correlations, which set the ensembling floor of Corollary 6.11 and, through Corollary 6.12, bracket how close the fitted ensemble comes to it, and the member error ratio, which the second-moment identity turns into a keep-or-drop verdict for each candidate.

Against PARA-Net the five-member pipeline is level, 0.022 points or about one standard error, which we read as a tie; without the spectral member the same pipeline trails by 0.061 points, about three standard errors, so the tie belongs to the configuration that carries that member. The margin over the other published methods survives both error bars of Section 4.4. These are the last few learnable hundredths above what Section 4.6 reads as evidence of a floor in the benchmark data, not a step on the way to zero.

Relation to prior work. Neither component of the pairing is new. Mora et al. (2025) place a neural operator inside the mean of a Gaussian process and fit both jointly; deep kernel learning (Wilson et al., 2016; Calandra et al., 2016) puts a kernel on a network's features and trains through it; and the kernel-flow line of emulation work, of which the OCO-2 baseline of Lamminpää et al. (2025) is the closest instance, has shown that kernels with data-adapted hyperparameters replace physics codes within measurement precision once the input is in its physical parametrization and the target has been reduced to something smooth. The differences here are operational, but they matter at this scale: the mean is fit first and the kernel after, so the expensive stage is ordinary network training; the kernel is tuned by validation rather than by marginal likelihood; the correction is an exact solve at $n = 19000$ rather than a training-time approximation; and the feature-space stage is evaluated against the same kernel on the raw input under the same budget, which is what lets the representation effect be measured (Table 7 and Section 5). The theory of Section 6 applies verbatim to the jointly trained setting; the measured drop in the fitted RKHS norm is the quantitative reason residual corrections of accurate means are the right place to spend kernel capacity. The step this paper does not take is the retrieval-level one. The OCO-2 Level 1 and Level 2 products are publicly distributed through NASA's Earthdata archive, and evaluating how the surrogate's conformal bands and fallback behavior propagate through an actual retrieval, rather than through the emulation metrics alone, is the natural continuation.

Limitations. The benchmark has a one-dimensional input function and a fixed geometry; the exact kernel solve exploits both, and problems with high-dimensional input fields would need the usual approximations (inducing points, random features), which give up the exact solve and with it the residual bound of Theorem 6.18 in the form stated. That bound controls the error relative to the RKHS norm of the residual operator, a quantity we can only lower-bound empirically; the capacity bounds for the stacks and the correction have measured constants but crude ones, and neither they nor the bound should be read as a pointwise error bar — the conformal band is the only object here that plays that role. Training the means on the metric, the reflection steps and the stacking are benchmark-agnostic, but the error levels reported here are properties of these datasets and protocols.

Three runs are incomplete, and their stopping rules are stated here once. The Fourier neural operator was stopped at every seed by the campaign's 36-hour per-task wall clock and is finalized from the best-validation checkpoint written before the stop, between epochs 10 and 70 of a scheduled 200, with its cosine schedule never annealed; nothing was retrained. The UNet member, scheduled after it, completed at two of ten seeds, so the six-member configuration that included it is not among the reported results. The OCO-2 networks are trained at a matched 250-epoch budget that is short of convergence; a single 750-epoch run reaches lower levels on the O2 band without changing the orderings (Section 5.1). The kernel-ridge baseline of Table 4 is a single run. Our low-data protocol reproduces that of Mora et al. (2025) up to the ambiguity of which 1250 samples are used; we use the first 1250 of the training block and the same 20000-sample test set, and release the splits.

The reading of the structural-mechanics plateau as a data floor is an inference from surrogates alone. The experiment that would decide it is to re-solve a set of the hardest cases on a refined mesh or with higher-order elements and compare the solver discrepancy with the shared residual of Section 4.6; we have not run it. Proposition 6.16 is an informal rate calculation rather than a theorem with a matching lower bound, and the OCO-2 sweep supports only its constant half: at a rank ratio between 0.54 and 0.60 it would permit a rate gain from an anisotropic metric, and across a sixteenfold range of training sizes at ten seeds the metric delivers a constant between 1.1 and 1.5, with no rate gain on one band and small ones on the other two (Section 5.1).

Outlook. Three directions seem worth pursuing. First, the benchmark itself: the re-solve described above, or regenerated data with refined meshes at the corner singularities, would settle whether the plateau is a property of the data and, if it is, return the problem to being a test of surrogates. The diagnostic used here — cross-family residual correlation, the second-moment prediction of Proposition 6.8 and the scaling-in-n check — is cheap to run on any benchmark suspected of the same condition. Second, the same recipe on problems where the input functions are genuinely high-dimensional, where the interplay between neural means and kernel corrections should look different and the exact solve will not be available. Third, the uncertainty side: P_λ is a design quantity, so it can be optimized, and the connection of Lemma 6.21 between the nugget, the spectrum and the effective dimension suggests principled ways to spend a fixed computational budget on the correction stage.

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Declaration of competing interest

The author declares no competing interests.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author made substantial use of generative AI tools, and describes their role here. A large part of the software was written with Fable 5 (Anthropic): the implementation of the experiments and diagnostics, the downloading, assembly and preparation of the benchmark and OCO-2 datasets, and the adaptation of the publicly released emulator code of Lamminpää et al. — originally split between Julia and a ReFRACtor-driving Python layer — into a single Python pipeline for sampling states and reducing radiances. The main contributions are the author's: the residual coupling of neural means and exact kernel corrections, the two-regime framing that organizes the paper, and the supporting theory, all developed with AI assistance in calculation, drafting, and exposition. Each theorem and its proof was checked independently and more than once — by Fable 5, by ChatGPT Pro (OpenAI), and by the author — and the reported numbers were verified against the released per-run data. The author reviewed all of the above and takes full responsibility for the content of the manuscript.

Data and code availability

The code, the per-run summaries behind every reported number, and the exact configuration of each experiment are available at `https://github.com/yspennstate/neural-means-kernel-corrections`. The structural-mechanics, Helmholtz, Navier–Stokes and advection data are distributed through the data record accompanying de Hoop et al. (2022) (`data.caltech.edu`, record `20091`); the OCO-2 emulation data and the kernel-flow emulator's predictions through the OSF project `u2t8a`; and the ClimSim data through the LEAP repository `subsampled_low_res`.

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Appendix A. Proofs

A.1 Proof of Proposition 6.1

By convexity of $\ell(\cdot, v)$,

$$\ell\big(\widehat{G}_S(u), G(u)\big) \;\le\; \tfrac{1}{2}\,\ell\big(\widehat{G}(u), G(u)\big) + \tfrac{1}{2}\,\ell\big(T\widehat{G}(Su), G(u)\big).$$

Since T is an involution, the equivariance $G(Su) = TG(u)$ applied at u gives $TG(Su) = T^2G(u) = G(u)$, hence

$$\ell\big(T\widehat{G}(Su), G(u)\big) = \ell\big(T\widehat{G}(Su), TG(Su)\big) = \ell\big(\widehat{G}(Su), G(Su)\big),$$

using the T-invariance of the loss. Taking expectations and using the S-invariance of μ (so that $Su \sim \mu$ when $u \sim \mu$),

$$\mathbb{E}\,\ell\big(\widehat{G}_S(u), G(u)\big) \le \tfrac{1}{2}\,\mathbb{E}\,\ell\big(\widehat{G}(u), G(u)\big) + \tfrac{1}{2}\,\mathbb{E}\,\ell\big(\widehat{G}(Su), G(Su)\big) = \mathbb{E}\,\ell\big(\widehat{G}(u), G(u)\big). \quad \square$$

A.2 Proof of Proposition 6.2

Strict positive definiteness of the restricted kernel matrix makes I_C the (unique) interpolant of the pairs indexed by C, so $I_C(z_j) = v_j$ for $j \in C$ and the terms of (1) indexed by C vanish, which is the first claim. For the second, write

$$e_2(\theta; C) \;=\; \sum_{i \in B\setminus C} g(C, i), \qquad g(C, i) := \|v_i - I_C(z_i)\|_2^2.$$

Conditionally on C, the sum has exactly $b/2$ terms, so $e_2(\theta; C) = \frac{b}{2}\,\mathbb{E}_{i\sim\mathrm{Unif}(B\setminus C)}[g(C, i)]$, and taking the expectation over C proves the identity. The gradient claim is immediate: for fixed (B, C) the map $\theta \mapsto e_2(\theta; C)$ is differentiable wherever the Cholesky factorization in I_C is (the kernel matrix stays positive definite in a neighborhood), and under a local integrable Lipschitz bound the derivative passes under the expectation, so $\mathbb{E}_{B,C}[\nabla_\theta e_2] = \nabla_\theta\,\mathbb{E}_{B,C}[e_2]$. $\square$

A.3 Proof of Proposition 6.3

(i) Since $\sum_m w_m = 1$, $F_w(u) - G(u) = \sum_m w_m\,(f_m(u) - G(u))$, and the triangle inequality gives $\|F_w(u) - G(u)\|_2 \le \sum_m w_m \|f_m(u) - G(u)\|_2$. Dividing by $\|G(u)\|_2$ yields the pointwise claim; taking expectations gives the risk inequality, and bounding each $\ell(f_m(u), G(u))$ by B gives $\ell(F_w(u), G(u)) \le B$.

(ii) Write $\ell_u(w) = \ell(F_w(u), G(u))$. First, ℓ_u is Lipschitz on Δ for the ℓ^1 norm: for $w, w' \in \Delta$, the reverse triangle inequality and the argument of (i) give

$$|\ell_u(w) - \ell_u(w')| \;\le\; \frac{\big\|\sum_m (w_m - w'_m)(f_m(u) - G(u))\big\|_2}{\|G(u)\|_2} \;\le\; B\,\|w - w'\|_1.$$

Next, discretize the simplex. For $k \in \mathbb{N}$ let $\mathcal{G}_k = \{w \in \Delta : kw \in \mathbb{Z}^M\}$; its cardinality is the number of compositions of k into M nonnegative parts, $\binom{k+M-1}{M-1} \le (k+1)^{M-1}$. Given $w \in \Delta$, set $w'_i = \lfloor kw_i \rfloor / k$ for $i < M$ and $w'_M = 1 - \sum_{i<M} w'_i$; then $w' \in \mathcal{G}_k$ (the last coordinate is a multiple of $1/k$ and is $\ge w_M \ge 0$ because the first $M-1$ coordinates only decreased), and

$$\|w - w'\|_1 = \sum_{i<M} (w_i - w'_i) + \big(w'_M - w_M\big) = 2 \sum_{i<M} (w_i - w'_i) \;\le\; \frac{2(M-1)}{k}.$$

By (i) the per-sample loss lies in $[0, B]$, so for each fixed w' Hoeffding's inequality gives $\mathbb{P}\big(|\widehat{R}(w') - R(w')| > t\big) \le 2e^{-2mt^2/B^2}$, and a union bound over $\mathcal{G}_k$ shows that with probability at least $1 - \delta$,

$$\sup_{w' \in \mathcal{G}_k} \big|\widehat{R}(w') - R(w')\big| \;\le\; B\sqrt{\frac{\log\big(2|\mathcal{G}_k|/\delta\big)}{2m}}.$$

On this event, for every $w \in \Delta$, approximating by its grid point and using the Lipschitz property for both R and $\widehat{R}$,

$$\big|\widehat{R}(w) - R(w)\big| \;\le\; B\sqrt{\frac{\log(2|\mathcal{G}_k|/\delta)}{2m}} + \frac{4B(M-1)}{k}.$$

If $w^\star$ minimizes R over Δ, the empirical minimizer satisfies $R(F_{\widehat{w}}) \le \widehat{R}(\widehat{w}) + \sup_\Delta |\widehat{R} - R| \le \widehat{R}(w^\star) + \sup_\Delta |\widehat{R} - R| \le R(F_{w^\star}) + 2\sup_\Delta |\widehat{R} - R|$. Choosing $k = m(M-1)$ and using $|\mathcal{G}_k| \le (m(M-1)+1)^{M-1}$ gives

$$R(F_{\widehat{w}}) \;\le\; \min_{w \in \Delta} R(F_w) + \frac{8B}{m} + B\sqrt{\frac{2\big((M-1)\log(m(M-1)+1) + \log(2/\delta)\big)}{m}},$$

which is the claim. $\square$

A.4 Proof of Lemma 6.4

For each k the validation scores $\ell(g_k(\tilde{u}_i), G(\tilde{u}_i))$, $i \le m'$, are i.i.d., take values in $[0, B]$, and have mean $R(g_k)$; the maps g_k are fixed relative to this sample, which is the only place the independence assumption enters. Hoeffding's inequality gives, for each k,

$$\Pr\Big(\big|\widehat{R}(g_k) - R(g_k)\big| \ge t\Big) \;\le\; 2\exp\big(-2m't^2/B^2\big),$$

so with $t = B\sqrt{\log(2K/\delta)/(2m')}$ and a union bound over $k \le K$, all K deviations are below t simultaneously with probability at least $1 - \delta$. On that event, if k^* attains $\min_k R(g_k)$,

$$R(g_{\widehat{k}}) \;\le\; \widehat{R}(g_{\widehat{k}}) + t \;\le\; \widehat{R}(g_{k^*}) + t \;\le\; R(g_{k^*}) + 2t,$$

and $2t = B\sqrt{2\log(2K/\delta)/m'}$. $\square$

A.5 Proof of Proposition 6.5

Fix a coordinate j and abbreviate $\varphi = \varphi_j$, $Y(u) = G(u)_j$, $D = \sqrt{M+1}$. Each entry of $\varphi(u)$ is bounded by b in absolute value (the members by assumption, the intercept entry by construction), so $\|\varphi(u)\|_2 \le bD$, and for $\|w\|_2 \le W$ Cauchy–Schwarz gives $|w^\top \varphi(u)| \le WbD$ and $|w^\top\varphi(u) - Y(u)| \le b(1+WD)$. The losses $\ell_w(u) = (w^\top\varphi(u) - Y(u))^2$ therefore take values in $[0, L_\infty]$ with $L_\infty = b^2(1+WD)^2$.

Uniform deviation. Consider the one-sided suprema $Z^+ = \sup_{\|w\|_2\le W} \big(\widehat{R}_j(w) - R_j(w)\big)$ and $Z^- = \sup_{\|w\|_2\le W} \big(R_j(w) - \widehat{R}_j(w)\big)$ over the validation sample of size m. Changing one sample moves either supremum by at most L_∞/m, so McDiarmid's inequality gives $Z^\pm \le \mathbb{E}Z^\pm + L_\infty\sqrt{\log(4q/\delta)/(2m)}$, each with probability at least $1-\delta/(2q)$. By symmetrization, $\mathbb{E}Z^\pm \le 2\,\mathfrak{R}_m(\mathcal{L})$, the Rademacher complexity of the loss class. The loss is the composition of $t \mapsto t^2$, restricted to $|t| \le b(1+WD)$ where it is $2b(1+WD)$-Lipschitz, with the class $\{u \mapsto w^\top\varphi(u) - Y(u)\}$; translation by the fixed function Y leaves the Rademacher average unchanged (the translated term does not involve w and has zero mean), and for the linear class over the ℓ^2 ball,

$$\mathfrak{R}_m \;\le\; \frac{W}{m}\,\mathbb{E}\Big\|\sum_{i=1}^m \sigma_i\varphi(\tilde u_i)\Big\|_2 \;\le\; \frac{W}{m}\Big(\sum_{i=1}^m \mathbb{E}\|\varphi(\tilde u_i)\|_2^2\Big)^{1/2} \;\le\; \frac{WbD}{\sqrt{m}},$$

by Jensen's inequality. Talagrand's contraction lemma then gives $\mathfrak{R}_m(\mathcal{L}) \le 2b(1+WD)\cdot WbD/\sqrt{m}$, hence

$$Z^\pm \;\le\; \frac{4Wb^2D(1+WD)}{\sqrt{m}} + L_\infty\sqrt{\frac{\log(4q/\delta)}{2m}} \qquad \text{jointly with probability} \ge 1-\delta/q.$$

Empirical risk minimization. On this event, for any minimizer w^* of R_j over the ball, $R_j(\widehat{w}_j) \le \widehat{R}_j(\widehat{w}_j) + Z^- \le \widehat{R}_j(w^*) + Z^- \le R_j(w^*) + Z^+ + Z^-$, and $Z^+ + Z^-$ is bounded by the displayed excess. A union bound over the two tails of each of the q coordinates makes all events hold simultaneously with probability at least $1-\delta$, and averaging over j preserves the (coordinate-uniform) bound. This proves (i).

For (ii), the ridge minimizer satisfies $\widehat{R}_j(\widehat{w}_j) + \lambda\|\widehat{w}_j\|_2^2 \le \widehat{R}_j(w^*) + \lambda\|w^*\|_2^2$, so $\widehat{R}_j(\widehat{w}_j) \le \widehat{R}_j(w^*) + \lambda\|w^*\|_2^2 \le \widehat{R}_j(w^*) + \lambda W^2$. Since by assumption $\|\widehat{w}_j\|_2 \le W$, the uniform event of part (i) applies to $\widehat{w}_j$ and w^* alike, and the chain above goes through with the extra λW^2. $\square$

A.6 Proof of Corollary 6.12

The lower bound is Corollary 6.11. For the upper bound, evaluate the quadratic form at the uniform weights $w_m = 1/M$:

$$\frac{1}{\bar e^{\,2}}\, w^\top S w = \frac{1}{\bar e^{\,2}}\Big(\frac{1}{M^2}\sum_m S_{mm} + \frac{1}{M^2}\sum_{m\ne k} S_{mk}\Big) \;\le\; \frac{1+\delta_e}{M} + \frac{M-1}{M}\,\big(\bar\varrho + \delta_\varrho\big),$$

using the entrywise upper bounds and the counts M and $M(M-1)$ of the two sums. The right side equals $\bar\varrho + \delta_\varrho + \big(1 + \delta_e - \bar\varrho - \delta_\varrho\big)/M$, and the minimum over the simplex is at most the value at any feasible point. $\square$

A.7 Proof of Corollary 6.6

Fix a coordinate j and a grid point γ. Substituting the affine form of the correction,

$$h_{w,\gamma}(u)_j = w_j^\top \widetilde{\varphi}_{j,\gamma}(u) + o_{j,\gamma}(u), \qquad \widetilde{\varphi}_{j,\gamma}(u) = \varphi_j(u) - \Phi_j^\top c_\gamma(u), \quad o_{j,\gamma}(u) = c_\gamma(u)^\top Y_j,$$

and neither $\widetilde{\varphi}_{j,\gamma}$ nor $o_{j,\gamma}$ depends on w or on the validation sample: c_γ is built from the training design alone. Each component of $\Phi_j^\top c_\gamma(u)$ is a c_γ-weighted sum of member values bounded by b, so its absolute value is at most $b\,\|c_\gamma(u)\|_1 \le b\kappa$, whence $\|\widetilde{\varphi}_{j,\gamma}(u)\|_2 \le bD + bD\kappa = bD(1+\kappa)$ and $|o_{j,\gamma}(u)| \le b\kappa$. For $\|w\|_2 \le W$ the prediction deviates from the target by at most $WbD(1+\kappa) + b\kappa + b \le b(1+WD)(1+\kappa) =: \tilde{c}$, so the squared losses lie in $[0, \tilde{c}^{\,2}]$ and are $2\tilde{c}$-Lipschitz in the prediction.

The argument of Proposition 6.5 now applies verbatim to the class indexed by w in the W-ball, for this fixed (j,γ): the Rademacher complexity of the affine part is at most $W\,bD(1+\kappa)/\sqrt{m}$, contraction multiplies it by $2\tilde{c}$, symmetrization doubles it, and McDiarmid at level $\delta/(2q|\Gamma|)$ adds $\tilde{c}^{\,2}\sqrt{\log(4q|\Gamma|/\delta)/(2m)}$. A union bound over the two tails of the $q|\Gamma|$ pairs makes, with probability at least $1-\delta$, each one-sided deviation $Z^{\pm}_{j,\gamma}$, defined as in the proof of Proposition 6.5, at most

$$\frac{4Wb^2D(1+WD)(1+\kappa)^2}{\sqrt{m}} + b^2(1+WD)^2(1+\kappa)^2\sqrt{\frac{\log(4q|\Gamma|/\delta)}{2m}}.$$

On that event, write $\bar{R}(\gamma) = \frac{1}{q}\sum_j R_j(h_{\widehat{w},\gamma})$ and $\widehat{\bar{R}}(\gamma)$ for its empirical version; averaging the $Z_{j,\gamma}$ over j bounds $|\widehat{\bar{R}}(\gamma) - \bar{R}(\gamma)|$ by the same quantity for every γ simultaneously, including at the data-dependent $\widehat{w}$, because the deviations are uniform over the ball. The selection chain $\bar{R}(\widehat{\gamma}) \le \widehat{\bar{R}}(\widehat{\gamma}) + Z \le \widehat{\bar{R}}(\gamma^*) + Z \le \bar{R}(\gamma^*) + 2Z$ with γ^* the aggregate oracle then gives the display, whose constant is $2Z$ in the factored form stated. When the c_γ are kernel ridge weights, Lemma 6.7 supplies the hypothesis with $\kappa = \kappa_{\mathrm{an}}$. $\square$

A.8 Proof of Lemma 6.7

Fix u and write $k_u = k(X,u)$, $A = K + n\lambda I$ and $c = A^{-1}k_u$. The proof of Theorem 6.18 below shows that $\rho(u)^2 = k(u,u) - k_u^\top A^{-1}k_u - n\lambda\|c\|_2^2 = P_\lambda(u)^2 - n\lambda\|c\|_2^2$ is the squared $\mathcal{H}_k$-norm of $k(u,\cdot) - \sum_i c_i k(u_i,\cdot)$, hence nonnegative, so $n\lambda\|c\|_2^2 \le P_\lambda(u)^2 \le k(u,u) \le k_{\max}$, the middle inequality because $k_u^\top A^{-1}k_u \ge 0$. Cauchy–Schwarz gives $\|c\|_1 \le \sqrt{n}\,\|c\|_2$, and combining, $\|c\|_1^2 \le n\|c\|_2^2 \le P_\lambda(u)^2/\lambda \le k_{\max}/\lambda$. For the Matérn kernel of Section 3.3, $k(u,u) = 1$ and the smallest nugget on the grid is 10^{-6}, so $\sup_{u,\gamma}\|c_\gamma(u)\|_1 \le \max_\gamma \lambda_\gamma^{-1/2} = 10^3$ irrespective of the length scale and of the design. $\square$

A.9 Proof of Proposition 6.8

Since $\sum_m w_m = 1$, the stack's normalized residual is $\rho_{f_w}(u) = \sum_m w_m \rho_{f_m}(u)$, hence

$$\mathbb{E}\,\ell\big(f_w(u), G(u)\big)^2 = \mathbb{E}\Big\|\sum_m w_m \rho_{f_m}(u)\Big\|_2^2 = \sum_{m,k} w_m w_k\,\mathbb{E}\,\langle \rho_{f_m}(u), \rho_{f_k}(u)\rangle = w^\top S w.$$

For (i), parameterize $w = (1-t, t)$; $q(t) = w^\top S w$ is a convex quadratic in t with $q'(0) = 2(\varrho e_1 e_2 - e_1^2)$, negative precisely when $\varrho < e_1/e_2$. In that case the unconstrained minimizer lies in $(0,1)$ and gives the stated interior value; otherwise $q'(0) \geq 0$, the minimum over $[0,1]$ is at $t=0$ with value e_1^2, and the second member is dropped. For (ii), $S = e^2\big((1-\varrho)I + \varrho\,\mathbf{1}\mathbf{1}^\top\big)$, so $w^\top S w = e^2\big((1-\varrho)\|w\|_2^2 + \varrho\big)$ on the simplex, minimized by the uniform weights, where $\|w\|_2^2 = 1/M$. $\square$

A.10 Proof of Corollary 6.11

Convex weights are nonnegative, so each term of $w^\top S w = \sum_m w_m^2 S_{mm} + \sum_{m\neq k} w_m w_k S_{mk}$ can be bounded below entrywise:

$$w^\top S w \;\geq\; \bar{e}^{\,2} \sum_m w_m^2 + \bar{\varrho}\,\bar{e}^{\,2} \sum_{m\neq k} w_m w_k = \bar{e}^{\,2}\Big(\|w\|_2^2 + \bar{\varrho}\,(1 - \|w\|_2^2)\Big),$$

using $\sum_{m\neq k} w_m w_k = (\sum_m w_m)^2 - \|w\|_2^2 = 1 - \|w\|_2^2$. The bracket is a convex combination of 1 and $\bar{\varrho}$, hence at least $\bar{\varrho}$. $\square$

A.11 Proof of Proposition 6.13

The metric is diagonal, so the squared norm splits over coordinates and the j-th summand $w_j^2\,\mathbb{E}\big(\sum_m v_{mj} f_m(u)_j - G(u)_j\big)^2$ depends on the weights only through the column $v_{\cdot j}$. Minimizing the sum is therefore q independent minimizations, and the positive factor w_j^2 does not move any of the q argmins, so the optimizer is the same for every positive w. A combination with shared weights is the special case $v_{mj} = v_m$ for all j, a subset of the feasible set of each coordinate problem, which gives the domination. $\square$

A.12 Proof of Corollary 6.14

The integrand is nonnegative, so Tonelli's theorem permits exchanging the expectation with the coordinate sum:

$$\mathbb{E}\Big[a(u)\sum_j w_j^2\big(f_v(u)_j - G(u)_j\big)^2\Big] = \sum_j w_j^2\,\mathbb{E}\Big[a(u)\big(f_v(u)_j - G(u)_j\big)^2\Big].$$

The j-th summand depends on v only through its j-th column $v_{\cdot j}$, exactly as in the proof of Proposition 6.13, so minimizing over v splits into q independent problems; the factors w_j^2 rescale the summands without moving any per-coordinate minimizer, which gives the simultaneous optimality over all diagonal w. The choice $a(u) = \|G(u)\|_2^{-2}$, $w = \mathbf{1}$ identifies the objective with $\mathbb{E}\,\|\rho_{f_v}\|_2^2$, the mean squared relative error. Finally $\mathbb{E}\|\rho\|_2 \leq (\mathbb{E}\|\rho\|_2^2)^{1/2}$ is Jensen's inequality. $\square$

A.13 Proof of Proposition 6.15

Fix a coordinate j and a member m. The validation scores $a(\tilde{u}_i)(f_m(\tilde{u}_i)_j - G(\tilde{u}_i)_j)^2$, $i \leq m_v$, are i.i.d., take values in $[0,L]$ by assumption, and have mean $R_j(m)$; the members and the weight are fixed relative to the validation sample. Hoeffding's inequality gives

$$\Pr\Big(\big|\widehat{R}_j(m) - R_j(m)\big| \geq t\Big) \;\leq\; 2\exp\big(-2m_v t^2/L^2\big)$$

for each of the Mq pairs (m, j). With $t = L\sqrt{\log(2Mq/\delta)/(2m_v)}$ and a union bound, all Mq deviations are below t simultaneously with probability at least $1-\delta$. On that event, for every j, writing $m^*(j)$ for a population minimizer,

$$R_j\big(\widehat{m}(j)\big) \le \widehat{R}_j\big(\widehat{m}(j)\big) + t \le \widehat{R}_j\big(m^*(j)\big) + t \le R_j\big(m^*(j)\big) + 2t.$$

The metric statement follows by multiplying the coordinate-wise inequality by $w_j^2 \ge 0$ and summing; the selection $\widehat{m}(\cdot)$ does not depend on w, so one event serves every diagonal metric. $\square$

A.14 Proof of Proposition 6.17

Let $T : \mathcal{H}_k \to \mathbb{R}^{\mathcal{X}}$ be the composition map $Tf = f \circ \varphi$. For $x \in \mathcal{X}$ the reproducing property gives $(Tf)(x) = f(\varphi(x)) = \langle f, k(\cdot, \varphi(x))\rangle_{\mathcal{H}_k}$, so evaluation of Tf at x is a bounded functional, and the image $\mathcal{H}_{k_\varphi} := T(\mathcal{H}_k)$ carries the quotient norm $\|g\|_{\mathcal{H}_{k_\varphi}} = \min\{\|f\|_{\mathcal{H}_k} : Tf = g\}$, the minimum over the affine subspace $T^{-1}(g)$ (nonempty exactly when g is a pullback). This normed space is an RKHS: its reproducing kernel is $k_\varphi(x, x') = \langle k(\cdot, \varphi(x)), k(\cdot, \varphi(x'))\rangle_{\mathcal{H}_k} = k(\varphi(x), \varphi(x'))$, since the minimum-norm representer of evaluation at x is $k(\cdot, \varphi(x))$. Taking $f = h$ in the minimum gives $\|h \circ \varphi\|_{\mathcal{H}_{k_\varphi}} \le \|h\|_{\mathcal{H}_k}$, and substituting this bound into Theorem 6.18 applied with the kernel k_φ replaces $\|G\|$ by $\|h\|_{\mathcal{H}_k}$. $\square$

A.15 Heuristic derivation of Proposition 6.16

The proposition is labelled informal, and this appendix records the calculation behind it at the level of the idealization stated with it — projected image marginal bounded above and below, exact rank at the working resolution, and the interpolation limit standing in for the validated nugget — as a sketch, without tracking constants; no lower bound is claimed. The argument has five steps: a pointwise error bound whose design dependence is the power function; a fill-distance bound for random designs; an upper bound on the warped target's native norm; the two substitutions; and the reduction from the validated nugget to the interpolation scale. Throughout, k is a Matérn kernel of smoothness s whose native space on a bounded Lipschitz domain $\Omega \subset \mathbb{R}^{d'}$ is norm-equivalent to $H^{s+d'/2}(\Omega)$ (Wendland, 2004, Ch. 10); we follow the convention of the statement and write s for the resulting approximation order.

Step 1: pointwise bound through the power function. For any target F with components in the native space and the interpolant $\widehat{F}_0$ on the design X, the argument that proves Theorem 6.18 (with $\lambda = 0$ and $m = 0$) gives, for every u,

$$\|F(u) - \widehat{F}_0(u)\|_2 \;\le\; \|F\|_K \, P_0(u),$$

and the classical zeros lemma for functions with many zeros on a dense set converts the power function into the fill distance $h_X = \sup_{u\in\Omega} \min_i \|u - u_i\|$: $\sup_u P_0(u) \le C\, h_X^s$ for h_X below a domain constant (Wendland, 2004, Ch. 11). Hence $\sup_u \|F(u) - \widehat{F}_0(u)\|_2 \le C\, h_X^s \,\|F\|_K$.

Step 2: fill distance of an i.i.d. design. Let $u_1, \ldots, u_n$ be i.i.d. from a density bounded below by $p_- > 0$ on Ω. Fix $\delta > 0$ and a $\delta/2$-net of Ω of cardinality $N(\delta) \le C_\Omega\, \delta^{-d'}$. A net ball of radius $\delta/2$ has μ-mass at least $c\,p_-\delta^{d'}$, so the probability that some net ball receives no sample is at most $C_\Omega\, \delta^{-d'}(1 - c\,p_-\delta^{d'})^n \le C_\Omega\, \delta^{-d'} e^{-c\,p_-\, n\delta^{d'}}$. Choosing

$\delta^{d'} = (2/(c\,p_-))\,(\log n)/n$ makes this at most $C'\,n^{-1}$, and on the complement every point of Ω lies within δ of a sample: with probability at least $1 - C'/n$,

$$h_X \;\le\; C''\left(\frac{\log n}{n}\right)^{1/d'}.$$

This is the high-probability statement in the proposition; the logarithm is absorbed into the logarithmic factors there.

Step 3: the warped target's norm, from above. Let $F = g(A\,\cdot)$ with $\|g\|_{H^s} = 1$ (native norm of the isotropic kernel in the image variables). For integer m, the chain rule gives, for any multi-index α with $|\alpha| = m$, $D^\alpha\{g(Au)\} = \sum_{|\beta|=m} c_{\alpha\beta}(A)\,(D^\beta g)(Au)$ with $\sum_\beta |c_{\alpha\beta}(A)| \le \|A\|^m = \sigma_1^m$, so by the change of variables $y = Au$ (Jacobian $|\det A|$),

$$\|D^\alpha\{g(A\cdot)\}\|_{L^2(\mathcal{U})} \;\le\; \sigma_1^m\,|\det A|^{-1/2}\,\|g\|_{H^m(A\mathcal{U})}.$$

Summing over $|\alpha| \le m$ bounds $\|g(A\cdot)\|_{H^m}$ by $C\,\sigma_1^m|\det A|^{-1/2}\|g\|_{H^m}$ for integer m, and the fractional case follows by interpolation between consecutive integers (Adams and Fournier, 2003, Ch. 7). Only this upper bound is used; the $|\det A|^{-1/2}$ and the domain constants are among the singular-value constants absorbed into the statement, leaving the displayed σ_1^s.

Step 4: the two designs. For the isotropic kernel on the raw input the domain has dimension $d' = d$, so Steps 1–3 give the first display: $C\,h_X^s\,\sigma_1^s \lesssim \sigma_1^s n^{-s/d}$ up to logarithms. For the adapted kernel $k_A(u,u') = k(Au, Au')$, interpolating F at $\{u_i\}$ is interpolating g at the image design $\{Au_i\}$, and $\|F\|_{K_A} = \|g\|_K \le 1$ by the pullback identity (Proposition 6.17). The image measure has a density bounded above and below on its support by the idealization of the statement. Split $A = A_r + A_\perp$ along the leading-r right singular subspace. The projected points $\Pi_r A u_i$ are i.i.d. on an r-dimensional bounded set, so Step 2 (with $d' = r$) puts every projected image point within $C(\log n/n)^{1/r}$ of a projected sample; and for every u in the domain, $\|Au - \Pi_r Au\| \le \sigma_{r+1}\operatorname{diam}(\mathcal{U})$. The triangle inequality then bounds the image fill distance by $C(\log n/n)^{1/r} + 2\sigma_{r+1}\operatorname{diam}(\mathcal{U})$, and the approximate-rank hypothesis $\sigma_{r+1} \lesssim n^{-1/r}$ makes the second term of the same order as the first. Steps 1 and 3 with $d' = r$ and $\|g\|_K \le 1$ give the second display. If instead σ_{r+1} is not below $n^{-1/r}$, the trailing directions are resolved by the design, the image fill distance is that of a d-dimensional set, and the two rates coincide; only the constant σ_1^s separates the bounds, as claimed.

Step 5: from the interpolant to the validated ridge estimator. The pipeline fits with a nugget from a grid containing values down to $10^{-8}n$ and selects on validation. For any fixed λ the pathwise bound of Theorem 6.18 controls the ridge estimator by $\|F\|_K \sup_u P_\lambda(u)$, and a two-line spectral computation compares the regularized power function with the interpolation one: with $k_u = k(X,u)$ and $A_\lambda = K + n\lambda I$,

$$P_\lambda(u)^2 - P_0(u)^2 = k_u^\top\big(K^{-1} - A_\lambda^{-1}\big)k_u = n\lambda\,k_u^\top K^{-1}A_\lambda^{-1}k_u \;\le\; \frac{n\lambda\,\|k_u\|_2^2}{\lambda_{\min}(K)\,\big(\lambda_{\min}(K) + n\lambda\big)},$$

which vanishes as $\lambda \to 0$. Validation selection over the grid can only improve on its smallest-nugget arm up to the selection cost priced by Lemma 6.4, which is of lower order than the rates displayed. Both displays are upper bounds under the stated idealization, and no matching lower bound is claimed. □

A.16 Proof of Theorem 6.18

Fix u and write $k_u = k(X,u) \in \mathbb{R}^n$, $A = K + n\lambda I$, $w = A^{-1}k_u$. For each component j, membership $r_j \in \mathcal{H}_k$ and the reproducing property give

$$r_j(u) - \widehat{r}_{\lambda,j}(u) = \Big\langle r_j,\ k(u,\cdot) - \sum_{i=1}^{n} w_i\, k(u_i,\cdot) \Big\rangle_{\mathcal{H}_k},$$

because $\widehat{r}_{\lambda,j}(u) = k_u^\top A^{-1} r_j(X) = \sum_i w_i\, r_j(u_i)$ and $r_j(u_i) = \langle r_j, k(u_i,\cdot)\rangle$. Cauchy–Schwarz yields

$$\big|r_j(u) - \widehat{r}_{\lambda,j}(u)\big| \ \le\ \|r_j\|_{\mathcal{H}_k} \Big\|k(u,\cdot) - \sum_i w_i k(u_i,\cdot)\Big\|_{\mathcal{H}_k},$$

and the second factor is independent of j; call it $\rho(u)$. Expanding,

$$\rho(u)^2 = k(u,u) - 2\, w^\top k_u + w^\top K w = k(u,u) - k_u^\top A^{-1}\big(2A - K\big)A^{-1}k_u.$$

Since $2A - K = A + n\lambda I$,

$$\rho(u)^2 = k(u,u) - k_u^\top A^{-1} k_u - n\lambda\, \|A^{-1}k_u\|_2^2 = \widetilde{P}_\lambda(u)^2 \ \le\ P_\lambda(u)^2.$$

Summing the squared componentwise bounds,

$$\big\|r(u) - \widehat{r}_\lambda(u)\big\|_2^2 = \sum_j \big(r_j(u) - \widehat{r}_{\lambda,j}(u)\big)^2 \le \rho(u)^2 \sum_j \|r_j\|_{\mathcal{H}_k}^2 = \widetilde{P}_\lambda(u)^2\, \|r\|_K^2. \quad \square$$

Note that the argument does not use interpolation: it holds for every $\lambda \ge 0$, with the (slightly sharper) factor $\widetilde{P}_\lambda$ showing that smoothing can only tighten this particular bound relative to the posterior standard deviation P_λ that we report.

A.17 Proof of Proposition 6.19

Write $s_i = \|e(\tilde{u}_i)\|_2 / P_\lambda(\tilde{u}_i)$ for the validation scores and $s^\star = \|e(u^\star)\|_2 / P_\lambda(u^\star)$ for the test score. Because m, $\widehat{r}_\lambda$, X and P_λ are fixed and the inputs $\tilde{u}_1, \dots, \tilde{u}_m, u^\star$ are exchangeable and independent of them, the scores $s_1, \dots, s_m, s^\star$ are exchangeable. The event $G(u^\star) \in C(u^\star)$ is exactly $\{s^\star \le q\}$, where $q = s_{(\lceil (1-\alpha)(m+1)\rceil)}$ is the $k := \lceil (1-\alpha)(m+1)\rceil$-th order statistic of the validation scores.

Consider the augmented sample $s_1, \dots, s_m, s^\star$ of size $m+1$ and let $\mathrm{rank}(s^\star)$ be its rank (ties broken uniformly at random, which only helps). By exchangeability the rank is uniform on $\{1, \dots, m+1\}$, so $\mathbb{P}(\mathrm{rank}(s^\star) \le k) = k/(m+1)$. If $s^\star$ is among the k smallest of the $m+1$ values then at most $k-1$ of the s_i are below it, so $s^\star \le s_{(k)} = q$; conversely $s^\star \le q$ forces $\mathrm{rank}(s^\star) \le k$ except possibly through ties. Hence

$$\mathbb{P}(s^\star \le q) \ \ge\ \mathbb{P}(\mathrm{rank}(s^\star) \le k) = \frac{k}{m+1} = \frac{\lceil (1-\alpha)(m+1)\rceil}{m+1} \ \ge\ 1 - \alpha,$$

which is the lower bound. When the scores are almost surely distinct there are no ties, $\{s^\star \le q\} = \{\mathrm{rank}(s^\star) \le k\}$ exactly, and $k/(m+1) < ((1-\alpha)(m+1)+1)/(m+1) = 1 - \alpha + 1/(m+1)$, giving the upper bound. $\square$

A.18 Proof of Corollary 6.22

The nonzero eigenvalues of $K/n = XX^\top/n$ coincide with those of the sample covariance matrix $S = X^\top X/n \in \mathbb{R}^{p\times p}$, and zero eigenvalues contribute nothing to $d_{\text{eff}}(\lambda) = \sum_i \lambda_i(K)/(\lambda_i(K)+n\lambda)$, so

$$\frac{d_{\text{eff}}(\lambda)}{n} = \frac{p}{n}\cdot\frac{1}{p}\sum_{j=1}^{p}\frac{\lambda_j(S)}{\lambda_j(S)+\lambda} = \frac{p}{n}\Big(1-\lambda\,\widehat{m}_S(-\lambda)\Big),$$

by the computation of Lemma 6.21 applied to S, with $\widehat{m}_S$ the Stieltjes transform of the empirical spectral distribution of S. By the Marčenko–Pastur theorem (Marčenko and Pastur, 1967), this distribution converges weakly, almost surely, to the law with ratio γ and scale σ^2, and since $x \mapsto (x+\lambda)^{-1}$ is bounded and continuous on $[0,\infty)$, $\widehat{m}_S(-\lambda) \to m(-\lambda)$. It remains to evaluate $m(-\lambda)$. The Stieltjes transform of the Marčenko–Pastur law satisfies the self-consistent equation

$$m(z) = \frac{1}{\sigma^2\big(1-\gamma-\gamma z\,m(z)\big)-z},$$

i.e. $\gamma\sigma^2 z\,m^2 + \big(z-\sigma^2(1-\gamma)\big)m + 1 = 0$. At $z=-\lambda$ the discriminant is $\big(\lambda+\sigma^2(1-\gamma)\big)^2 + 4\gamma\sigma^2\lambda > 0$, and of the two real roots

$$m(-\lambda) = \frac{\sigma^2(1-\gamma)+\lambda \pm \sqrt{\big(\lambda+\sigma^2(1-\gamma)\big)^2+4\gamma\sigma^2\lambda}}{-2\gamma\sigma^2\lambda}$$

only the one with the minus sign in the numerator is positive, as $m(-\lambda) = \int (x+\lambda)^{-1}dF(x) > 0$ requires; rearranging gives the stated expression. □

A.19 Proof of Lemma 6.21

Diagonalize K with eigenvalues λ_i. Then

$$\operatorname{tr}\big(K(K+n\lambda I)^{-1}\big) = \sum_{i=1}^{n}\frac{\lambda_i}{\lambda_i+n\lambda} = \sum_{i=1}^{n}\Big(1-\frac{n\lambda}{\lambda_i+n\lambda}\Big) = n-\lambda\sum_{i=1}^{n}\frac{1}{\lambda_i/n+\lambda} = n\big(1-\lambda\,\widehat{m}(-\lambda)\big),$$

using $\widehat{m}(-\lambda) = \frac{1}{n}\sum_i(\lambda_i/n+\lambda)^{-1}$. The limit statement follows from the weak convergence of the empirical spectral distribution together with the boundedness and continuity of $x \mapsto (x+\lambda)^{-1}$ on $[0,\infty)$ for fixed $\lambda > 0$. □

Appendix B. Implementation details

The pipeline was developed on a single laptop (one NVIDIA RTX PRO 2000, 8 GB; 16 CPU cores, 64 GB RAM); the structural-mechanics numbers were then recomputed from scratch on CPU-only cloud instances under the seed protocol below, in single precision for network training and double precision for all kernel solves and error computations, and it is those recomputed values that the tables report. One exception is stated where it occurs: the six-member configuration of Table 6 completed at two campaign seeds and is reported from those. The run records carry no device field, but each network trainer prints the device it selected when it starts, and the ten structural-mechanics pipeline logs hold that line for every launch, restarts included: 54 launches, all `cpu`. The kernel stages are NumPy and SciPy and run on the CPU by construction. The dataset is the distributed `StructuralMechanics` file pair (40000 samples); inputs are reduced to $\mathbb{R}^{41}$ after verifying the broadcast structure exactly. Splits: the training block is samples 1–20000, the test set is samples 20001–40000. High-data protocol: a fixed permutation of the training block reserves 1000 samples for validation, 19000 for fitting. Low-data protocol: the first 1250 samples of the training block, of which the last 250 form the validation split. The test set is never touched during development; each configuration is evaluated on it once, after selection on validation.

FNO. Width 64, 14 modes per dimension, 4 spectral layers with pointwise linear skips and GELU, lift from 3 channels (broadcast load, two coordinates), projection $64 \to 128 \to 1$. AdamW, learning rate 1.5×10^{-3} (2×10^{-3} with batch 256), weight decay 10^{-6}, cosine schedule to 10^{-6}, 200 epochs, batch 256. 6.45M parameters.

Transformer (implemented, not in the reported ensemble). Encoder: 41 tokens (load value and 10 sine/cosine pairs of the coordinate), 5 pre-norm blocks, dimension 192, 4 heads, MLP ratio 4. Decoder: grid queries from Fourier features of both coordinates, two cross-attention blocks, pointwise head $192 \to 192 \to 1$; 3.16M parameters. The cross-attention over 1681 grid queries is the most compute-intensive of our means, and the hardware available for this study (a single laptop GPU that proved unstable under sustained load) did not let us train it to the level of the other members; we therefore exclude it from the reported ensemble and note it as the natural fourth architecture family for a follow-up on stable hardware.

UNet. Three convolutional scales ($41 \to 21 \to 11$ by average pooling, ceil mode) and a bottleneck at 6×6; two 3×3 convolutions with GroupNorm(8) and SiLU per block; widths 48/96/192/384; bilinear upsampling with skip concatenation; 1×1 output head. Input channels as for the FNO. AdamW, learning rate 1.5×10^{-3}, weight decay 10^{-5}, cosine schedule, 200 epochs, batch 256. 4.38M parameters.

MSE variant. The residual MLP trained with mean squared error on standardized targets in place of the metric loss, otherwise identical recipe but for the schedule: the campaign ran it for 120 epochs; it enters the ensemble as a loss-diversity member and is also the strongest plain MLP we obtain.

MLP. Input layer $41 \to 1024$, three residual SiLU blocks of width 1024, output $1024 \to 1681$. AdamW, learning rate 10^{-3}, weight decay 10^{-5}, cosine schedule, 400 epochs, batch 256. 4.91M parameters. A wider variant (width 1536, five residual blocks, 14.5M parameters) is trained under the same recipe.

Refiner. The same residual trunk with input layer $(41 + 1681) \to 1024$: the load is concatenated with the kernel method's stress prediction for that load. Training uses four-fold out-of-fold kernel predictions for the input channel and full-data kernel predictions at evaluation; otherwise identical to the MLP but run for 100 epochs (the trainer's own default of 300 is not what the campaign used). 6.64M parameters. Reflection augmentation flips the load and permutes both the kernel channel and the target by the row-reversal index.

Common training details. Loss: mean over the batch of $\|\hat{v} - v\|_2/\|v\|_2$ in original units (predictions denormalized inside the loss; targets standardized by the per-pixel training mean and global standard deviation), except for the MSE variant above. Reflection augmentation with probability 1/2; reflection averaging at evaluation. Model selection: validation error computed every 10 epochs, best checkpoint kept. The ensemble's diversity comes from the architectures and losses rather than from reseeding; the correlation analysis of Section 4.6 indicates same-architecture reseeds would be more correlated still.

Seed protocol. One seed value enters every stochastic component at once: the permutation that draws the validation split from the training pool, each member's initialization and batch order, the half-split of the stacking comparison, the subsamples used to tune the kernel stages, and the conformal calibration draw; the pool/test boundary itself never moves. ClimSim is the one exception, and deliberately so: that benchmark fixes no test split of its own, so the seed there also draws the 20000-row test block out of the full set (`campaign/climsim_seeded.py`). Its seed spreads therefore include test-sampling variation, which the structural-mechanics and OCO-2 spreads exclude by construction; that makes them the more conservative of the two conventions where ClimSim is concerned, and it is why the ClimSim seed scatter is quoted as a check on the crossing rather than as a precision claim. Tables report the mean and standard deviation across seeds. What the repository carries is the per-seed result record for every stage of every run — the JSON files the tables and this appendix are computed from, together with the scripts that compute them. The network checkpoints and the full prediction arrays those records were derived from are too large for the repository; what is released in their place, for the structural-mechanics chain, is the residual of every member at every seed, in the form described under "Released residuals" below. A reader can therefore recheck every reported number against the records, rescore a new combination rule on the stored residuals, and rerun the pipeline from the code. Where a statement below turns on an array rather than on a record, it is marked.

What the campaign completed. The structural-mechanics chain was enqueued at ten seeds on four CPU instances and the members completed at all ten: the kernel ridge, the metric-loss MLP, the normalized-MSE MLP, and the refiner, forty trained artifacts in total. The campaign's 36-hour per-task wall clock then landed inside the FNO stage at every seed, which is why the two remaining members of the published configuration are in different states. The FNO has a best-validation checkpoint at all ten seeds, written by the trainer each time validation improved; those checkpoints are finalized here by running the tail of the trainer on them (evaluate, then save the named artifact), so the FNO is a ten-seed member and nothing about it is retrained. The UNet was scheduled after the FNO, and the wall clock reached it at only two of the ten seeds; it is the one member of Table 6's six-member row that the campaign supplies at two seeds rather than ten, and completing that row would require training eight more UNets. The FNO's own training curves are worth recording because they bound what the truncation cost: its best validation epoch falls between 10 and 70 of the scheduled 200, and at every seed validation was already rising again by the last epoch reached, in several cases sharply (seed 3, 0.0489 at its best against 0.0550 at epoch 150; seed 5, 0.0483 against 0.0557). The member is therefore the early-stopped optimum a full run would also have kept, with the caveat that the cosine schedule never annealed and a late anneal might have found more. With the FNO finalized, all ten seeds carry the five-member stack, the residual correction, the second-moment measurement and the conformal calibration; the four-member versions of each are retained and reported so that the FNO's contribution is a measurement rather than an assumption. Two tasks were lost to conditions worth recording because they shape what a reader should expect from a rerun: one instance was killed by the operating system while forming the 19000×19000 Gram matrix through full-size temporaries (the released code fills it in row blocks instead, Section 4.10), and four seeds hit the wall clock during stack stages and were completed on a larger instance afterwards. The OCO-2 and ClimSim sweeps of Section 5.3 report their own seed counts, which are ten per band and five, three and three by training size respectively. The low-data chain completed at ten seeds. The OCO-2 head pipeline is replicated at ten seeds on each of the three bands — thirty band-seeds at a matched 250-epoch budget, with one further run at 750 epochs to separate budget effects from ordering effects. Each band-seed writes its member predictions and per-sample errors, which is what makes the combination rules, the floors and the scoreboard of Section 5 recomputable without retraining; those arrays stay with the run and are not part of the distributed release. Seven of the thirty band-seeds were additionally run a second time, on different hardware, as a queue was being drained. That accident is a useful control, and we report it as measured: six of the ten rows reproduce to the printed precision across the two runs, including the kernel rows, the flat-metric network, its feature head and the combination that the paper reports, while the three weighted-loss members and their heads move by up to 0.39 points. Seeding fixes the initialization and the data order but not every reduction the training performs, so a seeded rerun of those members should be expected to land nearby rather than exactly; the tables' standard deviations, which are over seeds, are larger than this in every case.

Reconciling the stored arrays. Every member's saved prediction arrays are checked against the metric recorded in its own training record before any ensemble is formed: the array is rescored under the project's metric and the two must agree. The check matters because each seed has its own validation partition, and scoring one seed's predictions against another's split produces errors near 28% that are otherwise easy to mistake for a training failure — and, worse, produces no discrepancy at all on the one seed whose split happens to be the reference. Reconciliation is run at every seed and every member. On the test block, which no seed moves, all forty pairs agree with their recorded reflection-averaged error to between 10^{-14} and 6×10^{-8}, the precision at which the arrays are stored.

Released residuals. The residual and selection analyses of Sections 4.6 and 4.3 consume the member residuals $r_m = f_m - y$ rather than the predictions, and the residuals are what we release. For each of the ten seeds one compressed NumPy archive, `sm_residuals_s`k`.npz`, holds the test-block residual of each of the five members and of the reported pipeline (per-pixel stack plus kernel correction) as a 20000×1681 array, the validation-split residual of each member as a 1000×1681 array, the target norms $\|y\|$ of every sample on both splits, and the index arrays of both splits; the residuals are stored in IEEE half precision (`float16`), the norms and indices at full width. The archives were written from the reconciled arrays above by `campaign/export_residuals.py`, which also recomputes every member's validation and test error from the half-precision residual and compares it with the stage record the tables are built from: over all 110 arrays the two agree to within 5.4×10^{-8} in the relative-error unit, and the largest residual in any array is 192 against target norms of 3.7×10^3 to

1.7×10^4, so the rounding is far below the seed-to-seed spread of any quantity in the paper. At 384 MB per seed, 3.84 GB in all, the archives exceed what the code repository can carry; they are to be deposited with the published version, and the repository keeps the manifest (`runs/residuals_manifest.json`: per array the seed, the split seed, member, split, shape, dtype, the SHA-256 of its bytes, the largest residual and the three errors just described; per archive its size and SHA-256), so a deposited file can be checked byte for byte against the version the paper was computed from. From these files alone, without the dataset, a reader can recompute every member error of Table 5 as the mean of $\|r_m\|/\|y\|$ over samples; the second-moment matrix S, the simplex weights, the predicted and realized risks and the dispersion factor of Proposition 6.8 (`campaign/secmom_seeded5.py` does nothing else with the arrays than form $r_m/\|y\|$ on the validation split and evaluate $\|\sum_m w_m r_m\|/\|y\|$ on test, which is exact because the weights sum to one); the pairwise residual correlations and the equicorrelated floor; the equal-weight and fitted ensembles, the spatial error maps and the hard-sample comparison of Section 4.6; and the raw conformal scores $\|r\|$ of the pipeline, calibrated on the same seed-drawn subset of the test block. The per-pixel affine refit and the kernel correction need the predictions themselves, $f_m = y + r_m$, and the kernel's posterior scale needs the Gram matrix; both follow from the distributed dataset and the released code. The network checkpoints, the training-split predictions and the OCO-2 and ClimSim arrays remain with the runs, as stated where they arise.

Compute record. The campaign recorded wall clock but not memory, and Table 10 reports what it recorded; nothing in it is a new measurement. Each member's trainer writes its own elapsed minutes and the epoch of the checkpoint it kept into its run record, prints a cumulative timer every ten epochs, and prints the device it selected; the kernel stage prints the elapsed seconds of every grid cell, one Cholesky factorization and solve of the 19000×19000 system followed by the validation prediction, and the out-of-fold stage the seconds per fold at $n = 14250$. Every task ran six threads, five tasks to an instance, so the figures are wall clock under the campaign's own load rather than the cost of an isolated job, and the spread across seeds is mostly the spread across hosts: the four seeds the larger instance lists as its own in the campaign's task records (2, 4, 8 and 9) are separated from the other six by the kernel's per-cell time without overlap, 20–35 s against 42–113 s, and the table keeps the two classes apart. The FNO is the one member without a completed record. Its rate is read from the ten-epoch timer prints: the uninterrupted intervals run at 467–625 s per epoch on the standard instances and 348–410 s on the larger one, and the remaining intervals, which reach 10^4 s per epoch, record contention and suspension on the shared hosts rather than the method, which is why the table gives the median of all intervals alongside the minimum. The checkpoints that were kept sit at epochs 10 to 70 and were reached after 1.3–8.0 h of training at the five seeds whose timer ran uninterrupted to that epoch, and after 9.0–16.9 h at the other five, whose timers include such periods. The inference time for the test block was not recorded for any member (the prediction logs carry the score only), and no stage recorded peak memory; the out-of-memory marker at seed 4 is the Gram-matrix failure described above. The harvest, with the per-seed values behind every entry, is `runs/timing_harvest.json`, produced by `campaign/harvest_timing.py` from the run records and logs.

Table 10: Compute record of the structural-mechanics members, as written by the campaign's own trainers and logs (all stages on CPU, six threads per task, five tasks sharing each instance). Network rows: the trainer's elapsed minutes from the first epoch to the final evaluation, as the range over seeds, with the scheduled epochs and the epoch of the kept checkpoint (range over the ten seeds). The FNO row gives seconds per epoch from the ten-epoch timer prints, minimum and median over all intervals (count in parentheses), since no seed completed the schedule. Kernel rows: the whole stage (fifteen grid cells and the refit) and the single factor-and-solve it repeats, as ranges, with the number of timed cells or folds in parentheses. The two columns are the two instance classes of the campaign, with the seed count behind each.

Stage	Epochs: scheduled / kept	Standard instances (6 seeds)	Larger insta
Metric-loss MLP	400 / 39–79	129–158 min	26–3
MSE MLP	120 / 99–119	9.7–11.9 min	6.6–1
Refiner	100 / 99	9.9–13.1 min	7.9–3
FNO, per epoch	200 / 10–70	467 s min, 581 s median (88)	348 s min, 43
Kernel ridge, grid and refit	—	17.0–26.0 min	6.1–7
one factor-and-solve, $n = 19000$	—	42–113 s (90)	20–3
out-of-fold fold, $n = 14250$	—	32–66 s (24)	13–2

Stacking. Global weights: convex, initialized at the simplex minimizer of the measured second-moment matrix S (Proposition 6.8) and polished by a short random search on the validation metric. Per-pixel weights: affine in the members at each grid point, ridge parameter 10^{-3}, fitted on half the validation split and accepted only if they beat the global weights on the held-out half, then refitted on the full split.

Kernel stages. Matérn-5/2 on standardized inputs. The residual correction searches the scale grid $\{1, 2, 4\}\times$ the median pairwise distance (estimated on 2000 points) and the nugget grid $\{10^{-6}, 10^{-5}, 10^{-3}\}$ (scaled by n); the low-data chain of Table 4 uses $\{0.5, 1, 2, 4\}$ with nuggets $\{10^{-7}, 10^{-5}, 10^{-3}\}$. Both are tuned on validation using an 8000-sample subsample of the training set, refit at the chosen pair on the full training set in double precision. The pure kernel baseline (Table 3) uses the grid $\{0.5, 0.75, 1, 1.5, 2\}\times$median and nuggets $\{10^{-8}, 10^{-6}, 10^{-4}\}$ directly at $n = 19000$. Feature stage: penultimate activations of the ensemble members (FNO: spatially averaged pre-projection channels, 128; transformer: mean-pooled encoder tokens, 192; MLP: trunk output, 1024), concatenated, standardized, same kernel family and tuning.

Uncertainty. Posterior standard deviation (2) computed from the Cholesky factor of $K + n\lambda I$ by triangular solves. Conformal calibration uses a random 1000-sample subset of the test block that no training, checkpointing, stacking, or tuning step ever touched; the exact split-conformal quantile of the scores on that subset sets the band, and coverage is evaluated on the disjoint remainder. The hypotheses of Proposition 6.19 therefore hold as stated, with no approximate-exchangeability caveat.

Appendix C. Numerical verification of the stated results

Every result stated in Section 6 is checked numerically, and the checks are released with the code. They come in two kinds. The first kind builds a synthetic object whose ground truth is known exactly and tests the stated inequality or identity on it: the certified bound of Theorem 6.18 is evaluated on a residual constructed inside the native space with exactly known norm, at several nuggets, where any pointwise violation beyond numerical precision would falsify the statement; the leave-half-out identity and unbiasedness of Proposition 6.2 are checked by direct enumeration; the effective-dimension identity of Lemma 6.21 is checked to machine precision and the closed form of Corollary 6.22 against the empirical spectrum of an i.i.d.-entry Gram matrix; and the rate mechanism of the heuristic calculation in Proposition 6.16 is exercised on a controlled warped target $G = g(A\,\cdot)$ with a known singular-value profile, where the adapted metric must dominate at every sample size and improve the empirical rate. All four pass: the largest relative violation of the optimal-recovery bound is -3.3×10^{-2}, that is, the bound holds with margin everywhere it was evaluated; the kernel-flow identity closes to 2.8×10^{-14}; the effective-dimension identity to 3.7×10^{-15} and its Marčenko–Pastur closed form to 4.4×10^{-5} across five nuggets spanning four orders of magnitude; and the adapted metric dominates the isotropic one at every sample size of the warped construction.

The second kind evaluates the statements whose subject is the pipeline itself on the pipeline's own released artifacts, at every seed the campaign completed. The symmetrization inequality of Proposition 6.1 is checked on every trained member. It is a statement about the expected loss, so a finite test sample may go the other way, and at ten seeds one does: over the thirty members that carry a reflection, averaging lowers test error in twenty-nine cases, by 0.16 points on the metric-trained network, 0.07 on the FNO and 0.02 on the normalized-MSE one, and raises it in the thirtieth by 0.001 points. That single reversal is the sampling noise the proposition permits and a single-seed check cannot see. The second-moment identity of Proposition 6.8 is measured on the validation residual matrix at each seed and its predicted risk compared with the realized risk of the weights it implies; the ratio is 0.938 ± 0.003 over ten seeds. The simplex minimization behind that comparison is solved by sequential quadratic programming under an explicit sum-to-one constraint. The routine released with the earlier version of this work used projected gradient with clipping and renormalization, which is not a projection onto the simplex; on the five-member matrix it terminated 0.8–3.6% above the true minimum, at near-vertex points, and the weight vectors it reported were wrong in a way that read as a finding rather than as a numerical failure. The check that caught it is the cheapest one available and is now part of the suite: evaluate the objective at the returned point, at every pure-member vertex, and at a properly solved optimum, and require the first to be no worse than the others. The bracket of Corollaries 6.11 and 6.12 is evaluated against the realized convex stack at each seed, and the two-member rule of Proposition 6.8(i) is scored prospectively on every member pair of every seed rather than on selected examples — decided on the validation split and checked on the test block, 840 pairs over the thirty OCO-2 band-seeds. Section 5.2 reports what that scoring found, which is not a uniform verdict: the rule's accuracy is governed by the margin between the two quantities it compares. The selection conditions of Propositions 6.13, 6.15 and 6.5 are checked against the realized selections, with the measured constants b and W of

Proposition 6.5 and Corollary 6.6 released next to the realized excesses of the affine stack and of the corrected surrogate. Both are capacity bounds: the affine bound evaluated at the measured b and W exceeds its realized excess by many orders of magnitude, and the corollary's right-hand side, whose remaining constant is the analytic $\kappa_{\rm an} = 10^3$ of Lemma 6.7 (the smallest nugget on the grid to the power $-1/2$), is larger still; the constants are released for what they show about the bounds rather than as a guarantee. The realized scale of the correction weights is measured separately: the largest ℓ^1 norm of $c_\gamma(u)$ over the grid and over a fixed 500-point test probe is 31.2 ± 0.9 at the three seeds at which it was evaluated at the working configuration (29.3 at a single-seed probe run outside the campaign), falling to 3.0 at the largest nugget and length scale of the grid. Being a supremum over a finite probe, that value is a lower bound on $\sup_{u,\gamma} \|c_\gamma(u)\|_1$ and not a constant the corollary can be evaluated at; the collection script is written to insert it where it assembles the bound, which would understate the right-hand side by a factor of close to a thousand through the $(1+\kappa)^2$ term and leave it vacuous all the same. The probe value sits a factor of about thirty below $\kappa_{\rm an}$, and every per-configuration probe value respects the lemma's bound $\lambda^{-1/2}$ at its own nugget. The coverage of Proposition 6.19 is computed at every seed and each observed value placed against the exact Beta law of Remark 6.20: at the 90% level ten of ten fall inside the central band $[88.1\%, 91.8\%]$, and at the 95% level ten of ten inside $[93.6\%, 96.3\%]$. Finally the data-scaling sweep of the two raw-input kernel rows is refit at nested training sizes and at ten seeds per band. The approximate rank of the sensitivity map is 12 of 20 on O2 and 13–14 of 24 on the two CO_2 bands, a rank ratio between 0.54 and 0.60, so the heuristic calculation of Proposition 6.16 would permit a rate gain here and not merely a constant; the sweep finds the constant and not the rate, and Section 5.1 reports it that way.

The bound comparisons are reported as measured constants, bound value and realized excess side by side; where an oracle risk is needed it is estimated by a deterministic refit on the test arrays, which can only make the reported ratios conservative. The synthetic checks report pass or fail against explicit tolerances; the artifact checks report the measured quantities themselves, together with the number of seeds behind each.